IMPROVED ZERO-FREE HALF-PLANES FOR THE QUASI-RIEMANN HYPOTHESIS

BAIYING LIU

Abstract. Recently, a uniform $\frac{7}{8}$ zero-free half-plane for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and its transfer to Dirichlet L-functions was given by OpenAI in [Ope26a]. In its proof, there are two chosen parameters $b = \frac{1}{8}, \ell = \frac{1}{6}$. In this note, via varying b and ℓ, following the same strategy, we slightly improve the bound $\frac{7}{8} = 0.875$ to

$$B_{\mathrm{r}} = \frac{34999}{40000} = 0.874975, (\text{when } b_{\mathrm{r}} = \frac{1}{8}, \ell_{\mathrm{r}} = \frac{1}{6} + \frac{1}{10000});$$

and

$$B_{\mathrm{new}} = \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\ldots,$$
$$(\text{when } b_{\mathrm{new}} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}, \ell_{\mathrm{new}} = \frac{33 + 8\sqrt{921}}{1653}).$$

We further vary the moment parameter κ below $3/4$, after extending the unweighted moment to a smaller fixed interval. Reoptimizing the lengths and iterating the resulting zero-free bounds gives

$$B_\infty = \frac{18\xi_\infty + 7}{18\xi_\infty + 9} = 0.874957019420098946\ldots,$$

where ξ_∞ is the unique root in $(0.38858, 0.38859)$ of $1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0$. Explicitly, with $\arccos \in [0, \pi]$,

$$\xi_\infty = \frac{20}{33} + \frac{\sqrt{1391}}{33} \cos\left(\frac{1}{3} \arccos \frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right).$$

The limiting parameters are

$$b_\infty = \frac{8(3\xi_\infty - 1)(12\xi_\infty^2 - 40\xi_\infty + 15)}{9(2\xi_\infty + 1)(24\xi_\infty^2 - 20\xi_\infty + 5)},$$
$$\ell_\infty = \frac{5 - 6\xi_\infty}{9 + 18\xi_\infty}, \qquad \kappa_\infty = \frac{18\xi_\infty + 5}{18\xi_\infty + 9} = 2B_\infty - 1.$$

We also prove optimality within the explicitly specified exponent comparison system, including free total length and detector cutoff. The main results in Parts I–IV have been formalized and verified in Lean 4, including the zero-free bounds, the extended moment estimate, the limiting argument and optimality within the stated comparison system [Liu26].

Contents

Introduction 2

Part I. Summary of the 7/8 proof 6
1. The source argument and its two-stage reduction 6
2. Arithmetic conventions 8
3. A criterion for zero-free continuation 10
4. Cubic theta series and the smoothed average 11
5. Zero detection by two character polynomials 16
6. The preliminary 11/12 bound 17

Date: 10 October 2026.

7. The modified average and its direct estimate 18
8. Local cancellation and principal residues 20
9. The two moment estimates 21
10. Counting elements from simultaneous large values 24
11. Contour estimates and the endpoint inequality 26
12. Parameter choices and the final comparison 27
13. Mellin continuation and contradiction 28
14. Transfer to Dirichlet L-functions 29

Part II. A rational extension of the bound 29
15. Statement and choice of parameters 29
16. The direct estimate at balanced scales 30
17. Euler products and extraction of the principal term 32
18. The fixed moment parameter and the nonprincipal estimate 34
19. A rational polynomial inequality and the remaining norm ranges 36
20. Combining the estimates and the continuation contradiction 38

Part III. An algebraic extension of the bound 39
21. Algebraic parameters and the scope of the result 39
22. The direct estimate with free total length 40
23. Comparison of the character-count estimates 42
24. An exact polynomial proof of the endpoint 42
25. An algebraic lower bound for the exponent constraints 44
26. The auxiliary estimates and the final contradiction 44

Part IV. A moment-parameter extension and the optimal comparison bound 46
27. Statement of the limiting zero-free bound 46
28. Extending the unweighted moment below three quarters 46
29. The same-witness count with a variable moment parameter 49
30. A family of balanced geometries 50
31. An exact uniform endpoint certificate 51
32. The one-step zero-free bound via the continuation criterion 53
33. Proof of the new zero-free bound by iteration 56
34. Proof of optimality in the full comparison system 57
Appendix A. Theta coefficient conventions 64
Appendix B. Index of cited results 64
References 66

Introduction

We study how varying the parameters in the cubic-theta argument of [Ope26a] improves its zero-free half-plane. After recalling the source argument, we derive three successive improvements and determine the optimal boundary within the specified system of estimates.

The quasi-Riemann hypothesis asks whether there is a fixed $B < 1$ such that $\zeta(s) \neq 0$ for $\operatorname{Re} s > B$, with its pole at $s = 1$ allowed. The uniform character version asks for the same B independently of conductor and height. A fixed half-plane supplies a uniform gap between zeros and the line $\operatorname{Re} s = 1$; classical zero-free regions instead narrow towards that line as conductor or height increases. For general analytic background, see [IK04,

Chapter 5]; for the Dedekind zeta function of an abelian extension, and hence its Hecke L-function factors, a precise zero-free region allowing an exceptional real zero is [TZ19, Theorem 3.1]. Zero-density estimates bound the number of zeros in a region; for example, see [GM26, Theorem 1.2].

We follow the 30 September 2026 preprint [Ope26a], retaining its numbering in all citations to it. The companion [Ope26b] gives a different 11/12 argument. OpenAI's public `math` repository has a formalization scope note [Ope26c, Scope] that lists the 7/8 statements for zeta, Dirichlet and finite-order Hecke functions, with principal poles excluded; the paper's later applications are outside that scope.

Let $F = \mathbb{Q}(\sqrt{-3})$. Let $\mathcal{Z}_F$ be the set of nontrivial zeros of all primitive finite-order Hecke L-functions over F, of arbitrary conductor; poles are excluded. Put

$$\beta_* := \sup\big(\{1/2\} \cup \{\operatorname{Re}\rho : \rho \in \mathcal{Z}_F\}\big).$$

We study the dependence of the cubic-theta argument on its length parameters and on the moment parameter κ. The stated 7/8 bound of [Ope26a] is the starting point for three deductions from its intermediate estimates. The strongest deduction is the following.

Main theorem. *The family supremum satisfies*

$$\beta_* \le B_\infty := \frac{18\xi_\infty + 7}{18\xi_\infty + 9} = 0.874957019420098946\ldots,$$

where $\xi_\infty \in (0.38858, 0.38859)$ *is the unique root there of*

$$1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0.$$

Consequently every finite-order Hecke L-function over F *and every Dirichlet L-function, including* ζ*, is zero-free for* $\operatorname{Re} s > B_\infty$*. Principal poles at* $s = 1$ *are allowed.*

This is Theorem 27.1. Part III first establishes

$$B_{\text{new}} = \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\ldots.$$

Its improvement is small:

$$\frac{7}{8} - B_{\text{new}} = 4.293020083\ldots \times 10^{-5}.$$

The original 7/8 parameters therefore leave room for improvement. For the uniform exponent comparison system specified in Part IV, B_{new} is optimal under the restriction $3/4 \le \kappa \le 1$. Proposition 25.1 gives the affine lower bound; Part IV proves necessity of its three constraints while allowing the auxiliary parameters to vary. The extension below 3/4 then gives the smaller optimum B_∞.

The choices of b *and* ℓ*.* In the original 7/8 argument the lengths are chosen with $b = 1/8$ and $\ell = 1/6$. Here $b = l_y - l_x$ is the difference between the two summation exponents, and ℓ is the total exponent of the selected prime lengths. On balanced scales,

$$X = Z^{l_x}, \qquad Y = Z^{l_y}, \qquad l_x = \frac{1 - b - \ell}{2}, \qquad l_y = \frac{1 + b - \ell}{2}.$$

The selected prime scales P_i satisfy $\prod_i P_i = Z^\ell$. For the admissible geometries used below and every $\varepsilon > 0$, the direct bound and the exponent of the principal Mellin term are respectively

$$|J_\eta(Z)| \ll_\eta Z^{l_x/2 + b/12 + \varepsilon}, \qquad C_b(s) = s - \frac{4 + b}{6},$$

where J_η is the normalized cubic-theta average constructed in Part I and generalized in Part II. Matching these exponents gives

$$C_b(B) = \frac{l_x}{2} + \frac{b}{12} \quad \Longleftrightarrow \quad B = B(\ell) := \frac{11}{12} - \frac{\ell}{4}. \tag{0.1}$$

Thus increasing ℓ lowers the comparison boundary. The allowable increase is constrained by the nonprincipal terms, whose exponents also depend on b. For the following parameter comparisons, write $E(b, \ell; \delta, x) = E_{B(\ell)}(h)$ for the relative nonprincipal exponent in (18.5), with $h = (1 + b + 3\ell)/2$ and the character-count exponent R_* of (18.2). Here $\delta = 2a - 1$ is the zero-class parameter, and $x = \bar{g}/\delta$ measures the normalized prime-sum size, as defined in Section 10; the polynomial comparison uses $(\delta, x) \in [0, 5/6] \times [0, 1/2]$. These comparisons use the fixed moment parameter $\kappa = 3/4$. The notation $\partial_\ell E$ denotes partial differentiation of $E(b, \ell; \delta, x)$ with respect to ℓ, holding b, δ, x fixed; the dependence of $B(\ell)$ and $h = (1 + b + 3\ell)/2$ on ℓ is included in this differentiation.

For the first improvement we keep the original length difference and increase the total prime length by $1/10000$:

$$b_{\mathrm{r}} = \frac{1}{8}, \qquad \ell_{\mathrm{r}} = \frac{1}{6} + \frac{1}{10000}, \qquad B_{\mathrm{r}} = B(\ell_{\mathrm{r}}) = \frac{7}{8} - \frac{1}{40000} = \frac{34999}{40000}.$$

The rational choice is deliberately conservative: b_{r} retains the original difference between the summation exponents, while $1/10000$ is a convenient perturbation rather than a distinguished or maximal value. Lemma 19.1 establishes the uniform bounds

$$E(1/8, 1/6; \delta, x) \le -\frac{1}{5000}, \qquad \partial_\ell E(1/8, \ell; \delta, x) \le \frac{45}{32}.$$

Consequently, at $\ell = \ell_{\mathrm{r}}$,

$$E(1/8, \ell_{\mathrm{r}}; \delta, x) \le -\frac{1}{5000} + \frac{45}{32 \cdot 10000} = -\frac{19}{320000} < 0.$$

Thus the improvement spends only part of the available error margin. Part II proves these bounds by an exact calculation in the Bernstein basis polynomials (defined in (19.3), in the proof of Lemma 19.1; see also [Ber12, Far12]) and verifies the remaining contour and length conditions.

For the second improvement we vary both parameters:

$$b_{\mathrm{new}} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}, \qquad \ell_{\mathrm{new}} = \frac{33 + 8\sqrt{921}}{1653},$$
$$B_{\mathrm{new}} = B(\ell_{\mathrm{new}}) = \frac{1507 - 2\sqrt{921}}{1653}.$$

The algebraic values arise from the critical error class and a tangency condition. At $x = 1/2$, the count exponent is

$$R_*(\delta, 1/2) = \frac{185 - 288\delta + 96\delta^2}{185 - 138\delta}, \qquad \partial_b E = \frac{R_*}{2} - \frac{1}{3}.$$

Hence a class with $R_* = 2/3$ imposes a constraint that cannot be relaxed by changing b. Its admissible parameter is the smaller root

$$288\delta^2 - 588\delta + 185 = 0, \qquad \delta_{\mathrm{c}} = \frac{49 - \sqrt{921}}{48}.$$

In particular, $\sqrt{921}$ comes from the discriminant $588^2 - 4 \cdot 288 \cdot 185 = 144 \cdot 921$, rather than from an arbitrary choice. At this class, $E = 0$ determines

$$\ell = \frac{5 - 6\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}} = \ell_{\mathrm{new}},$$

and the stationary condition

$$\partial_\delta E(b, \ell_{\mathrm{new}}; \delta_{\mathrm{c}}, 1/2) = 0$$

determines $b = b_{\mathrm{new}}$. These values balance the direct comparison with the largest nonprincipal exponent. The balancing equations locate the candidate; their solution alone does not establish the estimate for other classes. Part III derives the critical zero class, proves the endpoint inequality for every permitted zero and amplitude class by an exact quadratic identity, and verifies all auxiliary strict margins. The dual argument in Proposition 25.1 explains the algebraic obstruction suggested by these parameter choices. The notation new distinguishes this bound from the rational one.

Varying the moment parameter. We vary the parameter κ governing the admissible lengths in the mixed fourth-moment estimate for unweighted character polynomials. Part IV extends the unweighted moment from $3/4 \le \kappa \le 1$ to $13/18 \le \kappa \le 1$, retaining the indispensable premise

$$\beta_* \le (1+\kappa)/2 \qquad (\kappa < 1).$$

The bound from Part III therefore permits $\kappa = 2B_{\mathrm{new}} - 1 < 3/4$. Decreasing κ increases the available selected-prime length in the unweighted moment, improves the nonprincipal count, and permits a further increase of ℓ. The direct boundary $B(\ell)$ would remain B_{new} if ℓ_{new} were kept fixed. Thus the improvement uses changed length parameters as well as a changed moment parameter.

For a critical class $\delta = \xi$ and $x = 1/2$, the count equation $R_\kappa = 2/3$ becomes

$$(216\kappa - 18)\xi^2 - (468\kappa - 57)\xi + 150\kappa - 20 = 0.$$

The equations $E = 0$ and $\partial_\delta E = 0$ then give

$$\ell(\xi) = \frac{5 - 6\xi}{9 + 18\xi}, \qquad b(\xi) = \frac{8(3\xi - 1)(12\xi^2 - 40\xi + 15)}{9(2\xi + 1)(24\xi^2 - 20\xi + 5)}, \qquad B(\xi) = \frac{18\xi + 7}{18\xi + 9}.$$

Part IV proves the nonprincipal inequality uniformly in every zero and amplitude class, checks the remaining analytic margins, and obtains a new bound from each previously established moment premise. Iteration reaches the limiting equality $\kappa_\infty = 2B_\infty - 1$; eliminating κ gives the cubic in the main theorem. Only the resulting real upper bounds are passed to their limit, so uniform analytic constants over the iteration are not required.

The limiting root has the explicit form

$$\xi_\infty = \frac{20}{33} + \frac{\sqrt{1391}}{33} \cos\left(\frac{1}{3} \arccos \frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right), \qquad \arccos \in [0, \pi].$$

Substitution in the balanced geometry gives the exact limiting parameters

$$b_\infty = \frac{8(3\xi_\infty - 1)(12\xi_\infty^2 - 40\xi_\infty + 15)}{9(2\xi_\infty + 1)(24\xi_\infty^2 - 20\xi_\infty + 5)} = 0.123405685419541002\ldots,$$

$$\ell_\infty = \frac{5 - 6\xi_\infty}{9 + 18\xi_\infty} = 0.166838588986270882\ldots,$$

$$\kappa_\infty = \frac{18\xi_\infty + 5}{18\xi_\infty + 9} = 2B_\infty - 1 = 0.749914038840197892\ldots.$$

The root formula is proved in Lemma 33.1; the parameter identities follow from (30.2) and (33.3). In particular,

$$B_{\mathrm{new}} - B_\infty = 5.037906979388974\ldots \times 10^{-8}.$$

The conservative rational consequence $B_{\mathrm{cp}} = 34998281/40000000 = 0.874957025$ gives the smaller improvement $B_{\mathrm{new}} - B_{\mathrm{cp}} = 4.479916874001834\ldots \times 10^{-8}$. The algebraic lower bound in Part III concerns the restricted moment range $3/4 \le \kappa \le 1$ and does not obstruct

this extension. The main structural contribution of Part IV is Theorem 34.4: it determines the exact optimum of the explicitly specified exponent comparison system, allowing free total length, every permitted detector cutoff and contour choice, and the full moment range $13/18 \le \kappa \le 1$. The proof isolates a critical class for which every permitted detector cutoff yields a certified count exponent at least $2/3$, forcing three affine inequalities for any putative boundary $B < B_\infty$. An explicit positive weighted combination eliminates the length parameters from these inequalities; the moment compatibility condition $B \le (1+\kappa)/2$ then forces $B \ge B_\infty$, and the limiting construction shows that this lower bound is sharp. This assertion concerns the supplied uniform upper bounds and does not exclude improvements from stronger analytic estimates. The limiting pair (b_∞, ℓ_∞) consists of cubic irrationals, whereas the rational checkpoint does not attain the limit.

All three improvements retain the exact arithmetic formulas and principal residues of [Ope26a]. Their Euler correction is shown to be holomorphic below $7/8$. In Parts II and III, the original conclusion $\beta_* \le 7/8$ permits $\kappa = 3/4$, removing the loss associated with the original varying parameter. In Part IV each new moment premise is supplied by the preceding proved bound. The comparison margins then give the same Mellin-continuation contradiction as in Part I. The analytic dependencies are indexed in Appendix B.

Here is the structure of the paper. Part I recalls the $7/8$ argument and fixes the arithmetic and analytic notation used throughout. Part II establishes the shared changed-parameter estimates and gives the simpler rational extension

$$B_{\mathrm{r}} = 34999/40000, \qquad 7/8 - B_{\mathrm{r}} = 1/40000.$$

Its calculation in the Bernstein basis polynomials is retained as a separate rational example; this bound is superseded by the stronger one in Part III. Part III varies both length parameters, proves the error-exponent inequality at the algebraic parameter choices, and discusses the resulting algebraic lower bound. Part IV proves the smaller-κ moment estimate, derives the changed count and length parameters, and iterates the analytic comparison to B_∞. All three extensions use the same continuation criterion and Dirichlet transfer from Part I.

Acknowledgments. OpenAI Codex (GPT 6.1 Sol Medium-Ultra) assisted with drafting, source checking, and the search for improved bounds. Anthropic Claude (Opus 5.5 Medium) assisted with auditing. The zero-free conclusions for B_{r}, B_{new} and B_∞ have Lean 4 formalizations for the finite-order Hecke family over F and its Dirichlet and zeta consequences. The Lean formalization, exact-arithmetic certificates and verification instructions are available in the accompanying GitHub repository [Liu26].

Part I. Summary of the 7/8 proof

1. The source argument and its two-stage reduction

The argument compares two descriptions of one cubic-theta average: reflection bounds it directly, while Poisson summation isolates a principal Mellin integral involving $1/L_F(s,\eta)$. A zero makes two character polynomials simultaneously large; their moment bounds control the remaining Poisson terms. Part I makes those dependencies, the common-character hypotheses and the numerical exponent comparisons explicit. Its statements record the source argument using the proved results indexed in Appendix B. The accompanying explanations retain the decisive identities, exponent comparisons and contradiction, including calculations used in Parts II–IV. Complete proofs of the imported results are given at the cited locations in [Ope26a]. Numbered statements have this paper's numbering; locators inside citations always refer to the cited work.

Theorem 1.1 (Main statement of [Ope26a, Theorem 1.1]). *Every finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ has no zeros in*

$$\operatorname{Re} s > 7/8. \tag{1.1}$$

The same holds for every Dirichlet L-function, including ζ. Principal poles at $s = 1$ are allowed; the boundary $\operatorname{Re} s = 7/8$ is not included.

The Hecke and Dirichlet conclusions are proved below in Theorem 13.1 and Corollary 14.1. Here $L_F(s,\eta)$ denotes the Hecke L-function over F, and $\zeta(s)$ the Riemann zeta function; η denotes the character fixed in each application.

For the primitive Hecke family define

$$\beta_* = \sup\left(\{1/2\} \cup \left\{\operatorname{Re}\rho : \begin{array}{l} L_F(\rho,\eta) = 0, \quad 1/2 \le \operatorname{Re}\rho \le 1, \\ \eta \text{ primitive and finite-order} \end{array}\right\}\right). \tag{1.2}$$

Poles are excluded. The inserted value $1/2$ gives $\beta_* \ge 1/2$; absolute convergence of the Euler product gives $\beta_* \le 1$. The family in (1.2) comprises *all primitive finite-order Hecke characters over F, of arbitrary conductor*. Indeed, Poisson summation produces denominators $L_F^S(s, \eta\overline{\chi_\bullet(u)})$ in (4.7). The primitive inducing characters of these twists belong to that family, so the same β_* controls their zeros.

The two stages. Stage 1 proves $\beta_* \le 11/12$. Stage 2 assumes, for contradiction, $\beta_* > 7/8$ and sets

$$\Delta = \beta_* - 7/8 \in (0, 1/24], \qquad \kappa = 2\beta_* - 1 = 3/4 + 2\Delta \le 5/6. \tag{1.3}$$

Group the nonprincipal terms by the location of the zeros of their associated twists; these groups are the zero classes of Section 5. In the second stage they satisfy

$$\delta = 2a - 1 \le \kappa \le \alpha := 5/6.$$

This restriction is essential in the source's character-count argument: the long-polynomial bound contains $(\alpha - \delta)r$, and its comparison with $r \le t$ uses $\alpha - \delta \ge 0$. Thus the preliminary theorem cannot be omitted from this proof without replacing that input.

The parameter choice giving $7/8$. The normalized modified average has a direct bound $Z^{L+\varepsilon}$, while its principal residue has Mellin exponent $C(s)$, where

$$L = \frac{l_x}{2} + \frac{b}{12}, \qquad b = l_y - l_x, \qquad C(s) = s + \frac{l_x}{3} - \frac{5}{6} + \frac{\ell}{6}.$$

Under $l_x + l_y = 1 - \ell$, matching these exponents gives

$$C(B) = L \quad \Longleftrightarrow \quad B = \frac{11}{12} - \frac{\ell}{4}.$$

The reflected mean-square calculation in [Ope26a, Lemma 15.1, proof] contains the algebraic excess

$$\left(\frac{1 + 3\ell' - 2M'}{4}\right)_+ = \left(\frac{d_J + 5\ell - 1}{4}\right)_+, \qquad \ell' = \ell - d_J, \quad M' = 1 - \ell - 2d_J.$$

For $0 \le d_J \le \ell$, this term vanishes for every subset exactly when $6\ell \le 1$. The convenient choice $\ell = 1/6$ therefore gives $B = 7/8$ in the exponent matching above. With the paper's scales (7.1),

$$B = 7/8, \qquad C(s) = s - 11/16, \qquad C(7/8) = L = 3/16.$$

This exponent matching motivates the parameters; proving a strict saving for every nonprincipal contribution requires both large values from the zero detector and the two moment estimates. That saving continues every reciprocal $1/L_F(s,\eta)$ to the left of β_*, producing the contradiction.

Sections 2–4 introduce the characters, continuation criterion and theta average. Sections 5–6 give the detector and the 11/12 stage. Sections 7–12 develop the modified average, moments, counts and exponent savings. Sections 13–14 complete the continuation and Dirichlet transfer.

2. Arithmetic conventions

We fix the arithmetic conventions used in every subsequent construction, including character presentations, zero extensions and reciprocity phases. These choices determine the local factors of the theta average in Section 4 and must be retained when its parameters vary.

Set

$$\omega = e^{2\pi i/3}, \quad \mathcal{O} = \mathbb{Z}[\omega], \quad \lambda = \sqrt{-3}, \quad q_a = |a|^2, \quad q_{\mathfrak{a}} = N\mathfrak{a}, \quad \varpi(a) = a/|a|.$$

$\mathcal{O}$ is norm-Euclidean. Ideals coprime to 3 have unique primary generators $a \equiv 1 \pmod 3$. Ideal sums use these generators; sums over elements may also include units. For an integral ideal $\mathfrak{a}$, $N\mathfrak{a} = |\mathcal{O}/\mathfrak{a}|$. Write $v_p(a)$ for its prime valuation, $\mathrm{rad}(a) = \prod_{p|a} p$, and μ for the ideal Möbius function. The notation sf in a sum means squarefree, and 1_E is the indicator of a condition E. Thus $(a, b) = 1$ means that the corresponding ideals are coprime. A prime outside S is called good.

Unless specified otherwise, $Z \geq 2$ is the asymptotic scale; $\varepsilon, \varepsilon_1 > 0$ are arbitrarily small losses. The symbols $\ll, \gg, \asymp$ refer to inequalities up to constants, and $x_+ = \max(x, 0)$. Auxiliary letters in separately stated lemmas have local scope: in particular, K in (4.4) is a norm bound, whereas K in Section 7 counts prime intervals. An annular weight is a smooth function supported on a fixed compact subinterval of $(0, \infty)$; its logarithmic derivatives are bounded. Constants A, J in height estimates denote finite nonnegative exponents or derivative orders fixed before the height cutoff.

Fix a finite set S containing primes above 6, primes in the full defining moduli of fixed character presentations, and primes of norm at most P_0. Unless stated otherwise, ideal sums avoid S. The datum is fixed independently of the scale and the current summation variables; it may depend on an already fixed character η. Denote these fixed moduli, characters, ray group, exclusions and normalizations collectively by $\mathcal{A}$. The positive real P_0 is the fixed prime-norm cutoff.

Symbol	Meaning and scope
β_*, Δ, κ	Family zero supremum; $\Delta = \beta_* - 7/8$; $\kappa = 2\beta_* - 1$.
$\alpha = 5/6,\ \varpi(a)$	Fixed amplification slope; unit phase $a/\lvert a\rvert$ (written $\alpha(a)$ in [Ope26a]).
$\epsilon_{\mathrm{b}},\ \varepsilon$	Fixed zero-class width (written e in [Ope26a]); arbitrarily small exponent loss.
$\mathfrak{a}(r)$	Amplified moment exponent $\max\{1, (1+5r)/6\}$ (written $e(r)$ in [Ope26a]).
$Z, U = Z^d, T_1$	Asymptotic scale; element norm scale; zero-detection height.
l_x, l_y, b, h, ℓ	Averaging exponents; $b = l_y - l_x$; combined frequency exponent; total auxiliary prime length.
z_{cap}, z_0	Upper bound for total prime length in (9.2); real contour value $z_0 = 17/50$ in Section 11.
$\mathfrak{r}$	Coprimality modulus in (9.2) (written R in [Ope26a]); distinct from a counting exponent R.
$\mathcal{R}, \mathcal{M}$	Reciprocity phase; Mellin transform of the Fourier-transformed weight.
H_η, J_η, f_η	Nonvanishing correction; normalized average; principal-residue Mellin integral.
$\Gamma_1(3), \Gamma_2;\ \Gamma(s)$	Matrix groups; Euler gamma function.

Other indices and length parameters are local to the statement in which they are introduced. A *zero class* is a set of elements satisfying the same pair of zero-location inequalities (5.1); an *element index* is the outer summation variable in a family of character polynomials.

Characters and their presentations. For an integral modulus $\mathfrak{f}$, the ray class group is $I(\mathfrak{f})/P_1(\mathfrak{f})$: $I(\mathfrak{f})$ consists of fractional ideals coprime to $\mathfrak{f}$, and $P_1(\mathfrak{f})$ of principal ideals generated by elements congruent to 1 modulo $\mathfrak{f}$. A finite-order Hecke character is a homomorphism from this finite group to roots of unity. Its conductor is the smallest modulus through which it factors; it is primitive when the displayed modulus is its conductor. A presentation includes a possibly larger defining modulus and the zero extension at its nonunits, as in [Ope26a, Section 4]. Initially,

$$L_F(s,\eta) = \sum_{\mathfrak{a}\neq 0} \eta(\mathfrak{a})(N\mathfrak{a})^{-s} = \prod_p (1-\eta(p)q_p^{-s})^{-1}, \qquad \operatorname{Re} s > 1;$$

outside this region the notation refers to its meromorphic continuation. A norm twist multiplies a character by $\mathfrak{a} \mapsto (N\mathfrak{a})^{it}$, $t \in \mathbb{R}$.

For a prime $p \notin S$, the sextic residue character is

$$\chi_p(a) = (a/p)_6, \qquad \chi_p(a) \equiv a^{(q_p-1)/6} \pmod p \quad (p \nmid a), \qquad \chi_p(a) = 0 \quad (p \mid a).$$

For primary $c = \prod p^{v_p(c)}$ put $\chi_c(a) = \prod_{p\mid c} \chi_p(a)^{v_p(c)}$. Throughout, all integer powers, including negative ones, are zero on nonunits, including powers divisible by six:

$$\chi_c(a)^6 = 1_{(a,c)=1}. \tag{2.1}$$

Passing to a primitive inducing character therefore requires retaining the removed coprimality conditions. Coefficients and such conditions are independent of the variable being averaged, except for the explicitly displayed residue symbols.

The additive normalization is

$$e(z) = \exp\bigl(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda)\bigr), \qquad d\mu(z) = \frac{2}{\sqrt{3}}\, d\xi_1\, d\xi_2.$$

Here $z = \xi_1 + i\xi_2$ with $\xi_1, \xi_2 \in \mathbb{R}$; for $z \in \mathbb{C}$ the trace in $e(z)$ means $z/\lambda + \overline{z/\lambda}$. The pairing $(z_1, z_2) \mapsto e(z_1 z_2)$ makes $\mathcal{O}$ self-dual. For squarefree c define

$$\gamma_j(c) = q_c^{-1/2} \sum_{v \bmod c} \chi_c(v)^j e(v/c).$$

Fixed-numerator symbols $n \mapsto \chi_n(u)$ are finite-order ray characters [Ope26a, Lemma 4.1]. Prime Gauss identities and sextic reciprocity provide finite-ray phases $\mathcal{R}, G$. Precisely, $\mathcal{R}$ is the bicharacter on residue square classes modulo 4 defined in [Ope26a, Lemma 4.4]; on coprime primary pairs it satisfies

$$\chi_b(a) = \mathcal{R}(a, b)\chi_a(b).$$

The same lemma defines the modulus-one finite-ray function G on all primary indices; on squarefree c, $G(c) = \overline{\chi_c(4)}\gamma_3(c)$. Its extension to nonsquarefree indices is that finite-ray function, rather than a Gauss sum, and satisfies

$$G(AB) = G(A)G(B)\mathcal{R}(A, B)$$

on the admissible primary elements [Ope26a, Lemmas 4.2–4.4]. These phases cancel cross-prime factors in the later Euler product.

3. A criterion for zero-free continuation

We isolate the analytic implication that converts the two estimates for an average into a zero-free half-plane. All four parts use this criterion; their remaining task is to construct the average and verify its hypotheses with a saving independent of the target character.

For a finite excluded set,

$$L_F^S(s, \eta) = L_F(s, \eta) \prod_{p \in S} (1 - \eta(p) q_p^{-s}).$$

The deleted factors are nonzero for $\operatorname{Re} s > 0$.

The formulation below is taken from [Ope26a, Proposition 2.1]. Its proof uses the classical principle that a growth bound for an inverse Mellin transform yields holomorphic continuation of a reciprocal L-function, and hence zero-freeness; compare Riesz's criterion [Rie16, Sections 3–4]. Here the positive saving must be uniform over the target characters, whereas the implied constants may depend on the character.

Proposition 3.1 (Continuation criterion, [Ope26a, Proposition 2.1])**.** *Fix $1/2 < \sigma_0 < 1$ and $C(s) = s + c$, $c \in \mathbb{R}$. Suppose that, whenever $\beta_* > \sigma_0$, the following construction is available. For every primitive η, suppose H_η is holomorphic on $\operatorname{Re} s > \sigma_0$ and*

$$\sup_{\operatorname{Re} s > \sigma_0} |H_\eta(s) - 1| \le 1/2. \tag{3.1}$$

Define

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)}\, ds. \tag{3.2}$$

Here (v) denotes a vertical line of real part v. Assume there is a normalized average J_η satisfying

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega_0}, \qquad 0 < \omega_0 < \beta_* - \sigma_0, \tag{3.3}$$

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma}, \qquad \sigma > 0, \tag{3.4}$$

The positive numbers ω_0, σ are independent of η; constants, finite excluded sets and lower thresholds may depend on η. Then $\beta_* \le \sigma_0$.

Idea of the continuation argument. Assume $\beta_* > \sigma_0$ and put

$$\epsilon_c = \min\{\beta_* - \sigma_0 - \omega_0,\ \sigma,\ (\beta_* - \sigma_0)/2\} > 0.$$

The two estimates give $f_\eta(Z) \ll_\eta Z^{C(\beta_*)-\epsilon_c}$ for $Z \geq 1$. Moving (3.2) to any fixed line $\operatorname{Re} s = b_c > 2$ gives $f_\eta(Z) \ll_{\eta,b_c} Z^{b_c+c}$ for $0 < Z \leq 1$. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s-c}\,\frac{dZ}{Z}$$

is holomorphic on $\operatorname{Re} s > \beta_* - \epsilon_c > \sigma_0$. Mellin inversion and analytic continuation identify

$$F_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s,\eta) \qquad (\operatorname{Re} s > 1) \quad \text{[Ope26a, Proposition 2.1]}.$$

Since $|H_\eta| \geq 1/2$, the function

$$e^{-(s-5/6)^2} F_\eta(s)/H_\eta(s)$$

is a holomorphic continuation of $1/L_F^S(s,\eta)$ to $\operatorname{Re} s > \beta_* - \epsilon_c$, for every primitive η. The supremum definition supplies some η_0 and a zero ρ there; attainment of β_* is unnecessary. Locally,

$$L_F^S(s,\eta_0) = (s-\rho)^{k_\rho} g(s), \qquad k_\rho \geq 1, \quad g(\rho) \neq 0.$$

Its reciprocal has a pole, contradicting this holomorphic continuation. Neither the Gaussian nor H_{η_0} can cancel it. Principal poles, in contrast, give zeros of the reciprocal and cause no obstruction.

The two applications in Part I use

σ_0	$C(s)$	$C(\sigma_0)$
11/12	$s - 2/3$	1/4
7/8	$s - 11/16$	3/16

and different normalized averages.

4. Cubic theta series and the smoothed average

We construct the cubic-theta average whose two representations enter Proposition 3.1. Theta reflection supplies the mean squares for its direct estimate, while Poisson summation exposes the reciprocal L-function and the nonprincipal terms to be controlled in the following sections.

4.1. **The cubic theta function and its Fourier coefficients.** The function used here is *Kubota's cubic metaplectic theta function* θ_{cub} on $\mathbb{H}^3 = \mathbb{C} \times \mathbb{R}_{>0}$. Kubota developed the underlying metaplectic construction [Kub69]; Patterson determined the explicit cubic theta coefficients [Pat77, Theorem 8.1]. We use the normalization and cusp formulas recorded by Dunn–Radziwiłł [DR24, Section 5.1 and Appendix A], as implemented in [Ope26a, Proposition 5.1]. Here *cubic* refers to the cubic residue symbol in its automorphy multiplier. Set

$$\Gamma_1(3) = \{g \in \mathrm{SL}_2(\mathcal{O}) : g \equiv I \pmod 3\}, \qquad \Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma_1(3)\rangle.$$

For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{C})$, its action on $w = (z,v) \in \mathbb{H}^3$ is

$$g(z,v) = \left(\frac{(az+b)\overline{(cz+d)} + a\bar{c}\,v^2}{|cz+d|^2 + |c|^2 v^2},\ \frac{v}{|cz+d|^2 + |c|^2 v^2} \right).$$

The transformation law is

$$\theta_{\mathrm{cub}}(gw) = \kappa_{\mathrm{cub}}(g)\theta_{\mathrm{cub}}(w), \qquad \kappa_{\mathrm{cub}}(g) = \begin{cases} (c/a)_3, & g \in \Gamma_1(3),\ c \neq 0, \\ 1, & g \in \Gamma_1(3),\ c = 0, \text{ or } g \in \mathrm{SL}_2(\mathbb{Z}). \end{cases}$$

The multiplier extends to Γ_2 as a homomorphism [Pat77, Section 2]; the action and multiplier above are [DR24, equations (5.1)–(5.4)]. The symbol $(c/a)_3$ is the cubic residue symbol, multiplicative in its primary denominator; at good primes $(x/p)_3 = \chi_p(x)^2$. The notation κ_{cub} distinguishes this multiplier from the real zero-bound parameter κ.

The Fourier expansions needed in the proof are those at the three cusps represented by

$$H_j = \begin{pmatrix} 1 & 0 \\ t_j & 1 \end{pmatrix}, \qquad (t_0, t_1, t_2) = (0, \omega^2, \omega), \qquad j = 0, 1, 2.$$

Each H_j sends ∞ to a boundary cusp in $F \cup \{\infty\}$; expanding $\theta_{\mathrm{cub}} \circ H_j$ gives its Fourier expansion there. With $\check{e}(x) = \exp(2\pi i(x + \bar{x}))$, these expansions are

$$\overline{\theta_{\mathrm{cub}}(H_j(z, v))} = C_j(v) + \sum_{\substack{\mu \in \lambda^{-4}\mathcal{O} \\ \mu \neq 0}} d_j(\mu)\, v K_{1/3}(4\pi|\mu|v)\check{e}(\mu z).$$

Here

$$C_0(v) = \frac{3^{5/2}}{2} v^{2/3}, \qquad C_1(v) = C_2(v) = 0.$$

The symbol d_j denotes the fixed arithmetic Fourier-coefficient function, and $K_{1/3}$ the modified Bessel function of the second kind. The precise correspondence with the notation of [DR24] is given in Appendix A. In particular, for primary squarefree c and primary n,

$$d_0(\lambda^{-3} c n^3) = 3^{5/2}|n|\, \widetilde{g}_3(c), \qquad \widetilde{g}_3(c) = q_c^{-1/2} \sum_{x \bmod c} (x/c)_3 \check{e}(x/c).$$

When $(c, S) = 1$ this becomes

$$\widetilde{g}_3(c) = \chi_c(\lambda)^{-2} \gamma_2(c).$$

Thus the squarefree Gauss coefficients and the cube multiples in the following Dirichlet series come directly from these Fourier coefficients. The automorphy law supplies the reflection formula; its explicit coefficients supply the mean-square estimates. These are the theta inputs used below; Appendix A checks the constants directly from [DR24, equations (5.6), (5.7), (5.15), (5.16)].

4.2. **Completed theta coefficients and mean squares.** For a ray character Ψ with the conventions of Section 2, and a weight $V \in C^\infty(0, \infty)$ satisfying, for every $C > 0$ and $j \geq 0$,

$$(x\partial_x)^j V(x) = O_{C,j}(x^C) \quad (x \to 0), \qquad (x\partial_x)^j V(x) = O_{C,j}(x^{-C}) \quad (x \to \infty),$$

put

$$\begin{aligned} D(s, \Psi) &= \sum_{c \text{ sf}} \gamma_2(c) \overline{\varpi(c)} \Psi(c) q_c^{-s}, \\ L_0(s, \Psi) &= \sum_{n} \overline{\varpi(n)}^3 \Psi(n)^3 q_n^{-3s+1/2}, \\ T(s, \Psi) &= D(s, \Psi) L_0(s, \Psi). \end{aligned}$$

The series $T(s, \Psi)$ is the character-twisted Dirichlet series built from the theta coefficients after the fixed rescaling and phase conversion in [Ope26a, Proposition 5.1]; it converges absolutely for $\mathrm{Re}\, s > 1$. Its smoothed expression is

$$\mathcal{C}_V(X; \Psi) = \sum_{c \text{ sf}, n} \frac{\gamma_2(c) \overline{\varpi(cn^3)} \Psi(cn^3)}{\sqrt{q_c}\, q_n} V(q_c q_n^3 / X). \tag{4.1}$$

No condition $(c,n)=1$ is imposed. The cube factor is included because the natural theta support consists of the completed indices cn^3. Here *completed* means that all cube multiples of the squarefree coefficient are included. For a primary A,

$$A=cn^3 \quad\Longleftrightarrow\quad v_p(A)\equiv 0 \text{ or } 1 \pmod 3 \quad \text{for every } p,$$

with c squarefree and c,n uniquely determined primary generators. It is an arithmetic completion of the indices; the gamma factors enter separately through reflection.

Proposition 4.1 (Cubic theta reflection, [Ope26a, Proposition 5.1]). *Fix a finite family of ray characters with conductors supported on S, and a common periodicity modulus L_{per} supported on S, divisible by 18 and their defining moduli, as specified in* [Ope26a, Proposition 5.1, hypotheses]. *Write $\Psi=\Psi_0\prod_{p\in\mathcal{P}}\chi_p^{j_p}$ on primary elements, with $\mathcal{P}$ a finite set of distinct primes outside S and $j_p\in\{0,\dots,5\}$; zero extensions are retained. The product $T(s,\Psi)$ continues to an entire function with polynomial growth on each fixed vertical strip, with constants depending on the fixed arithmetic data and the strip. For $X>0$ and the test V specified above, the precise reflection identity is*

$$\mathcal{C}_V(X;\Psi)=\sum_{\mathfrak{t}}\zeta_{\mathfrak{t}}\sum_{0\neq\mu\in\lambda^{-4}\mathcal{O}}\frac{d_{\mathfrak{t}}(\mu)\varpi(\mu)\vartheta_{\mathfrak{t}}(\lambda^4\mu)}{\sqrt{q_\mu}}\prod_{p\text{ active}}B_p(\lambda^4\mu)V^\sharp(q_\mu X/q_{c_{\mathfrak{t}}}^2). \tag{4.2}$$

Here $\mathfrak{t}$ runs over the terms indexed by $h_0 \bmod L_{\mathrm{per}}$ and an active subset of $\mathcal{P}$ [Ope26a, Proposition 5.1, item (1)]. *Every prime with $j_p\neq 0$ is active. The denominator is $c_{\mathfrak{t}}=c_F\prod_{p\text{ active}}p$, where c_F is the reduced denominator of $\lambda^2h_0/L_{\mathrm{per}}$, with the normalizing unit fixed as in that item. The function $d_{\mathfrak{t}}$ is one of d_0,d_1,d_2; $\vartheta_{\mathfrak{t}}$ is an additive character modulo a fixed modulus depending on L_{per}; the scalar $\zeta_{\mathfrak{t}}$ is independent of μ and has $|\zeta_{\mathfrak{t}}|\le 1/81$. Their exact phases are specified in* [Ope26a, Proposition 5.1, items (2)–(3)]. *The local factors, all zero on nonunits when a character is evaluated, are*

$$B_p(x)=\begin{cases}\chi_p(x)^{-j_p-2}, & j_p\neq 0,4,\\ q_p^{-1/2}(-1+q_p1_{p|x}), & j_p=4,\\ q_p^{-1/2}\chi_p(x)^{-2}, & j_p=0 \text{ and } p \text{ active.}\end{cases}$$

Its radial transform is

$$V^\sharp(x)=\frac{1}{2\pi i}\int_{(v)}\widehat{V}(-t)\prod_{\pm}\frac{\Gamma(1+t\pm 1/6)}{\Gamma(1-t\pm 1/6)}\left(\frac{(2\pi)^4x}{27}\right)^{-t}dt,\qquad v\ge 0,$$

where Γ is the Euler gamma function and $\widehat{V}(t)=\int_0^\infty V(x)x^t\,dx/x$. The dual coefficients have support $u\lambda^k cn^3$, where $u\in\mathcal{O}^\times$ is a unit, $k\in\mathbb{Z}$, $k\ge -4$, c is squarefree, and c,n are primary. At such an index each of the three fixed cusp coefficient functions d_j of Section 4.1 has absolute value $\le 27\,3^{k/6}|n|$ [Ope26a, Proposition 5.1]. *The formula retains the local character zeros, including the coprimality conditions of imprimitive presentations.*

The quadratic large sieve over number fields is [GL13, Definition 1 and Theorem 1.1].

Lemma 4.2 (Completed theta mean square, [Ope26a, Lemma 5.8]). *Fix the arithmetic data, bounded ranges for M,N and dyadic support comparisons. For every $\varepsilon>0$, there exist finite seminorm orders A,J such that, for all sufficiently large Z and every nonempty powerful-part dyadic range O, one has*

$$\sum_{\substack{0<q_m\ll Z^M\\ q_{m_{\mathrm{pow}}}\asymp Z^O}}|\mathcal{C}_V(Z^N;\nu\chi_\bullet(m))|^2\ll Z^{O/2+\max(M-O,2M-O-N)+\varepsilon}\mathfrak{M}(V)^2. \tag{4.3}$$

Here $M, O \geq 0$ and $N \in \mathbb{R}$ are norm-range exponents, ν is a fixed finite ray character, and $\chi_\bullet(m)$ means $n \mapsto \chi_n(m)$. The ideal $m_{\mathrm{pow}} = \prod_{v_p(m)\geq 2} p^{v_p(m)}$ is the powerful part. The seminorm is $\mathfrak{M} = \mathfrak{M}_{A,J}$ for sufficiently large fixed A, J, where [Ope26a, Proposition 5.1] *defines*

$$\mathfrak{M}_{A,J}(V) = \sup_{-A\leq\sigma\leq 1/4} \int_{\mathbb{R}} (1+|t|)^J |\widehat{V}(\sigma+it)|\, dt.$$

Fixed ray decompositions and kernel decay justify the specialization to the Gaussian test [Ope26a, Lemmas 5.2–5.7].

For the later divisibility restrictions, the corresponding mean-square estimate is [Ope26a, Lemma 14.3]. Its arithmetic input [Ope26a, Lemma 14.2] combines the quadratic sieve [GL13, Theorem 1.1] and the cubic sieve [HB00, Theorem 2].

Lemma 4.3 (Quadratic–cubic sieve, [Ope26a, Lemma 14.2])**.** *Let K, N, B, L be independent positive real norm scales:*

$$K, N, B, L \geq 1, \qquad q_k \ll K, \quad q_n \asymp N, \quad q_b \asymp B, \quad q_P \asymp L.$$

Here k, n, P are squarefree primary ideals and b is any primary ideal. Thus $KNBL$ is the product of the four norm scales, not a further variable. The indices k, P avoid S; n, b may meet S except at λ, as specified in [Ope26a, Lemma 14.2]. *For every $\varepsilon > 0$,*

$$\sum_{k \text{ sf},\ q_k\ll K} \left| \sum_{P \text{ sf},\ q_P\asymp L} \frac{a(P)}{\sqrt{q_P}} 1_{(P,k)=1} \sum_{\substack{n \text{ sf},\ q_n\asymp N \\ q_b\asymp B}} \beta(n,b)\chi_k(nb)^3 \chi_P(n)^{-2} 1_{(P,b)=1} \right|^2 \tag{4.4}$$
$$\ll (KNBL)^\varepsilon (K+NB)B\{N+L+(NL)^{2/3}\}.$$

The coefficient functions satisfy $|a(P)|, |\beta(n,b)| \leq 1$; $a(P)$ is independent of k, n, b, and $\beta(n,b)$ of k, P.

The ideal P need not be prime. Inclusion-exclusion removes the moving coprimality condition before applying the sieves. The full reflected estimate follows by dyadic decomposition, the local theta coefficients, and decay of $V^\sharp$.

4.3. **Definition and separation.** Choose a fixed finite ray group $\mathcal{T}$ for the supplementary phases [Ope26a, Lemma 4.4]. Here $\widehat{\mathcal{T}}$ is its character group, and $1_{\mathcal{T}}$ its identity ray class. Membership $p \in 1_{\mathcal{T}}$ means that the primary generator of p has trivial image in $\mathcal{T}$. Write $\Theta = \langle \eta, \widehat{\mathcal{T}} \rangle$ for the finite group of Hecke characters generated by these characters; all its conductor primes are in S [Ope26a, Section 8]. Let b_* generate the squarefree product of primes in S and choose a primitive character $\xi \bmod b_*$ of order dividing six, nonprincipal at each local prime. For a residue character ψ modulo a primary modulus c and $h \in \mathcal{O}$, set

$$g_\psi(c,h) = \sum_{d \bmod c} \psi(d) e(hd/c), \quad \tau_* = q_{b_*}^{-1/2} g_\xi(b_*, 1), \quad |\tau_*| = 1, \quad \Xi(A) = \xi(A)\chi_A(b_*).$$

For the primary averaging variable v let

$$\Psi_{m,v,\eta}(A) = \eta(A)\Xi(A)^{-1}\mathcal{R}(A,v)\chi_A(m)$$

on primary elements coprime to S, extended by zero elsewhere, and define

$$T_{m,v,\eta}(t) = \sum_{c \text{ sf}, n} \gamma_2(c)\overline{\varpi(cn^3)}\Psi_{m,v,\eta}(cn^3)\overline{G(cn^3)} q_c^{-1/2-t} q_n^{-1-3t}.$$

For fixed nonnegative nonzero annular W_0, W_1 and $\Phi(t) = e^{t^2}$, the source's average is

$$I_\eta(X,Y,Z) = \frac{1}{Y}\sum_v \frac{W_1(q_v/Y)\chi_v(b_*)}{\tau_*\xi(v)\sqrt{q_{b_*}q_vX}}\sum_m \xi(m)q_v^{-1/2}g_{\chi_v}(v,-m) \times W_0(q_m/(q_{b_*}q_vX))\frac{1}{2\pi i}\int_{(4)} Z^t\Phi(t)T_{m,v,\eta}(t)dt. \tag{4.5}$$

The m, v sums are finite, and the completed series converges absolutely on the initial line.

Expand $\overline{G(A)} = \sum_{\theta\in\widehat{\mathcal{T}}} a_\theta\theta(A)$, where $\sum|a_\theta| \le |\mathcal{T}|^{1/2}$. For a class $\sigma \in \mathcal{T}$ and representative v_σ, put $\nu_\sigma(A) = \eta(A)\Xi(A)^{-1}\mathcal{R}(A, v_\sigma)$ and

$$B_{m,\sigma}(Z) = \sum_\theta a_\theta\frac{1}{2\pi i}\int_{(4)} Z^t\Phi(t)T(t+1/2, \nu_\sigma\theta\chi_\bullet(m))dt,$$

$$A_{m,\sigma,t_0}(Y) = \frac{1}{Y}\sum_{v\in\sigma}\frac{W_1(q_v/Y)\chi_v(b_*)}{\tau_*\xi(v)}(q_v/Y)^{-1/2+it_0}q_v^{-1/2}g_{\chi_v}(v,-m).$$

Writing $Q = q_{b_*}XY$, Mellin inversion separates the average:

$$I_\eta = \frac{Q^{-1/2}}{2\pi}\sum_{\sigma\in\mathcal{T}}\int_{\mathbb{R}}\widehat{W}_0(it_0)\sum_{m\ne0}\Omega(q_m/Q)\xi(m)(q_m/Q)^{-it_0}A_{m,\sigma,t_0}(Y)B_{m,\sigma}(Z)\,dt_0. \tag{4.6}$$

Here $t_0 \in \mathbb{R}$ is the Fourier integration variable, $\widehat{W}_0(it_0) = \int_0^\infty W_0(r)r^{it_0}\,dr/r$, and Ω is a fixed smooth cutoff equal to one on the support of W_0 [Ope26a, Section 6]. Thus Cauchy–Schwarz reduces a direct estimate to two mean squares.

4.4. **Poisson summation and the scalar Euler factors.** Poisson summation in m removes zero frequency and frequencies meeting S, by primitivity of ξ. Every remaining element is uniquely ua^6, where u includes its unit and has prime valuations at most five, while a is primary. Let $\widetilde{W}_0(|y|^2)$ be the planar Fourier transform of $W_0(|y|^2)$ with the pairing $e(xy)$ and measure $d\mu$ defined in Section 2; this is the transform in [Ope26a, Section 7]. Set

$$\mathcal{M}(z) = \int_0^\infty \widetilde{W}_0(r)r^{z-1}dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r)r^{w-1}dr,$$

$$\mathcal{W} = X^{1/2-z}Z^{s+z-1}Y^{w-1}\Phi(s+z-1)\mathcal{M}(z)\widehat{W}_1(w).$$

The complete local calculation [Ope26a, Lemma 7.1] gives

$$I_\eta = \frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)}\mathcal{W}\sum_u^{(6)} q_u^{-z}\overline{\xi(u)}\frac{\zeta_F^S(6z)L_F^S(w,\chi_\bullet(u))}{L_F^S(s,\eta\overline{\chi_\bullet(u)})}\mathcal{H}_{\eta,u}(s,w,z)\,dz\,dw\,ds. \tag{4.7}$$

Here $\chi_\bullet(u) : n \mapsto \chi_n(u)$ and $\sum^{(6)}$ means sixth-power-free nonzero elements coprime to S.

Let $\zeta_F^S(s) = L_F^S(s,1)$ be the Dedekind zeta function with the primes of S deleted. The local coefficient $P_p = P_p(s,w,z;\eta,u)$ is the complete valuation series computed in [Ope26a, Lemma 7.1], immediately before and in its local identity: $v_p(c) \in \{0,1\}$ and $v_p(n), v_p(v), v_p(a) \ge 0$ are summed, with $v_p(u) \in \{0,\dots,5\}$ fixed. This summary's s is the variable called x in that lemma. For its scalar Euler factors write

$$V_p = q_p^{-6z}, \quad W_p = \chi_p(u)q_p^{-w}, \quad D_p = \eta(p)\overline{\chi_p(u)}q_p^{-s}, \qquad H_p = P_p\frac{(1-V_p)(1-W_p)}{1-D_p}.$$

Then $\mathcal{H}_{\eta,u} = \prod_{p\notin S}H_p$. Gauss identities, reciprocity and the inserted phases factor the coefficient into these local terms. The six cases $v_p(u) = 0,\dots,5$ are summed before

extracting the scalar quotient. The correction is holomorphic in the two regions

$$\mathcal{D}_1: \quad s_r \ge 51/100, \quad z_r \ge 17/50, \quad w_r \ge -1/100, \quad s_r + w_r \ge 1 + \varepsilon_0, \tag{4.8}$$

$$\mathcal{D}_2: \quad s_r \ge 7/8, \quad z_r \ge 33/200, \quad w_r \ge 19/20, \tag{4.9}$$

where $s_r = \operatorname{Re} s$, $w_r = \operatorname{Re} w$, $z_r = \operatorname{Re} z$, and $\varepsilon_0 > 0$ is a fixed margin, with $\mathcal{H}_{\eta,u} \ll_\varepsilon q_u^\varepsilon$ uniformly in heights.

The only principal numerator is $u = 1$: a good prime in u forces nontrivial ramification, and a nontrivial unit gives a nontrivial Kummer character. At $u = 1$ the poles are $w = 1, z = 1/6$. Define

$$H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6), \qquad c_S = \widehat{W}_1(1)\mathcal{M}(1/6)\frac{(\operatorname{Res}_{v=1}\zeta_F^S(v))^2}{6} > 0. \tag{4.10}$$

The $1/6$ is the residue Jacobian. A positive Gaussian-integral representation proves $\mathcal{M}(1/6) > 0$. Since $(s, 1, 1/6) \in \mathcal{D}_2$ for $\operatorname{Re} s > 7/8$, (4.9) and (4.10) give holomorphy of H_η there. The uniform local estimate $H_\eta(s) = 1 + O(P_0^{-4/5})$ [Ope26a, Lemma 7.1] allows P_0 to be chosen sufficiently large that

$$\sup_{\operatorname{Re} s > 7/8} |H_\eta(s) - 1| \le 1/2. \tag{4.11}$$

The cutoff is independent of the target phases; see also [Ope26a, Section 11, principal correction].

5. Zero detection by two character polynomials

We associate the nonprincipal terms from Section 4 with zero classes and detecting character polynomials. A selected zero forces two polynomials with the same character and height to be simultaneously large; the moment estimates of Section 9 will turn this information into a count.

For the zero-detection method, see [MP24, Appendix C, Zero detecting setup]. The simultaneous Hecke-character version used here is [Ope26a, Proposition 8.3].

For $q_u \asymp U = Z^d$ consider

$$\mathcal{X}_u = \{\nu(n)\chi_n(u)^{\pm 1} : \nu \in \Theta\}.$$

For a presentation ψ, let ψ^* denote its primitive inducing character, and $L(s, \psi)$ its Hecke series with the original zero extension; thus $L(s, \psi^*) = L_F(s, \psi^*)$. Primitive conductors have norm $\ll_{\mathcal{A}} U$; original presentations retain Euler deletions supported on S and u. Choose $0 < \epsilon_{\mathrm{b}} < 10^{-3}$ and $T_1 = Z^\tau \le U^{1/100}$. Set $I_{\epsilon_{\mathrm{b}}} = \lceil 1/\epsilon_{\mathrm{b}} \rceil + 2$. With $a_0 = 51/100$ and $j \in \{1, \dots, I_{\epsilon_{\mathrm{b}}}\}$, let

$$M_j(u) = \max\big(\{a_0\} \cup \{\operatorname{Re}\rho : \psi \in \mathcal{X}_u,\ L(\rho, \psi^*) = 0,\ \operatorname{Re}\rho \ge a_0,\ |\operatorname{Im}\rho| \le 3jT_1\}\big).$$

A pigeonhole argument gives $1 \le i < I_{\epsilon_{\mathrm{b}}}$ and $a \in (a_0 + \epsilon_{\mathrm{b}}\mathbb{Z}_{\ge 0}) \cap [a_0, 1]$ such that

$$a \le M_i(u) < a + \epsilon_{\mathrm{b}}, \qquad M_{i+1}(u) < a + 2\epsilon_{\mathrm{b}} \qquad \text{[Ope26a, Lemma 8.1]}. \tag{5.1}$$

Hence the associated functions have a zero-free rectangle to the right of $a + 2\epsilon_{\mathrm{b}}$ at the needed height. Concretely,

$$\mathcal{U}_{a,i} = \{s \in \mathbb{C} : a + 2\epsilon_{\mathrm{b}} \le \operatorname{Re} s \le 2,\ |\operatorname{Im} s| \le 3(i+1)T_1\}$$

contains no zero of $L_F(s, \psi^*)$ for $\psi \in \mathcal{X}_u$, by (5.1) and the definition of M_{i+1}. If $a > a_0$, a selected zero $\rho = \sigma + i\gamma$ satisfies $a \le \sigma < a + \epsilon_{\mathrm{b}}$, $|\gamma| \le 3iT_1$. Since these are actual family zeros, $a \le \beta_*$.

Let $\mathcal{B}$ denote a subset of sixth-power-free elements $q_u \asymp U$ in one fixed (i, a) class; later it may also fix prime-sum size ranges. All class counts are finite for fixed parameters

[Ope26a, Lemma 8.1]. For one presentation $\psi \in \mathcal{X}_u$, real length exponents $r, m \geq 0$, and annular weights W_M, W_S, define

$$M_r = U^{-r/2} \sum_n \mu(n)\psi(n) W_M(q_n/U^r), \qquad S_m = U^{-m/2} \sum_n \psi(n) W_S(q_n/U^m).$$

The smooth weights may contain pure norm twists. Polynomial bounds on fixed vertical strips, logarithmic bounds in zero-free disks and deleted Euler estimates [Ope26a, Lemmas 4.8–4.10] yield

$$|M_r|^2 \ll U^{\delta r+\varepsilon}, \qquad |S_m|^2 \ll U^{\delta \min(m,1-m)+\varepsilon}, \qquad \delta = 2a-1, \tag{5.2}$$

when $(1+T_1)^A \leq U^{\varepsilon/10}$ [Ope26a, Lemma 8.2]. The second estimate takes the better of the direct and reflected Hecke functional-equation bounds. For the classical functional equation in this setting, see [GZ20, equation (1.1)]; the zero-class bounds with deleted Euler factors are those of [Ope26a, Lemma 8.2].

Proposition 5.1 (Simultaneous large values, [Ope26a, Proposition 8.3])**.** *Fix $\varepsilon > 0$, a class (i, a) from (5.1) with $a > a_0$, and a sixth-power-free u of norm $\asymp U$ in that class. Assume $T_1 \leq U^{1/100}$ and the height bound $(1+T_1)^A \leq U^{\varepsilon/10}$ from* [Ope26a, Lemma 8.2]*, after choosing its preliminary losses sufficiently small. For every $t \in [1, 3/2]$ and all $U \geq U_0(\mathcal{A}, \varepsilon)$, there exist a common presentation $\psi \in \mathcal{X}_u$, a selected zero ρ of its primitive inducing function, and two polynomials with the same presentation and twist height such that*

$$r \leq t + O(1/\log U), \quad r + m \geq t - O(1/\log U), \tag{5.3}$$

$$t - \tfrac{1}{2} - O(\varepsilon) \leq r, \qquad 0 \leq m \leq \tfrac{1}{2} + O(\varepsilon), \tag{5.4}$$

$$|M_r|^2 \gg U^{\delta r-\varepsilon}, \qquad |S_m|^2 \gg U^{\delta m-\varepsilon} \qquad \text{[Ope26a, Proposition 8.3].} \tag{5.5}$$

Idea of zero detection. To obtain them, put

$$C_\psi(s) = \sum_l \mu(l)\psi(l) V_{\leq}(q_l/U^t) q_l^{-s}, \qquad V_{\leq} = 1 \text{ on } [0,1], \quad V_{\leq} = 0 \text{ on } [2,\infty),$$

where $V_{\leq}$ is a fixed smooth cutoff. Here l ranges over integral ideals and t is the real truncation exponent. Use this truncated reciprocal in

$$\frac{1}{2\pi i} \int_{(2)} U^{20z} \Gamma(z) L(\rho+z, \psi) C_\psi(\rho+z)\, dz.$$

Shifting to $-1/4$ gives a small integral; the residue at zero vanishes because $L(\rho, \psi) = 0$. Expansion of LC_ψ cancels the ideal coefficients up to length U^t except the unit term. A remaining dyadic product therefore has $|M_r S_m|^2 \gg U^{\delta(r+m)-\varepsilon}$. Comparison with (5.2) forces $m \leq 1/2 + O(\varepsilon)$ and then both individual lower bounds. At $a = a_0$ there need be no selected zero; that class is counted directly.

6. The preliminary 11/12 bound

We first apply the continuation criterion at 11/12, using the unmodified average and a preliminary character count. This bound restricts the possible zero classes and supplies the starting point for the refined 7/8 argument in the remaining sections of Part I.

Take $X = Y = Z^{1/2}$ and $Q = q_{b_*} Z$. For the number-field additive large sieve, see [Hux68]; its exact planar form and application here are [Ope26a, Lemmas 6.1–6.2]. It gives

$$\sum_{q_m \ll Q} |A_{m,\sigma,t_0}(Y)|^2 \ll (Q + Y^2)/Y \ll Q/Y \qquad \text{[Ope26a, Lemmas 6.1–6.2].}$$

Equation (4.3), at $M = N = 1$, gives $\sum |B_{m,\sigma}(Z)|^2 \ll Z^{1+\varepsilon}$. Thus (4.6) implies

(6.1) $$|I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll Q^{-1/2}(Q/Y)^{1/2} Z^{1/2+\varepsilon} = Z^{1/4+\varepsilon} \qquad \text{[Ope26a, Proposition 6.3].}$$

The antecedents are the cubic sieve [HB00, Theorem 2], the quadratic sieve [GL13, Theorem 1.1], and the higher-order norm recursion [BGL14, Theorem 1.3 and Section 3]. An Eisenstein-integer sextic formulation is also recorded in [GZ24, Lemma 2.9]; [Ope26a, Lemma 9.1] gives the primary-generator convention used here.

Lemma 6.1 (Sextic sieve, [Ope26a, Lemma 9.1]). *Let $K, N_{\mathrm{s}} \geq 1$ bound the norms of squarefree primary elements k, n coprime to S, and let c_n be finitely supported and independent of k. For every $\varepsilon > 0$,*

$$\sum_{k \text{ sf},\ q_k \leq K} \left| \sum_{n \text{ sf},\ q_n \leq N_{\mathrm{s}}} c_n \chi_n(k)^{\pm 1} \right|^2 \ll (KN_{\mathrm{s}})^{\varepsilon} \{K + N_{\mathrm{s}} + (KN_{\mathrm{s}})^{2/3}\} \sum_n |c_n|^2.$$

Proposition 6.2 (Preliminary character count, [Ope26a, Proposition 9.2]). *Let $\mathcal{B}$ be a zero class of sixth-power-free elements $q_u \asymp U$ as in Section 5, with the detector height assumptions and $\delta = 2a - 1$. For every $\varepsilon > 0$ and sufficiently large U,*

(6.2) $$\#\mathcal{B} \ll U^{R_{\mathrm{I}}(\delta)+\varepsilon}(1 + T_1)^A, \quad R_{\mathrm{I}}(\delta) = \min\{1, \max(1 - \delta/2, 4/3 - \delta)\}.$$

Proposition 6.3 (Preliminary zero-free bound, [Ope26a, Theorem 3.1]). *The family supremum in (1.2) satisfies*

(6.3) $$\beta_* \leq 11/12.$$

The preliminary exponent comparison. Assume $\beta_* > 11/12$ and put $\Delta_1 = \beta_* - 11/12 > 0$. The principal residue in (4.7) has exponent $X^{1/3} Z^{s-5/6} = Z^{s-2/3}$. Hence

$$C_{\mathrm{I}}(s) = s - 2/3, \qquad J_{\mathrm{I},\eta}(Z) = I_\eta(Z^{1/2}, Z^{1/2}, Z)/c_S,$$

with $f_{\mathrm{I},\eta}$ defined by (3.2). The preliminary character count, the contour estimates [Ope26a, Lemmas 10.2–10.6], and the height choice in [Ope26a, Proposition 11.2] give

$$|J_{\mathrm{I},\eta} - f_{\mathrm{I},\eta}| \ll_\eta Z^{C_{\mathrm{I}}(\beta_*) - 1/4800}.$$

Apply Proposition 3.1 with

$$\sigma_0 = 11/12, \qquad c = -2/3, \qquad \omega_0 = \Delta_1/2, \qquad \sigma = 1/4800.$$

The correction hypotheses follow from (4.9), (4.10) and (4.11). The displayed comparison bound and (6.1), with its loss chosen below $\Delta_1/2$, verify the two estimates, since $C_{\mathrm{I}}(11/12) = 1/4$. Thus the preliminary argument reaches the contradiction through the same continuation criterion used for the 7/8 bound.

7. The modified average and its direct estimate

Starting from the preliminary bound, we introduce selected prime factors and unequal averaging lengths to obtain the stronger direct estimate needed at 7/8. The resulting modified average is also the construction retained in Parts II–IV; the next section identifies its principal Mellin term.

Assume $\beta_* > 7/8$ and use (1.3). The averaging scales are

(7.1) $$\begin{gathered} b = 1/8, \quad h = 13/16, \quad \ell = 1/6, \quad l_x = 17/48, \quad l_y = 23/48, \\ X = Z^{l_x}, \quad Y = Z^{l_y}, \quad l_x + l_y = 5/6, \quad b = l_y - l_x = 1/8, \quad h = 1 - l_x + \ell. \end{gathered}$$

Choose a positive integer K and K disjoint prime intervals with smooth nonnegative weights W_i, norms $q_p \asymp P_i = Z^{\ell_i}$, and $\sum_i \ell_i = \ell$. The allowed set $\mathcal{P}_i$ also imposes $p \notin S$ and $p \in 1_{\mathcal{T}}$. The underlying intervals are disjoint before these restrictions.

For a squarefree ideal D outside S, let $I_{\eta;D}$ denote (4.5) with $1_{D|cn^3}$ inserted. Write $J^c = \{1,\dots,K\} \setminus J$ and $p_J = \prod_{i\in J} p_i$ (empty product 1). All tuple sums have $p_i \in \mathcal{P}_i$. Define

$$I_{\eta,\mathrm{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J\subseteq\{1,\dots,K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \times I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \tag{7.2}$$

The one-prime operation is $\overline{\eta(p)} I_{\eta;p}(X,Y,Zq_p) - q_p^{-3/2} I_\eta(X/q_p, Y/q_p, Z)$. Its purpose is to cancel an unwanted scalar Euler contribution, leaving a sextic-character prime sum. All prime intervals retain their original scales.

Proposition 7.1 (Modified direct bound, [Ope26a, Proposition 15.3]). *For the fixed arithmetic data, prime intervals and smooth weights specified above, with the scales (7.1), one has*

$$|I_{\eta,\mathrm{mod}}(Z)| \ll_{\eta,\varepsilon} Z^{3/16+\varepsilon}. \tag{7.3}$$

For a fixed J put

$$d_J = \sum_{i\in J} \ell_i, \quad X' = X/q_{p_J}, \quad Y' = Y/q_{p_J}, \quad M' = 5/6 - 2d_J, \quad Q = q_{b_*} X'Y'.$$

Let $B^J_{m,\sigma}$ be the completed factor in the separated expression, with the primes $i \notin J$ already summed and their divisibility restrictions imposed. The theta functional equation with the divisibility restrictions [Ope26a, Corollary 14.1] and its reflected mean square give the following input.

Lemma 7.2 (Modified theta mean square, [Ope26a, Lemma 15.1]). *For every $\varepsilon > 0$, fixed subset J and ray class σ, uniformly in the fixed rescaled prime tuple,*

$$\sum_{q_m \ll Q} |B^J_{m,\sigma}|^2 \ll Z^{M'+((d_J-1/6)/4)_+ +\varepsilon} = Z^{M'+\varepsilon} \qquad \text{[Ope26a, Lemma 15.1].} \tag{7.4}$$

The extra term vanishes because $0 \le d_J \le \ell = 1/6$. The remaining prime sums are included in $B^J_{m,\sigma}$.

Proposition 7.3 (Additive mean square, [Ope26a, Proposition 15.2]). *Fix $S, \mathcal{T}, b_*, \xi, W_1$ and a ray class σ. Let $Q, Y' \ge 1$ range over fixed polynomial ranges in Z, and suppose $P_a = Y'^2/Q \ge 1$. For every $\varepsilon > 0$ and real t_0, with the coefficient A defined in Section 4,*

$$\sum_{q_m \ll Q} |A_{m,\sigma,t_0}(Y')|^2 \ll (1+|t_0|)^{J_{\mathrm{Gr}}} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'}\right) Z^{\varepsilon} \qquad \text{[Ope26a, Proposition 15.2].} \tag{7.5}$$

Here J_{Gr} is a fixed seminorm order, independent of the subset J, moving scales and t_0.

Idea of the additive estimate. Expansion of the square and Poisson summation reduce (7.5) to complete finite correlations. The common-factor identities and local prime-power formulas are [Ope26a, Lemmas 13.2–13.4]; the final mean-square estimate is [Ope26a, Proposition 15.2]. Both error terms $P_a^{1/6}$ and P_a^2/Y' in (7.5) are retained. Their comparison at the chosen scales is the calculation needed below.

The direct exponent calculation. With the fixed scales (7.1),

$$P_a = q_{b_*}^{-1} Z^{1/8}, \qquad l_y - d_J - 11b/6 \ge 1/12, \qquad P_a^2/Y' \ll P_a^{1/6}.$$

Thus $P_a \ge 1$ once $Z^{1/8} \ge q_{b_*}$, which holds for all sufficiently large Z with the arithmetic data fixed. Equations (7.4)–(7.5) and Cauchy–Schwarz therefore bound each separated

term by $(X')^{1/2}P_a^{1/12}Z^{\varepsilon}$. The rescaled prime tuples have total weight

$$Z^{d_J+\varepsilon}O(Z^{-3d_J/2}), \qquad (X')^{1/2} \asymp X^{1/2}Z^{-d_J/2}.$$

Consequently the exponent in (7.3) is $l_x/2 + b/12 = 3/16$, since $d_J - 3d_J/2 - d_J/2 = -d_J \le 0$. The asymmetry in (7.1) was chosen to obtain this exponent while keeping the additive error under control.

8. Local cancellation and principal residues

We compute the effect of the modification in Section 7 on the local Euler factors and principal residues. This gives the normalized average and its nonvanishing principal correction, together with the joint local error estimate used in Section 11.

8.1. **Holomorphic correction and local errors.** Let P_p^* be the part of P_p with positive valuation of the completed index. For $Q_p = q_p$, the selected-prime operation replaces P_p by

$$Q_p^{z-1}\{\overline{\eta(p)}Q_p^s P_p^* - Q_p^{-w}P_p\}.$$

After extracting the same scalar Euler factors as in (4.7), define

$$G_p = \{\overline{\eta(p)}Q_p^s P_p^* - Q_p^{-w}P_p\}\frac{(1-V_p)(1-W_p)}{1-D_p}. \tag{8.1}$$

The local identity of [Ope26a, Lemma 7.1] shows that G_p continues holomorphically in $\mathcal{D}_1, \mathcal{D}_2$. It does not require division by P_p. The correction is

$$\mathfrak{H}_{\eta,u,Z} = \sum_{(p_i)}\prod_i [W_i(q_{p_i}/P_i)q_{p_i}^{z-1}G_{p_i}] \prod_{\substack{p\notin S\\ p\notin\{p_i\}}} H_p. \tag{8.2}$$

Replacing $\mathcal{H}$ by $\mathfrak{H}$ in (4.7) gives the exact modified average.

On the contours used for the nonprincipal contributions, $s_r = a + 16\epsilon_{\mathrm{b}}, w_r = 1 - a - 6\epsilon_{\mathrm{b}}, z_r = 17/50$, [Ope26a, Section 16] decomposes

$$\mathfrak{H} = \mathcal{H}\prod_i(\mathcal{Q}_i + \mathcal{D}_i), \qquad \mathcal{Q}_i = -\sum_{p\in\mathcal{P}_i} W_i(q_p/P_i)q_p^{z-1}\overline{\chi_p(u)}.$$

Precisely, on these contours define

$$\mathcal{B}_i = \sum_{p\in\mathcal{P}_i} W_i(q_p/P_i)q_p^{z-1}\frac{G_p}{H_p}, \qquad \mathcal{D}_i = \mathcal{B}_i - \mathcal{Q}_i.$$

Proposition 8.1 (Joint local error, [Ope26a, Proposition 16.1])**.** *Let $a_0 \le a \le 1$, $0 < \epsilon_{\mathrm{b}} \le 10^{-3}$, $q_u \asymp U$, with u sixth-power-free and $(u,S) = 1$. Put $\delta = 2a - 1$. Suppose the primitive character ψ_u^* inducing $\chi_\bullet(u)$ is nonprincipal. On the displayed contours restrict to $|\Im w| \le T_1$, and assume $|L_F(1-w, \overline{\psi_u^*})| \ll U^{\varepsilon}$ uniformly on the points under consideration. For sufficiently large P_0, the local H_p are nonzero on these contours. Moreover, for every subset $I \subseteq \{1,\dots,K\}$, the joint error estimate of* [Ope26a, Proposition 16.1] *is*

$$\left|L_F^S(w,\chi_\bullet(u))\prod_{i\in I}\mathcal{D}_i\right| \ll U^{\delta/2+O(\epsilon_{\mathrm{b}})+\varepsilon}(3+T_1)^A\prod_{i\in I}P_i^{17/50-1/2+O(\epsilon_{\mathrm{b}})+\varepsilon}. \tag{8.3}$$

Let $\mathfrak{f}_u$ be the conductor ideal of ψ_u^*, and I_0 any set of distinct selected primes dividing u. With $j_p = v_p(u) \in \{1,\dots,5\}$,

$$q_{\mathfrak{f}_u} \ll_S q_{\mathrm{rad}(u)} \le q_u\prod_{p\in I_0} q_p^{-(j_p-1)}.$$

Applying this conductor deficit once to the primitive numerator's functional equation pays for all strict ramified local errors simultaneously. Absolute bounds on the other contours use [Ope26a, Lemma 16.2].

8.2. **The principal coefficient and the normalized average.** At $u = 1$ in $\mathcal{D}_2$, $G_p/H_p = -1 + O(q_p^{-7/8})$ uniformly in heights.

Lemma 8.2 (Prime asymptotic in a fixed ray class, [Ope26a, Lemma 13.1])**.** *Fix the ray quotient $\mathcal{T}$, the excluded set S, and a nonnegative nonzero smooth annular weight W_i. As $P_i \to \infty$, with $\mathcal{P}_i$ restricted to the identity class of $\mathcal{T}$ and primes outside S, one has*

$$S_i(Z) := \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{-5/6} \sim \frac{P_i^{1/6}}{|\mathcal{T}| \log P_i} \int_0^\infty W_i(y) y^{-5/6} dy > 0. \tag{8.4}$$

Fixed-ray prime counting follows by applying Chebotarev to the fixed ray-class field; see [TZ19, Theorem 1.1]. Smooth partial summation gives (8.4).

For large Z set

$$A_{\mathcal{T}}(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z), \qquad J_\eta(Z) = \frac{I_{\eta,\mathrm{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}. \tag{8.5}$$

Then $|A_{\mathcal{T}}| \asymp (\log Z)^{-K}$ and $|A_{\mathcal{T}}|^{-1} \ll_\varepsilon Z^\varepsilon$, so

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16+\varepsilon}. \tag{8.6}$$

At $w = 1, z = 1/6$ the correction equals

$$H_\eta(s) Z^{\ell/6} A_{\mathcal{T}}(Z) \{1 + O(Z^{-\kappa_P})\}, \qquad 0 < \kappa_P < (7/8) \min_i \ell_i.$$

The residue exponent is

$$C(s) = s + l_x/3 - 5/6 + \ell/6 = s - 11/16, \qquad C(7/8) = 3/16, \tag{8.7}$$

and the resulting integral is

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-11/16} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)} ds. \tag{8.8}$$

Crossing $w = 1$ on the way to $19/20$, then $z = 1/6$ on the way to $33/200$, gives principal remainder exponents

$$C(\beta_*) + (1+h)\epsilon_{\mathrm{b}} - m_w + \varepsilon, \quad C(\beta_*) + \epsilon_{\mathrm{b}} - m_z + \varepsilon, \quad C(\beta_*) + \epsilon_{\mathrm{b}} - \kappa_P + \varepsilon,$$

where

$$m_w = 23/960, \qquad m_z = 13/9600 \qquad [\text{Ope26a, Lemma 10.5}].$$

The reciprocal stays on $s_r = \beta_* + \epsilon_{\mathrm{b}}$ throughout.

9. The two moment estimates

We record the two moment estimates for the detecting polynomials of Section 5, one with Möbius coefficients and one without. Their different length restrictions are combined in Section 10 to bound each zero and prime-amplitude class.

All lengths in this section are logarithmic exponents in a common base Z. Products of auxiliary prime polynomials use disjoint prime intervals and one common character presentation. For example, length r means norms $\asymp Z^r$. A mean square is a sum $\sum_u |F_u|^2$; the fourth moment arises from $\sum_u |S_u S_u|^2 = \sum_u |S_u|^4$.

9.1. **Möbius-weighted second moment.** Let $\psi_u(n) = \nu(n)\chi_n(u)^{\pm 1}$, with fixed ray character ν, and

$$M_u(Z^r) = Z^{-r/2}\sum_n \mu(n)\psi_u(n)W(q_n/Z^r), \qquad Q_u = \prod_i Z^{-z_i/2}\sum_{p\in\mathcal{P}_i} a_i(p)\psi_u(p)W_i(q_p/Z^{z_i}).$$

Lemma 9.1 (Möbius-weighted moment, [Ope26a, Lemma 17.1])**.** *Let $r, m_0, z_i \geq 0$ be norm exponents: the ideal variable n has norm $\asymp Z^r$, u ranges up to Z^{m_0}, and the ith prime set has norms $\asymp Z^{z_i}$. The coefficients $a_i(p)$ are bounded and independent of u. Fix bounded ranges for these lengths and the number of prime factors. Writing $z_{\mathrm{p}} = \sum_i z_i$, assume for a fixed $c_m > 0$ that*

$$r + 2z_{\mathrm{p}} \leq m_0 - c_m, \qquad 2r + 8z_{\mathrm{p}} \leq 3m_0 - c_m.$$

Then for every $\varepsilon > 0$ and all sufficiently large Z,

$$\sum_{0<q_u\ll Z^{m_0}} |M_u(Z^r)Q_u|^2 \ll Z^{m_0+\varepsilon} \quad \text{[Ope26a, Lemma 17.1].} \tag{9.1}$$

The implied constant may depend on c_m, the bounded ranges and finitely many smooth seminorms, but is uniform in prime supports and bounded coefficients. There is no restriction on individual prime lengths in this lemma.

Structure of the inverse-moment induction. For the recursive Poisson large-sieve method, compare [HB95], [HB00, Section 4], and [GL13, Sections 4–7], as acknowledged in [Ope26a, Section 17, canonical estimate]. The precise moment estimates with the displayed auxiliary restrictions are those of [Ope26a, Sections 17–18]. One initial Poisson transform reduces the assertion to [Ope26a, Lemma 17.2]:

(9.2)

$$Z^{-V}\sum_{f\ \mathrm{sf},\ q_f\asymp Z^V} w(f)\sum_{0<q_k\ll Z^M}\left|Z^{-N/2}\sum_{n\ \mathrm{sf}}\overline{\varpi(n)}\gamma_2(n)\nu(n)1_{(n,\mathfrak{r})=1}\chi_n(kf^4)\mathfrak{d}(n)W(q_n/Z^N)\right|^2$$
$$\ll Z^{F_0+\varepsilon}, \qquad F_0 = N + V,$$

under

$$q_{\mathfrak{r}} \leq Z^{F_0-M-z_{\mathrm{cap}}-c_*}, \qquad 3M + 6z_{\mathrm{cap}} + c_* \leq 4F_0, \qquad c_* > 0.$$

The parameters $M, N, V, z_{\mathrm{cap}} \geq 0$ in this lemma are local norm exponents: f has norm $\asymp Z^V$, k norm $\ll Z^M$, and n norm $\asymp Z^N$. The squarefree ideal $\mathfrak{r}$ is the coprimality modulus, independent of k, f; $c_* > 0$ is a fixed margin and z_{cap} an upper bound for total prime length. Here $w(f)$ depends on f alone and is nonnegative and bounded by a fixed power of the divisor function. Also,

$$\mathfrak{d}(n) = \sum_{(p_i)}\prod_i a_i(p_i)1_{p_i|n}$$

has total prime length at most z_{cap}: its tuples run over disjoint fixed prime sets of norms $\asymp Z^{z_i}$, with $\sum_i z_i \leq z_{\mathrm{cap}}$. The coefficients a_i are bounded, with data independent of k, f. No additional arbitrary coefficient depending on k is admitted.

The inequalities in (9.1) supply the initial positive margins. Two finite Poisson transforms then decrease M while preserving the common character, coprimality condition and $4F_0 - 3M - 6z_{\mathrm{cap}}$ margin [Ope26a, Lemma 17.3]. The reflected mean square [Ope26a, Lemma 14.3] handles the final range of the induction, and [Ope26a, Corollary 17.4] bounds the seminorm and height orders. These preserved margins are the features needed for the later counts.

Lemma 9.2 (Sixth-power amplification, [Ope26a, Lemma 17.6]). *Fix a ray character ν, a common sign $\varepsilon_\chi \in \{-1,1\}$, and a smooth annular weight W, with the zero extensions of Lemma 9.1. Put $\psi_u(n) = \nu(n)\chi_n(u)^{\varepsilon_\chi}$ and use M_u as defined above, without auxiliary prime factors. Let $r \ge 0$ range over a fixed bounded interval and $q_u \asymp U$, with u sixth-power-free. For every $\varepsilon > 0$ and sufficiently large U,*

$$\sum_{q_u \asymp U}^{(6)} |M_u(U^r)|^2 \ll U^{\mathfrak{a}(r)+\varepsilon}(1+T_1)^A, \qquad \mathfrak{a}(r) = \max\{1, (1+5r)/6\}. \tag{9.3}$$

The estimate permits a smooth weight depending on the element u $W(y)y^{-\sigma_u+it_u}$, with σ_u in a fixed compact interval and $|t_u| \le T_1$. The factor $(1+T_1)^A$ is its smooth-seminorm cost; A depends only on the fixed ranges and arithmetic data.

Idea of sixth-power amplification. For a norm scale $D_{\mathrm{amp}} \ge 1$ and primary a, inclusion-exclusion gives

$$M_u(D_{\mathrm{amp}}) = \sum_{d \mid \mathrm{rad}\, a} \mu(d)\psi_u(d) q_d^{-1/2} M_{ua^6}(D_{\mathrm{amp}}/q_d).$$

Average over $q_a \le P_{\mathrm{amp}} = (H_{\mathrm{amp}}/U)^{1/6}$ with $H_{\mathrm{amp}} = \max(2U, D_{\mathrm{amp}}^{1+c})$. The map $(u,a) \mapsto ua^6$ is injective on sixth-power-free u, and the enlarged mean square costs $H_{\mathrm{amp}}/P_{\mathrm{amp}} = U^{1/6}H_{\mathrm{amp}}^{5/6}$. Here $c > 0$ is a small fixed enlargement parameter. Choosing c small proves (9.3), including $r = 1$ without a shrinking strict margin.

9.2. **Fourth moment of unweighted character polynomials.**

Lemma 9.3 (Unweighted product moment, [Ope26a, Lemma 18.1]). *For $\psi_k(n) = \nu_{\mathrm{fix}}(n)\chi_n(k)$ let m_0 be the element range exponent, and q_0 bound the radical of the displayed variable residue-symbol factors, meaning that the product of their distinct supporting primes has norm $\le Z^{q_0}$. The twist ν_{fix} is fixed during the k-sum. Put $M = m_0 + q_0$. Allow a common coprimality condition with modulus of norm $O(Z^B)$ for a fixed B, independent of k and common to every factor. Prime coefficients are fixed finite combinations of characters in Θ. For $\psi = \psi_k$ use the definitions*

$$S_\psi(n_j) = Z^{-n_j/2}\sum_a \psi(a) W_j(q_a/Z^{n_j}), \qquad Q_k = \prod_i Z^{-z_i/2}\sum_{p\in\mathcal{P}_i} a_i(p)\psi_k(p) W_i(q_p/Z^{z_i}),$$

where $n_1, n_2 \ge 0$ are length exponents, not ideals, and $z_{\mathrm{p}} = \sum_i z_i$. Every factor includes the common coprimality indicator just specified. All real length parameters range over prescribed bounded sets. Let $3/4 \le \kappa \le 1$; if $z_{\mathrm{p}} > 0$ and $\kappa < 1$, assume $\beta_* \le (1+\kappa)/2$. For every $\varepsilon > 0$ there exists $\ell_{\mathrm{mesh}} > 0$, depending only on these ranges and ε, such that $z_i \le \ell_{\mathrm{mesh}}$ for every i implies

$$\sum_{k\in\mathcal{R}_{z_{\mathrm{p}}}} |S_{\psi_k}(n_1)S_{\psi_k}(n_2)Q_k|^2 \ll Z^{M+\varepsilon}. \tag{9.4}$$

Here $\mathcal{R}_{z_{\mathrm{p}}}$ consists of the elements $0 < q_k \ll Z^{m_0}$ whose primitive inducing characters satisfy the exclusions below:

	length condition	*excluded inducing characters*
$z_{\mathrm{p}} = 0$	n_1, n_2 *arbitrary bounded*	*principal*
$z_{\mathrm{p}} > 0$	$n_1 + n_2 + 6\kappa z_{\mathrm{p}} \le M$	Θ

The permitted ℓ_{mesh} is independent of the number of prime factors and uniform for $\kappa \in [3/4, 1]$; constants and finite seminorm orders may depend on the fixed number of factors.

In this application $\beta_* = (1+\kappa)/2$ by (1.3), so the zero-free premise holds by equality.

Cancellation in the unweighted-moment argument. Ordinary Hecke reflection first reduces the length ranges. Two finite transforms then reduce the combined range exponent M. Transformed characters in Θ have possible main terms of the full summation size. For longer inputs the source uses the difference with fixed ideals $\mathbf{b} = (b_1, b_2)$, primary ideal variables l_1, l_2, and positive real scales X_i, Y_i:

$$D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^{2} W_i(q_{b_i} q_{l_i}/X_i) - \prod_{i=1}^{2} W_i(q_{b_i} q_{l_i}/Y_i), \qquad X_1 X_2 = Y_1 Y_2. \tag{9.5}$$

The two transforms preserve, on each side of the squared norm, one character, one coprimality condition and one norm twist on the full product of prime variables and unrestricted ideal variables [Ope26a, Lemma 18.2]. For an inducing character in Θ, fixing the prime variables leaves $\vartheta(l_1)\vartheta(l_2)1_{(l_1 l_2, \mathfrak{r}_*)=1} q_{l_1 l_2}^{it} D_{\mathbf{b}}(l_1, l_2)$.

Here $\vartheta \in \Theta$, $\mathfrak{r}_*$ is a squarefree coprimality modulus with $q_{\mathfrak{r}_*} \le Z^{B_{\mathrm{cop}}}$ for fixed B_{cop}, and $t \in \mathbb{R}$ is the common norm-twist height. The W_i in this argument are the two annular ideal weights, distinct from the earlier prime-interval weights. By [Ope26a, Lemma 18.3], the two product main terms have a common coefficient, independent of X_i, Y_i, multiplying respectively $(X_1 X_2)^{1+it}$ and $(Y_1 Y_2)^{1+it}$. Hence

$$(X_1 X_2)^{1+it} - (Y_1 Y_2)^{1+it} = 0,$$

which is the decisive cancellation in (9.5). The cited lemma supplies the uniform Poisson remainder. Direct volume bounds suffice first in the uncentered range $n_1 + n_2 + z_{\mathrm{p}} \le 5M/6$; the cancellation handles the remaining ranges. Increasing-width induction proves all zero-prime cases before the positive-prime cases. Uniform aggregation of support errors into $O_K(1/\log Z)$ keeps the maximum-length restriction independent of K. This cancellation is what permits the recursive estimate to close without discarding those transformed characters.

10. Counting elements from simultaneous large values

We combine the simultaneous large values of Section 5 with the two moments of Section 9. Optimizing the detecting lengths gives a character-count exponent for each zero and prime-amplitude class, which enters the contour estimate in Section 11.

10.1. **Prime sum size and admissible lengths.** For a fixed Mellin parameter z on those contours, put

$$Q_i(u; z) = P_i^{-1/2} \sum_{p \in \mathcal{P}_i} \overline{\chi_p(u)} W_i(q_p/P_i)(q_p/P_i)^{z-1}, \qquad \mathcal{Q}_i = -P_i^{z-1/2} Q_i.$$

The logarithmic-derivative estimate in the zero-free rectangle gives $|Q_i| \ll U^{\varepsilon_1} P_i^{\delta/2 + O(\epsilon_{\mathrm{b}})}$ [Ope26a, Lemma 19.1]. Choose a fixed small width for the exponent intervals $\theta_{\mathrm{bin}} > 0$. Partition by real exponents g_i so

$$|Q_i| \le P_i^{g_i + \theta_{\mathrm{bin}}}, \qquad g_i > 0 \Longrightarrow |Q_i| \ge P_i^{g_i}, \qquad 0 \le g_i \le \delta/2.$$

For an error factor set $g_i = 0$. Write

$$\bar{g} = \frac{\sum_i \ell_i g_i}{\ell}, \qquad 0 \le \bar{g} \le \delta/2, \qquad x = \bar{g}/\delta \in [0, 1/2]. \tag{10.1}$$

Ordering positive exponents decreasingly and selecting an initial set of prime sums of total length at most z_{p} gives

$$\left| \prod_{i \text{ selected}} Q_i \right|^2 \ge U^{2\bar{g} z_{\mathrm{p}} - \delta \max_i (\ell_i/d)}, \tag{10.2}$$

up to a fixed small length decrement. Fractional selection has gain at least $\bar{g}z_{\mathrm{p}}$; deleting its last partial interval gives the displayed error.

After selecting a presentation, take $\psi(n) = \nu(n)\overline{\chi_n(u)}$. The prime coefficient is $\overline{\nu(p)}1_{p\in 1_\mathcal{T}}$, a fixed finite combination of Θ-characters. Conjugate the whole unweighted product when using (9.4). An element with a good prime divisor induces outside Θ. The required moment hypotheses therefore hold; smooth parameter estimates handle the allowed element-dependent smooth weights with a fixed polynomial dependence on the height.

Measure lengths in base U: the variable u has norm $\asymp U$, so its range exponent is one. The largest total prime lengths allowed by the two moment inequalities are

$$z_M(r) = \frac{1-r}{2}, \qquad z_S(m) = \frac{1-2m}{6\kappa} = \frac{1-2m}{9/2+12\Delta}. \tag{10.3}$$

Using (5.5), (10.2) and the moments,

$$\#\mathcal{B} \ll \min\left\{U^{1-\delta r-2\bar{g}z_M+\varepsilon}, U^{1-2\delta m-2\bar{g}z_S+\varepsilon}\right\}(1+T_1)^A \tag{10.4}$$

where the respective lengths are admissible. The second bound uses two copies of S_m, hence $|S_m|^4$.

10.2. **Optimization of the two estimates.**

Proposition 10.1 (Refined character count, [Ope26a, Proposition 19.2])**.** *Assume the Stage-2 setting (1.3), so $\Delta = \beta_* - 7/8 \in (0, 1/24]$ and $\kappa = 3/4 + 2\Delta$. Let $\mathcal{B}$ be a fixed zero class and prime-sum size class, with $q_u \asymp U = Z^d$, $a > a_0$, $0 < \delta \le \alpha$, $0 \le \bar{g} \le \delta/2$ and $x = \bar{g}/\delta$. Assume the detector height hypotheses and that the available prime length ℓ/d exceeds $7/37$ by a fixed positive amount. Put*

$$D_x = 3 - 17x/9, \qquad P_x = (2 - 8x/9)(1-x).$$

For $t \in [1, 3/2]$ define

$$R_{\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t). \tag{10.5}$$

$$R_{\mathrm{long}}(t) = 1 - \delta + (\alpha - \delta)(t-1), \qquad \alpha = 5/6. \tag{10.6}$$

Then, for every $\varepsilon > 0$, the length decrements and maximum individual prime length can be chosen so that

$$\#\mathcal{B} \ll U^{\max(R_{\mathrm{short}}(t), R_{\mathrm{long}}(t))+\Delta/4+\varepsilon}(1+T_1)^A. \tag{10.7}$$

The constants and exponent A are uniform in u; their arithmetic dependence is on the fixed data. With no selected prime sums, $t = 1$ instead gives $\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon}(1+T_1)^A$ without any hypothesis on the available prime length.

The character-count exponent calculation. At $\delta = \alpha$, choose the detector parameter $t = 3/2$, so $r \ge 1 - O(\varepsilon)$. Sixth-power amplification then gives $\#\mathcal{B} \ll U^{1-\delta+\varepsilon}(1+T_1)^A$, which is at least as strong as (10.7); hence assume $\delta < \alpha$. For short inverse length $r < 1$ and $m < 1/2$, compare the unweighted estimate first with its $\Delta = 0$ prime length. Using $m \ge t - r - O(\varepsilon)$ gives

$$A_I(r) = 1 - \delta\{x + (1-x)r\}, \qquad A_S(r) = 1 - \delta\{4x/9 + (2-8x/9)(t-r)\}.$$

Let

$$r_*(t) = \frac{(2-8x/9)t - 5x/9}{D_x}.$$

Then $r_*(t) \ge 23/37$ and $1/3 \le t - r_*(t) \le 1/2$. The optimal common upper exponent is (10.5).

On the inverse side a fixed decrement $\nu_0 > 0$ ensures

$$1 - r - 2z_{\mathrm{p}} \ge 2\nu_0, \qquad 3 - 2r - 8z_{\mathrm{p}} = 4(1 - r - 2z_{\mathrm{p}}) + (2r - 1), \quad 2r - 1 \ge 9/37.$$

Its required prime length is at most $7/37$, whereas $\ell/h = 8/39 > 7/37$. On the unweighted side, restoring the actual length in (10.3) costs at most

$$\begin{aligned} 2\bar{g}(1 - 2m)\left(\frac{2}{9} - \frac{1}{9/2 + 12\Delta}\right) &\le \frac{5}{6}\left(\frac{1}{3} + O(\varepsilon)\right)\frac{16}{27}\Delta \\ &\le \frac{40}{243}\Delta + O(\varepsilon) < \Delta/4 + O(\varepsilon). \end{aligned}$$

Here $\bar{g} \le \delta/2 \le 5/12$, $m \ge t - r_*(t) - O(\varepsilon) \ge 1/3 - O(\varepsilon)$, and $\frac{2}{9} - \frac{1}{9/2+12\Delta} \le \frac{16}{27}\Delta$. For $r \ge 1$, (9.3) instead gives (10.6).

If an admissible prime length is zero or very small, the unweighted moments are used. These comparisons give the count exponent in (10.7). With no selected prime sums and $t = 1$, balance $1 - \delta r$ and $1 - 2\delta(1 - r)$ at $r = 2/3$ to obtain the stated unweighted count.

11. Contour estimates and the endpoint inequality

We insert the count from Section 10 into the nonprincipal contour integrals and compare their powers of Z with the principal term. The endpoint inequality and the bounds for the remaining frequency ranges provide the positive margins assembled in Section 12.

11.1. The exponent of a nonprincipal contribution.

Move the original zero-class integral with the holomorphic correction (8.2) to $s_r = a + 16\epsilon_{\mathrm{b}}, w_r = 1 - a - 6\epsilon_{\mathrm{b}}, z_r = z_0 := 17/50$. Only afterward subdivide pointwise by prime-sum sizes. For a subset with cardinality $\ll U^{R+\varepsilon}(1 + T_1)^A$, the factors, before small losses, are

$$\underbrace{X^{1/2-z_0}Z^{a+z_0-1}Y^{-a}}_{\text{Mellin powers}} \quad \underbrace{U^{R+\delta/2-z_0}}_{\text{count and numerator}} \quad \underbrace{Z^{\ell(z_0-1/2)+\bar{g}\ell}}_{\text{prime sums}}.$$

The numerator and all error factors are estimated jointly by (8.3). Subtracting $C(7/8) = 3/16$ gives

(11.1)

$$\begin{aligned} E(d) &= a - \frac{7}{8} + h(z_0 - 1/6) - al_y - \ell/2 + \bar{g}\ell + d(R + \delta/2 - z_0) \\ &= -\frac{1}{48} + \frac{2}{3}\delta + \frac{\bar{g}}{6} - h(1 - R) + (d - h)(R + \delta/2 - z_0) \qquad \text{[Ope26a, Lemma 20.1].} \end{aligned}$$

The integral is bounded by the supremum of this exponent over subsets defined by the sizes of the prime sums at each fixed Mellin parameter. The contour shift is made before this subdivision, so no holomorphic continuation of a subset is required.

11.2. Choice of the truncation exponent and positivity.

Lemma 11.1 (Endpoint saving, [Ope26a, Lemma 20.2]). *Let $1/50 < \delta < \alpha = 5/6$ and $0 \le x \le 1/2$, with D_x, P_x as above and the fixed scales (7.1). Choose*

$$\mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \qquad t = 1 + \frac{\delta P_x}{2\mathcal{J}}, \qquad R_* = R_{\mathrm{short}}(t) = R_{\mathrm{long}}(t). \tag{11.2}$$

Let $E_* = E(h)$ in (11.1) with $\bar{g} = x\delta$ and $R = R_*$. Then $-E_* \ge 49/440640$. If the count loss $\Delta/4$ is included and all remaining exponent losses have sum $\varepsilon_{\mathrm{total}}$, then

$$E_{\mathrm{actual}}(h) - \Delta \le -\frac{49}{440640} - \frac{51}{64}\Delta + \varepsilon_{\mathrm{total}}. \tag{11.3}$$

Here E_{actual} denotes the resulting exponent relative to $C(7/8)$.

The endpoint exponent calculation. Here $1 < t < 3/2$ and $35/54 \le \mathcal{J} \le 5/2$. Put

$$y = 1/2 - x \in [0, 1/2], \quad v_y = 51 + 41y, \quad p_y = 7 + 18y + 8y^2,$$
$$j_y = 185 + 170y + (-138 + 12y + 96y^2)\delta.$$

Then $P_x = p_y/9$ and $\mathcal{J} = j_y/108$. The source's polynomial identity is

$$\begin{aligned} 10368 v_y \mathcal{J}(-E_*) &= (3+5y)\{(4v_y\delta - 79)^2 + 49\} \\ &\quad + 4y\{4v_y\delta[(1+3y)(15+32y)\delta + 9 - 13y] + 265 + 3485y\}. \end{aligned} \tag{11.4}$$

Since $9 - 13y \ge 5/2$ and $v_y \le 17(3+5y)$,

$$-E_* \ge \frac{49}{176256\mathcal{J}} \ge \frac{49}{440640} > \frac{1}{10000} \qquad \text{[Ope26a, Lemma 20.2].} \tag{11.5}$$

Restoring the $\Delta/4$ in (10.7), the coefficient of R in (11.1) is h. Since $C(\beta_*) - C(7/8) = \Delta$, this gives (11.3). Here E_{actual} includes the $\Delta/4$ count loss, and $\varepsilon_{\text{total}}$ denotes the sum of the small contour, length, size-partition and smoothing losses. The subtraction of Δ is essential.

At $\delta = \alpha$, use $t = 3/2$ and unweighted amplification: $R = 1 - \delta$, whence $E(h) \le -1/48 - \delta/16$. At $a = a_0$ use the direct count $R = 1$, giving $E(h) \le -7/1200$. No selected zero or vanishing strict moment margin is needed in these respective boundary cases.

11.3. **All frequency ranges.** Let $d_{\min} = 1/100$ and $0 < \zeta < 1/48$. The estimates covering $U = Z^d$ are:

(1) $d < d_{\min}$: absolute tuple bounds on $(s_r, w_r, z_r) = (\beta_* + \epsilon_{\text{b}}, 1/2, 17/50)$ give saving $79/800 - 2d_{\min} = 63/800$ relative to $C(\beta_*)$. Here $2d_{\min}$ conservatively bounds the scale cost $63d/50$ for $d < d_{\min}$.

(2) $d_{\min} \le d \le 1/2$: using $R = 76/75 - 2\delta/3$ (also valid at $a = a_0$),

$$E(d) \le E(1/2) \le -529/2400 + (25/96)\delta \le -49/14400.$$

The slope in d is $R + \delta/2 - z_0 = 101/150 - \delta/6 > 0$, since $\delta \le 5/6$; this justifies $E(d) \le E(1/2)$.

(3) $1/2 \le d \le h$: the total available prime length is $\ell/d \ge 8/39 > 7/37$, and $R + \delta/2 - z_0 \ge 33/50 - \delta/2 \ge 4/25 > 0$. Thus (11.3) controls these ranges.

(4) $h < d \le h + \zeta$: the total available prime length remains $> 1/5 > 7/37$. The frequency slope is below two, so the additional cost is at most 2ζ.

(5) $d > h + \zeta$: absolute lines $(2, 2, z_\infty)$ give

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{53/32 + (h+\zeta)(1+\varepsilon) - \zeta z_\infty + \varepsilon}.$$

Here U runs over dyadic norm scales, and $\mathcal{R}_\eta(U; Z)$ denotes the modified nonprincipal contribution from $q_u \asymp U$ in (4.7), before normalization, on these absolute lines [Ope26a, Lemma 16.2]. Choose the fixed $z_\infty > 2$ sufficiently large.

The outer estimates are [Ope26a, Lemmas 10.6, 16.2]. Only the ranges $d \ge 1/2$ use selected prime sums in the character-count moments.

12. Parameter choices and the final comparison

We choose the auxiliary parameters in an order that preserves all margins obtained in Sections 8–11. The resulting direct and comparison estimates have precisely the uniformity required by Proposition 3.1, whose application completes the Hecke argument in the next section.

The lowest zero class and the intermediate norm range satisfy $E \le -7/1200$ and $E \le -49/14400$, respectively, relative to $C(7/8)$. Relative to $C(\beta_*) = C(7/8) + \Delta$, they therefore save at least Δ. Since $51\Delta/64 < \Delta$, both are already covered by m_{high} below. The common real margins are

$$m_{\mathrm{high}} = \min\{51\Delta/64, 63/800\}, \quad m_w = 23/960, \quad m_z = 13/9600.$$

Choose small moment errors and strict length decrements. Obtain the maximum permissible individual prime length $\ell_{\mathrm{mesh}} > 0$ from [Ope26a, Lemma 18.1] before choosing K. Then choose a fixed even K with equal $\ell_i = \ell/K$ and $2\ell/K < \ell_{\mathrm{mesh}}$ and smaller than the allowed exponent loss from discarding a partial prime interval. Smooth weights in disjoint subintervals of $(1, 2)$ give disjoint prime intervals at the common scale $Z^{\ell/K}$. Thus $\ell_i/d \le 2\ell/K$ throughout the selected ranges. Choose $0 < \kappa_P < (7/8)\ell/K$, then the width of the prime-sum size intervals, ϵ_{b}, ζ, z_∞ and P_0 within the remaining margins. All these choices precede the target.

One may retain

$$\mu_0 = \frac{1}{4}\min\{m_{\mathrm{high}}, m_w, m_z, \kappa_P\} > 0, \qquad \omega_0 = \Delta/2. \tag{12.1}$$

After fixing η and its arithmetic data, choose all internal seminorm orders. [Ope26a, Lemmas 10.2–10.4] and the preceding estimates give

$$|J_\eta - f_\eta| \ll_{\eta,N} Z^{C(\beta_*)-\mu_0}(1 + T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}. \tag{12.2}$$

A_η, B_η are finite nonnegative exponents fixed before the external decay order $N \in \mathbb{Z}_{\ge 0}$. The later N_η is a sufficiently large choice of this order. There is a fixed $\tau_{0,\eta} > 0$ ensuring all zero-detection height conditions for $T_1 = Z^\tau$, $0 < \tau \le \min(d_{\min}/100, \tau_{0,\eta})$.

Choose

$$0 < \tau_\eta \le \min\left\{d_{\min}/100, \tau_{0,\eta}, \frac{\mu_0}{4(A_\eta + 1)}\right\}, \qquad B_\eta - N_\eta\tau_\eta < C(\beta_*) - \mu_0/2.$$

The first term in (12.2) saves at least $3\mu_0/4$, and the second saves $\mu_0/2$. Increasing external derivative orders changes only seminorm constants and thresholds. This is the height choice of [Ope26a, Lemma 11.1]; the full order is [Ope26a, Proposition 20.3]. Together with (8.6),

$$|J_\eta(Z)| \ll_\eta Z^{3/16+\Delta/2}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\mu_0/2}. \tag{12.3}$$

The exponents are independent of η. Neither J_η nor f_η contains T_1.

13. Mellin continuation and contradiction

We now apply Proposition 3.1 to the estimates assembled in Section 12 and conclude the 7/8 bound for the Hecke family. The extension to imprimitive presentations is included here; the Dirichlet consequence follows in Section 14.

Theorem 13.1 (Hecke conclusion, [Ope26a, Theorem 1.1])**.** *The family supremum in (1.2) satisfies $\beta_* \le 7/8$. Consequently every finite-order Hecke L-function over F has no zeros in $\operatorname{Re} s > 7/8$, with principal poles allowed.*

The final continuation contradiction. Assume $\beta_* > 7/8$. Apply Proposition 3.1 with

$$\sigma_0 = 7/8, \qquad c = -11/16, \qquad \omega_0 = \Delta/2, \qquad \sigma = \mu_0/2.$$

Holomorphy of H_η follows from (4.9) and (4.10); the chosen cutoff gives (4.11). Together with (12.3), these verify the criterion's hypotheses; $C(7/8) = 3/16$ and $0 < \omega_0 < \beta_* - 7/8$. Its contradiction therefore gives

$$\boxed{\beta_* \le 7/8}. \tag{13.1}$$

For an imprimitive character η induced by η^*,

$$L_F(s,\eta) = L_F(s,\eta^*) \prod_{\substack{\mathfrak{p}\mid \mathfrak{f}_\eta \\ \mathfrak{p}\nmid \mathfrak{f}_{\eta^*}}} (1-\eta^*(\mathfrak{p})q_{\mathfrak{p}}^{-s}).$$

Here $\mathfrak{f}_\eta$ is the defining modulus and $\mathfrak{f}_{\eta^*}$ the primitive conductor. Each finite factor is nonzero for $\operatorname{Re} s > 0$, giving the same conclusion for every presentation.

14. Transfer to Dirichlet L-functions

We transfer the Hecke bound to Dirichlet L-functions by pulling a Dirichlet character back along the ideal norm and factoring its L-function over F. The same argument will transfer each improved Hecke bound in Parts II–IV, so it suffices to give the factorization and pole check once.

Corollary 14.1 (Dirichlet conclusion, [Ope26a, Proposition 11.3])**.** *Every Dirichlet L-function, including ζ, has no zeros in $\operatorname{Re} s > 7/8$, with principal poles at $s = 1$ allowed.*

The Dirichlet factorization. The identity below is the quadratic case of Artin formalism (compatibility with induction and direct sums); see [Mil20, Chapter VIII, Section 10]. Its application to the present zero-free bounds is [Ope26a, Proposition 11.3]. For a primitive Dirichlet character χ of conductor q, define $\eta(\mathfrak{a}) = \chi(N\mathfrak{a})$ on ideals coprime to $3q$, and let χ_{-3} be the primitive quadratic Dirichlet character associated with $F/\mathbb{Q}$, given by the Kronecker symbol $\chi_{-3}(n) = (-3/n)$. Deleting primes dividing $3q$ and their prime ideals,

$$L_F^{S_F}(s,\eta) = L^{S_{\mathbb{Q}}}(s,\chi) L^{S_{\mathbb{Q}}}(s,\chi\chi_{-3}) \qquad \text{[Ope26a, Proposition 11.3].} \tag{14.1}$$

The split and inert Euler factors give (14.1) in absolute convergence; meromorphic continuation extends the identity [Ope26a, Proposition 11.3]. Both Dirichlet factors are holomorphic on $7/8 < \operatorname{Re} s \le 1$, $s \ne 1$. A zero of either is therefore a zero of their product, contradicting the Hecke conclusion. Euler convergence handles $\operatorname{Re} s > 1$. At $s = 1$ a possible principal pole pairs only with χ_{-3}, whose value is

$$L(1,\chi_{-3}) = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

Finally, finite Euler deletions transfer the result to all imprimitive Dirichlet presentations. The character modulo one gives ζ.

Part II. A rational extension of the bound

15. Statement and choice of parameters

We state the rational improvement and choose its length parameters, which will be verified in Sections 16–20. The estimates established in this part also supply the common analytic construction for Parts III and IV. We vary the lengths in the modified average of Section 7; all arithmetic conventions, zero extensions, smooth tests and character families remain those of Part I. Throughout Parts II–IV we use the proved intermediate results of [Ope26a]. In particular, we apply Theorem 13.1 before making the improved comparison.

Theorem 15.1 (Rational extension)**.** *Every finite-order Hecke L-function over F is zero-free in*

$$\operatorname{Re} s > B_{\mathrm{r}}, \qquad B_{\mathrm{r}} = 34999/40000.$$

The same assertion holds for every Dirichlet L-function. Principal poles are allowed, and the boundary line is excluded.

For clarity, the parameters that change are displayed together:

$$\begin{gathered}\vartheta_{\mathrm{r}}=\frac{1}{10000},\quad b_{\mathrm{r}}=\frac{1}{8},\quad \ell_{\mathrm{r}}=\frac{1}{6}+\vartheta_{\mathrm{r}}=\frac{5003}{30000},\\ b=b_{\mathrm{r}},\qquad \ell=\ell_{\mathrm{r}},\\ l_x=\frac{1-b-\ell}{2}=\frac{21247}{60000},\quad l_y=\frac{1+b-\ell}{2}=\frac{28747}{60000},\quad h=1-l_x+\ell=\frac{16253}{20000},\\ X=Z^{l_x},\quad Y=Z^{l_y},\quad P_i=Z^{\ell_i},\quad \sum_{i=1}^{K}\ell_i=\ell.\end{gathered}\tag{15.1}$$

The pair $(b_{\mathrm{r}},\ell_{\mathrm{r}})$ specifies the rational extension. We use $b=b_{\mathrm{r}}$ and $\ell=\ell_{\mathrm{r}}$ at this geometry, and retain b,ℓ for the general parameter formulas shared with Part III. Here $b=l_y-l_x$ and K is a fixed even integer. Its disjoint prime sets $\mathcal{P}_i$ lie in the class $1_{\mathcal{T}}$ and outside S, as in Part I. The zero-class width ϵ_{b} is independent of ϑ_{r}. We retain

$$a_0=51/100,\quad \delta=2a-1,\quad x=\bar{g}/\delta\in[0,1/2],\quad z_0=17/50,\quad \alpha=5/6.$$

Thus x denotes an amplitude ratio, whereas l_x denotes a length. In all applications $\delta\geq 1/50$, so the ratio is defined. Its extension to $\delta=0$ is used only in polynomial inequalities.

The balanced condition $l_x+l_y+\ell=1$ gives

$$C_b(s)=s-\frac{4+b}{6},\quad B(\ell)=\frac{11}{12}-\frac{\ell}{4},\quad C_b(B(\ell))=\frac{l_x}{2}+\frac{b}{12}.\tag{15.2}$$

The shared balanced hypotheses for the extensions are

$$\begin{gathered}l_x+l_y+\ell=1,\qquad b=l_y-l_x>0,\qquad h=1-l_x+\ell,\\ l_x>\ell>0,\qquad l_y>\ell,\qquad l_y-\ell-11b/6>0,\qquad 1-3\ell>0,\qquad \ell<1/5,\\ \ell/h\geq 7/37+\rho_{\sup},\qquad \rho_{\sup}>0,\qquad 5\ell>h,\\ B=B(\ell)=11/12-\ell/4\in[437/500,7/8].\end{gathered}\tag{15.3}$$

The positive reserve $\rho_{\sup}$ is fixed independently of the target. The conditions on rescaled lengths and the additive estimate are separated from the numerical choice of geometry. For (15.1), $C_b(s)=s-11/16$ and $B(\ell)=B_{\mathrm{r}}$. The hypotheses are verified by

$$\begin{gathered}l_x-\ell=\frac{3}{16}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\quad l_y-\ell=\frac{5}{16}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\\ l_y-\ell-\frac{11b}{6}=\frac{1}{12}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\quad 1-3\ell>0,\quad \ell<\frac{1}{5},\\ \frac{\ell}{h}>\frac{8}{39}>\frac{7}{37},\qquad 5\ell-h=\frac{1}{48}+\frac{7}{2}\vartheta_{\mathrm{r}}>0.\end{gathered}\tag{15.4}$$

They ensure that the same finite modification defines an average at the new scales, that all rescaled lengths remain positive, and that there are enough prime factors for the moments with selected prime factors.

16. The direct estimate at balanced scales

We derive the direct estimate at variable balanced lengths from the general reflected estimate [Ope26a, Lemma 14.3] and the additive estimate (7.5). This supplies the direct hypothesis for Section 20; the specialized bound (7.4) alone does not justify changing ℓ.

If $\mathcal{I}\subset\{1,\dots,K\}$ is the subset of rescaled factors, put

$$d_{\mathcal{I}}=\sum_{i\in\mathcal{I}}\ell_i,\quad \ell'=\ell-d_{\mathcal{I}},\quad M'=1-\ell-2d_{\mathcal{I}},\quad X'=Z^{l_x-d_{\mathcal{I}}},\quad Y'=Z^{l_y-d_{\mathcal{I}}}.$$

The symbol $d_{\mathcal{I}}$ is unrelated to the norm exponent d used later. For a fixed ray class σ and rescaled prime tuple, let $B^{\mathcal{I}}_{a,\sigma}$ denote the completed coefficient $B^{J}_{m,\sigma}$ of (7.4), with $J=\mathcal{I}$, the outer element index renamed a, and the scales replaced by X',Y' above. The primes $i\notin\mathcal{I}$ are already summed, with their divisibility restrictions imposed. Put $Q=q_{b_*}X'Y'$.

Lemma 16.1 (Balanced reflected estimate)**.** *Under the coefficient and smooth-weight hypotheses of* [Ope26a, Lemma 14.3]*, the completed theta sum with remaining prime-length exponent ℓ' and outer element-range exponent M' has squared norm*

$$\sum_{q_a\ll Q}|B^{\mathcal{I}}_{a,\sigma}|^2\ll Z^{M'+(d_{\mathcal{I}}+5\ell-1)_+/4+\varepsilon}. \tag{16.1}$$

Here $(t)_+=\max(t,0)$ for a real argument t; the loss $\varepsilon>0$ is arbitrary.

Proof. We record the exponent calculation, retaining the normalization of (7.4). Write O for the length of the powerful part, H for the residual squarefree length, and $\Delta_H=M'-O-H$. The local auxiliary lengths in [Ope26a, Lemma 14.3] satisfy, up to losses that can be made arbitrarily small,

$$O,\Delta_H,S_0,B_0\ge 0,\quad 2A_0\le O,\quad 0\le N_0\le A_0,\quad z_a\le\ell',$$

$$T_d=2H+2A_0+2z_a-1-\ell'-N_0-3B_0,\qquad v+3\ell_b+e_\lambda\le T_d.$$

Here A_0,N_0,B_0,S_0 describe the local divisor decompositions, z_a is their retained primal prime length, v,ℓ_b,e_λ are the retained dual lengths, and T_d is the available dual length. These variables occur only in this reflection calculation. Since $M'+\ell'-1=-3d_{\mathcal{I}}$,

$$T_d\le H-3d_{\mathcal{I}}\le H.$$

The squared-energy exponent supplied by the cited lemma is

$$\begin{aligned}E_{\mathrm{ref}}&=O/2+H-S_0-B_0+z_a-u_{\mathrm{hyb}}-\ell_b-2e_\lambda/3\\&\quad-(T_d-v-3\ell_b-e_\lambda)_+/2,\qquad u_{\mathrm{hyb}}=\min\{v,z_a,(v+z_a)/3\}.\end{aligned}$$

If $u_{\mathrm{hyb}}=z_a$, then $E_{\mathrm{ref}}\le M'$. Otherwise $u_{\mathrm{hyb}}\ge v/2$, and

$$u_{\mathrm{hyb}}+\ell_b+2e_\lambda/3+(T_d-v-3\ell_b-e_\lambda)_+/2\ge T_d/4.$$

Substituting T_d, using $N_0\le A_0$ and $z_a\le\ell'$, gives

$$E_{\mathrm{ref}}\le M'+\frac{1+3\ell'-2M'-2\Delta_H}{4}\le M'+\frac{d_{\mathcal{I}}+5\ell-1}{4}.$$

The maximum of the two bounds proves (16.1). Fixed annular offsets are $O(1/\log Z)$; the Gaussian partition, margins in the prime-length restrictions and logarithmically many dyadic intervals fit into ε. Separation is performed on the joint weight before positivity or squaring, so the separated coefficients remain independent of the outer element. □

Proposition 16.2 (Balanced direct bound)**.** *Assume $l_x+l_y+\ell=1$, $b=l_y-l_x>0$, and*

$$l_x>\ell>0,\quad l_y>\ell,\quad l_y-\ell-11b/6>0,\quad 1-3\ell>0,\quad \ell<1/5.$$

Then

$$|I_{\eta,\mathrm{mod}}(Z)|\ll_{\eta,\varepsilon}Z^{l_x/2+b/12+\varepsilon}. \tag{16.2}$$

Proof. Apply the separation of (4.6) to each term of (7.2). In (7.5) the additive scale is $P_a=q_{b_*}^{-1}Z^b$, and the inequality $l_y-d_{\mathcal{I}}-11b/6>0$ implies $P_a^2/Y'\ll P_a^{1/6}$. Cauchy's inequality and Lemma 16.1 bound each unscaled tuple by

$$(X')^{1/2}P_a^{1/12}Z^{(d_{\mathcal{I}}+5\ell-1)_+/8+\varepsilon}.$$

The number of rescaled tuples is $\ll Z^{d_{\mathcal{I}}+\varepsilon}$. Their coefficient is $O(Z^{-3d_{\mathcal{I}}/2})$, and replacing X' by X contributes $Z^{-d_{\mathcal{I}}/2}$. Thus their net additional exponent is

$$-d_{\mathcal{I}} + (d_{\mathcal{I}} + 5\ell - 1)_+ /8 \le -7d_{\mathcal{I}}/8 \le 0.$$

Sum over the finitely many subsets. This proves the claim without invoking the specialized direct bound at altered scales. □

17. Euler products and extraction of the principal term

We extend the holomorphic Euler correction and extract the principal Mellin term at the changed lengths of Section 16. The resulting domain and normalization are used in the continuation arguments of Parts II–IV.

Let

$$\frac{437}{500} \le B \le \frac{7}{8}, \qquad \mathcal{D}_2(B) = \{\operatorname{Re} s \ge B,\ \operatorname{Re} w \ge 19/20,\ \operatorname{Re} z \ge 33/200\}. \tag{17.1}$$

The first domain (4.8) remains available, with a positive margin in $\operatorname{Re}(s+w) > 1$. All Euler products use the same set S as the modified average. A fixed prime cutoff is chosen first; the target's conductor primes may then be added to S.

Lemma 17.1 (Extended principal domain)**.** *The unselected and selected local expressions of Part I are holomorphic on $\mathcal{D}_2(B)$. Their full correction (8.2) is defined there by the same local holomorphic expressions. For sufficiently large fixed cutoff P_0, the principal correction satisfies*

$$H_\eta(s) = 1 + O(P_0^{-4/5}), \qquad \sup_{\operatorname{Re} s \ge B} |H_\eta(s) - 1| \le \tfrac{1}{2}. \tag{17.2}$$

At the principal residue,

$$G_p(s, 1, 1/6)/H_p(s, 1, 1/6) = -1 + O(q_p^{-B}). \tag{17.3}$$

The constants are uniform in imaginary parts and in the unit phases.

Proof. To check the changed boundary, we give the exact local formulas from [Ope26a, Lemma 7.1]. At a good prime put $Q = q_p$, $j = v_p(u) \in \{0, \dots, 5\}$, $\rho_p = \chi_p(u/p^j)$, and

$$a_p = \overline{\varpi(p)}^3 \eta(p)^3, \quad b_p = \overline{\varpi(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \quad b_p^3 = a_p^2.$$

Here ϖ is the phase defined in Part I; these a_p, b_p are local unit phases, unrelated to a, b above. Set

$$V = Q^{-6z}, \quad R = a_p^2 Q^{4-6s-6z}, \quad W = \chi_p(u) Q^{-w}, \quad D = \eta(p)\overline{\chi_p(u)} Q^{-s}, \quad W_{\mathrm{loc}} = \rho_p Q^{-w}.$$

At $p \mid u$, $W = D = 0$ by zero extension. With $\mathbf{1}$ denoting an indicator, the complete local formulas are

$$P_p^* = \frac{1}{1-R}\left\{ \frac{R(1-Q^{-1}) - \eta(p)(Q-1)Q^{-s-w}V^{\mathbf{1}_{j\le 1}}}{1-V} + J_j \right\},$$

$$P_p = \frac{1}{1-V} + \mathbf{1}_{j=0}\frac{W_{\mathrm{loc}}}{1-W_{\mathrm{loc}}} + P_p^*, \qquad H_p = P_p \frac{(1-V)(1-W)}{1-D},$$

where

j	J_j
0	$-\eta(p)\rho_p^{-1}Q^{-s} + W_{\mathrm{loc}}R$
1	$\eta(p)Q^{-s-w}$
2	$a_p\rho_p^{-3}Q^{3/2-3s}$
3	$-\eta(p)b_p\rho_p^{-2}Q^{2-3s-w} + b_p^2\rho_p^{-4}Q^{2-4s}$
4	$-\eta(p)a_p\rho_p^{-3}Q^{5/2-4s-w}$
5	$-a_p^2Q^{3-6s}$.

Writing $\mathcal{E}_p = P_p^* + D$, cancellation gives

$$H_p - 1 = \frac{D(V+W-VW) - VW + (1-V)(1-W)\mathcal{E}_p}{1-D}. \tag{17.4}$$

For $p \nmid u$, the possible exponents are

$$-s_r - w_r, \quad -s_r - 6z_r, \quad -w_r - 6z_r, \quad 4 - 6s_r - 6z_r, \quad 1 - s_r - w_r - 6z_r,$$

where $s_r = \operatorname{Re} s$, and similarly for w_r, z_r. Their maximum on (17.1) is $-B - 47/50 < -9/5$. For $p \mid u$, the table instead gives $H_p - 1 = O(Q^{1/20-B})$. The denominators $1-R, 1-V, 1-D$ are uniformly separated from zero; (17.4) has no denominator $1-W$. Normal convergence off u and a finite-product q_u^{ε} bound on its prime divisors follow. For $u = 1$, the tail saves $B - 3/50 > 4/5$, proving (17.2).

The selected factor is defined without dividing by H_p:

$$G_p = \frac{(\overline{\eta(p)}Q^s - Q^{-w})(1-V)(1-W)P_p^* - Q^{-w}(1-VW)}{1-D}. \tag{17.5}$$

It agrees with (8.1) on absolute lines, hence continues the finite modification by (8.2). For $u = 1$ the unselected factors are close to one, so division is permitted on the residue rectangle. The four error exponents are

$$-s_r, \quad -6z_r, \quad 4 - 5s_r - 6z_r, \quad 1 - w_r - 6z_r,$$

all at most $-B$ for the range (17.1). This proves (17.3). No nonvanishing of a nonprincipal local correction is needed. □

Define $A_{\mathcal{T}}(Z)$ and c_S by (8.5), with the present ℓ_i and the same radial Mellin transform $\mathcal{M}$. The prime asymptotic (8.4) gives, for even K,

$$A_{\mathcal{T}}(Z) > 0 \quad (Z \text{ sufficiently large}), \qquad |A_{\mathcal{T}}(Z)| \asymp (\log Z)^{-K}.$$

The poles remain $w = 1$ and $z = 1/6$; the latter is independent of ℓ. For arbitrary positive l_x, l_y, ℓ, put

$$h = 1 - l_x + \ell, \qquad C(s) = s + l_x/2 - 1 + h/6,$$

$$J_\eta(Z) = \frac{I_{\eta,\mathrm{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}, \qquad f_\eta(Z) = \frac{1}{2\pi i}\int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds. \tag{17.6}$$

These are the same constructions as (8.8), with the changed scales. Neither function depends on a later height truncation.

Proposition 17.2 (Principal extraction)**.** *Assume (15.3) and $\beta_* \ge B$. The individual prime lengths satisfy $\ell_i > 0$ and $\sum_{i=1}^K \ell_i = \ell$, as in (15.1). For every $0 < \kappa_P < B \min_i \ell_i$, the principal term $\mathcal{P}_\eta$ satisfies*

$$\left| \frac{\mathcal{P}_\eta(Z)}{c_S A_{\mathcal{T}}(Z)} - f_\eta(Z) \right| \ll Z^{C(\beta_*)+(1+h)\epsilon_{\mathrm{b}} - l_y/20 + \varepsilon} + Z^{C(\beta_*)+\epsilon_{\mathrm{b}} - h/600 + \varepsilon} + Z^{C(\beta_*)+\epsilon_{\mathrm{b}} - \kappa_P + \varepsilon}. \tag{17.7}$$

Proof. The identification of the principal frequency in Section 4 and [Ope26a, Lemmas 10.2–10.5] is arithmetic and unchanged: the only principal numerator occurs at $u = 1$. For this frequency the scalar quotient is $\zeta_F^S(6z)\zeta_F^S(w)/L_F^S(s,\eta)$. Start with $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$, $\operatorname{Re} w = 1 + \epsilon_{\mathrm{b}}$, $\operatorname{Re} z = 1/6 + \epsilon_{\mathrm{b}}$. Move w to $19/20$, then in its residue move z to $33/200$. Lemma 17.1 justifies both moves. The raw exponent is

$$l_x/2 + \operatorname{Re} s - 1 + h \operatorname{Re} z + l_y(\operatorname{Re} w - 1).$$

The two unresidued integrals therefore save $l_y/20$ and $h/600$. At the double residue, (17.3) and the nonnegative prime tests give

$$\mathfrak{H}_{\eta,1,Z}(s,1,1/6) = H_\eta(s)Z^{\ell/6}A_{\mathcal{T}}(Z)(1+O(Z^{-\kappa_P})).$$

The scalar residue is exactly c_S, including the factor $1/6$ from $\zeta_F^S(6z)$. Bound the error on its original global s line; move only the main Gaussian integral right to $\operatorname{Re} s = 2$. The reciprocal has no poles to the right of β_*, and principal L-poles become reciprocal zeros. Gaussian decay removes horizontal sides. This proves (17.7) without moving an approximation error through possible target zeros. □

18. The fixed moment parameter and the nonprincipal estimate

We use the already proved $7/8$ bound to keep $\kappa = 3/4$ fixed while varying the averaging lengths. This gives the character count and general nonprincipal exponent used by the rational certificate in Section 19 and the algebraic optimization in Part III.

Theorem 13.1 permits the fixed choice

$$\kappa = 3/4, \qquad \beta_* \le (1+\kappa)/2. \tag{18.1}$$

It is important to fix this value rather than substitute $\kappa = 2\beta_* - 1$ below the allowed moment range. All moments below have exactly the common-character, coefficient and smooth-weight hypotheses of Section 9.

Proposition 18.1 (Fixed-parameter count)**.** *Let $U = Z^d$ be a dyadic interval of original sixth-power-free elements in one zero and amplitude class. Suppose $0 < \delta \le \alpha$, $0 \le \bar{g} \le \delta/2$, and $\ell/d \ge 7/37 + \rho_{\sup}$ for a fixed $\rho_{\sup} > 0$. With the D_x, P_x of Part I set*

$$\mathcal{J} = (\alpha-\delta)D_x + \delta P_x, \quad t_* = 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha-\delta)\delta P_x}{2\mathcal{J}}. \tag{18.2}$$

For every requested loss $\varepsilon > 0$, sufficiently fine fixed prime lengths give

$$\#\mathcal{B} \ll U^{R_*+\varepsilon}(1+T_1)^A. \tag{18.3}$$

Without an assumption on the available prime length, the count without selected prime factors is $\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon}(1+T_1)^A$.

Proof. The simultaneously large detecting polynomials of (5.5) have lengths U^r, U^m, $r + m \ge t - O(\varepsilon)$, and satisfy

$$|M_r|^2 \gg U^{\delta r - O(\varepsilon)}, \qquad |S_m|^4 \gg U^{2\delta m - O(\varepsilon)}.$$

Here M_r, S_m are the Möbius-weighted and unweighted detecting polynomials defined in Section 5, with the same character and retained presentation. For total selected length U^z, the inverse moment with selected prime factors (9.1) and the unweighted moment (9.4) give the same set two count bounds:

$$U^{1-\delta r - 2\bar{g}z+\varepsilon}(1+T_1)^A, \qquad U^{1-2\delta m - 2\bar{g}z+\varepsilon}(1+T_1)^A.$$

At norm scale U, the upper bounds on the total selected prime length are

$$z_M = (1-r)/2, \qquad z_S = 2(1-2m)/9.$$

Weighted greedy selection achieves gain at least $\bar{g}z$; rounding one fractional prime factor loses at most $\delta \max_i(\ell_i/d)$ in the squared lower bound. Decrease the selected prime-length bound by a fixed $\nu > 0$ before applying a moment; when the remaining prime-length bound is nonpositive, use the version without selected prime factors. These errors fit the requested ε.

For $r<1, m<1/2$, the affine count bounds without an additional prime-length restriction are

$$A_I(r)=1-\delta\{x+(1-x)r\},\qquad S_t(r)=1-\delta\{4x/9+(2-8x/9)(t-r)\}.$$

They cross at

$$r_*(t)=\frac{(2-8x/9)t-5x/9}{D_x},\quad r_*(1)\ge 23/37,\quad r_*(3/2)=1.$$

Use the inverse estimate on $r\ge r_*$ and the unweighted estimate on $r\le r_*$. The prime-length bound for the inverse moment is at most $7/37$, the bound for the unweighted moment is at most $2/27+O(\varepsilon)$; the hypothesis on the available prime length suffices. On the inverse side the second strict margin is

$$3-2r-8z=4(1-r-2z)+(2r-1)\ge 9/37,$$

while the first margin is $1-r-2z\ge 2\nu$. For short detecting polynomials, the count exponent is consequently

$$R_{\mathrm{short}}(t)=1-\delta+\frac{\delta P_x}{D_x}(3/2-t).$$

For $r\ge 1$, use the amplification estimate without selected prime factors (9.3):

$$\mathfrak{a}(r)-\delta r\le R_{\mathrm{long}}(t)=1-\delta+(\alpha-\delta)(t-1).$$

For $m\ge 1/2$ the unweighted bound without selected prime factors is at most $1-\delta$. Balancing the short and long bounds gives (18.2). Since $D_x-P_x=1+x-8x^2/9>0$,

$$\mathcal{J}\ge 35/54,\qquad 1\le t_*\le 3/2.$$

At $\delta=\alpha$, the balanced values are $t_*=3/2$ and $R_*=1-\alpha=1/6$. At $r=1$ the inverse prime-length bound is $z_M=0$; use the estimate without selected prime factors (9.3). For $r\ge 1$, $\mathfrak{a}(r)-\alpha r=1/6$. Thus the endpoint uses a moment without selected prime factors, rather than a limit of moments with selected prime factors and vanishing reserve.

All retained presentations use the same detecting polynomials in both moment estimates; there are only logarithmically many dyadic presentations. Common smooth-weight separation puts height dependence into $(1+T_1)^A$. There is no $\Delta/4$ term from (10.7): that term compares a varying κ with $3/4$, whereas (18.1) is already exact. The common character can be written as $\psi(n)=\nu(n)\overline{\chi_n(u)}$, and each selected prime coefficient is $\overline{\nu(p)}1_{p\in 1_{\mathcal{T}}}$; it is a fixed finite combination of Θ-characters. Conjugate the whole unweighted product to match (9.4). Elements divisible by a good prime divisor induce outside Θ, as in Section 10; unit elements are covered by the small-norm estimate below. With no selected factors, $t=1$ and the crossing $r=2/3$ give the last assertion. □

Proposition 18.2 (Exponent at arbitrary scales)**.** *Let $B\le\beta_*$, and assume (17.1), positive lengths and the unchanged joint-error and moment hypotheses of Part I. For a class with element-norm exponent d, count exponent R, zero-class parameter a and amplitude mean $\bar g$, the normalized nonprincipal contribution is, for each prescribed loss $\varepsilon>0$ and sufficiently small fixed contour and zero-class widths, bounded by*

$$Z^{C(B)+E_B(d)+\varepsilon}(1+T_1)^A=Z^{C(\beta_*)+E_B(d)-(\beta_*-B)+\varepsilon}(1+T_1)^A,$$

where

$$E_B(d)=a-B+h(z_0-1/6)-al_y-\ell/2+\bar g\ell+d(R+\delta/2-z_0),\quad z_0=17/50. \tag{18.4}$$

Thus $E_B(d)$ is measured relative to $C(B)$; relative to $C(\beta_*)$, the exponent is $E_B(d)-(\beta_*-B)$. On balanced scales with $B=B(\ell)$,

$$E_B(h)=-\frac{1}{4}+\frac{b}{6}+\frac{5\ell}{4}+(1/2+\ell)\delta+\ell\bar{g}-h(1-R). \tag{18.5}$$

Proof. Use the exact Poisson identity (4.7), whose scale factors are valid at arbitrary positive lengths. Move the full holomorphic correction before partitioning it into pointwise amplitude classes. The contour with a positive margin is

$$(\operatorname{Re} s,\operatorname{Re} w,\operatorname{Re} z)=(a+16\epsilon_{\mathrm{b}},1-a-6\epsilon_{\mathrm{b}},z_0).$$

For example, first retain the global s line and move w so that $\operatorname{Re}(s+w)\ge 1+14\epsilon_{\mathrm{b}}$; after external height truncation move s to the displayed line, where $\operatorname{Re}(s+w)=1+10\epsilon_{\mathrm{b}}$. The same zero-free separation and first Euler domain used in Part I apply. The joint bound (8.3) consumes the conductor deficit once for the entire product and yields

$$\sum_{u\in\mathcal{B}}|L_F^S(w,\chi_\bullet(u))\mathfrak{H}_{\eta,u,Z}|\ll U^{R+\delta/2+\varepsilon}Z^{\ell(z_0-1/2)+\bar{g}\ell+\varepsilon}(1+T_1)^A.$$

The remaining factors contribute

$$Z^{l_x(1/2-z_0)+a+z_0-1-al_y}U^{-z_0}.$$

Subtracting $C(B)=B+l_x/2-1+h/6$ gives (18.4). Algebra using $l_x+l_y+\ell=1$ gives (18.5). Horizontal joins and discarded heights are bounded by $Z^{B_\eta}T_1^{-N}$: the fixed internal seminorm orders are chosen before the arbitrary external integration-by-parts order N. The class sets need no holomorphic dependence on the contour variables. □

19. A rational polynomial inequality and the remaining norm ranges

We verify that the nonprincipal exponent from Section 18 has a strict saving at the rational parameters, uniformly over all zero and amplitude classes. The polynomial certificate and the remaining norm estimates supply the margins needed to prove Theorem 15.1 in the next section.

Lemma 19.1 (Uniform rational saving)**.** *For (15.1), $R=R_*$ and $(\delta,x)\in[0,5/6]\times[0,1/2]$, one has*

$$E_{B_{\mathrm{r}}}(h)\le -19/320000. \tag{19.1}$$

Proof. At the Part I geometry $\ell=1/6,b=1/8,h=13/16$ denote the balanced exponent by E_0. Put $y=1/2-x$ and

$$j=185+170y+(-138+12y+96y^2)\delta=108\mathcal{J},$$

$$\begin{aligned}Q_0&=10(37+34y)-8(237+377y+26y^2)\delta\\&\quad+48(51+131y+94y^2+32y^3)\delta^2,\qquad Q=Q_0-96j/5000.\end{aligned}$$

Direct expansion gives

$$Q=10368\mathcal{J}(-E_0-1/5000). \tag{19.2}$$

To prove its sign on the whole rectangle, use the Bernstein basis, named after Sergei Natanovich Bernstein, who introduced it in his constructive proof of Weierstrass's approximation theorem [Ber12]; see [Far12] for a modern account. The Bernstein basis polynomials of degree n on $[0,1]$ are

$$\mathsf{B}_{i,n}(t)=\binom{n}{i}t^i(1-t)^{n-i},\qquad 0\le i\le n. \tag{19.3}$$

They satisfy

$$\mathsf{B}_{i,n}(t) \ge 0, \qquad \sum_{i=0}^{n} \mathsf{B}_{i,n}(t) = 1 \qquad (0 \le t \le 1).$$

On $[a_1, a_2] \times [b_1, b_2]$, substitute $\delta = a_1 + (a_2 - a_1)v$, $y = b_1 + (b_2 - b_1)w$ and write $Q = \sum_{p=0}^{2} \sum_{q=0}^{3} A_{pq} v^p w^q$. Its representation in the tensor-product Bernstein basis is

$$Q(\delta, y) = \sum_{i=0}^{2} \sum_{k=0}^{3} B_{ik} \mathsf{B}_{i,2}(v) \mathsf{B}_{k,3}(w).$$

The bidegree $(2, 3)$ Bernstein coefficients are

$$B_{ik} = \sum_{p \le i,\ q \le k} A_{pq} \frac{\binom{i}{p}}{\binom{2}{p}} \frac{\binom{k}{q}}{\binom{3}{q}}, \quad 0 \le i \le 2, \quad 0 \le k \le 3.$$

Nonnegativity and the partition-of-unity identity give

$$\min_{i,k} B_{ik} \le Q(\delta, y) \le \max_{i,k} B_{ik} \qquad ((\delta, y) \in [a_1, a_2] \times [b_1, b_2]).$$

The listed rectangles cover the original rectangle; their complete partition and coefficients are verified by the accompanying exact arithmetic checker. The last column is a coefficient minimum, not a sampled value:

δ interval	y interval	$\min B_{ik}$
$[0, 5/24]$	$[0, 1/4]$	1380407/18000
$[5/24, 5/16]$	$[0, 1/8]$	27677/2000
$[5/16, 35/96]$	$[0, 1/16]$	12437/8000
$[35/96, 25/64]$	$[0, 1/32]$	2301/16000
$[25/64, 5/12]$	$[0, 1/32]$	12581/32000
$[35/96, 5/12]$	$[1/32, 1/16]$	2952669/1024000
$[5/16, 5/12]$	$[1/16, 1/8]$	85933/192000
$[5/24, 5/16]$	$[1/8, 1/4]$	53811/2560
$[5/16, 5/12]$	$[1/8, 1/4]$	8213/800
$[0, 5/24]$	$[1/4, 1/2]$	336467/4000
$[5/24, 5/16]$	$[1/4, 3/8]$	946273/24000
$[5/16, 5/12]$	$[1/4, 3/8]$	666871/16000
$[5/24, 5/12]$	$[3/8, 1/2]$	70651/4000
$[5/12, 5/6]$	$[0, 1/2]$	319/125

Thus $Q > 0$ throughout, so $-E_0 \ge 1/5000$. With R_*, δ, x fixed, changing ℓ at fixed b gives

$$\partial_\ell E(h) = -1/4 + \delta + \bar{g} + 3R_*/2 = 5/4 + \delta(x - 1/2) + \frac{3(\alpha - \delta)\delta P_x}{4\mathcal{J}} \le 45/32.$$

We used $\mathcal{J} \ge \alpha P_x$ and $\delta(\alpha - \delta) \le \alpha^2/4$. Therefore $E(h) \le -1/5000 + 45/(32 \cdot 10000) = -19/320000$. □

The endpoint inequality alone is insufficient: elements of very small and very large norm require different contours. We record a reusable division of the norm ranges.

Lemma 19.2 (Remaining ranges)**.** *Assume the general hypotheses (15.3), $B \le \beta_* \le 7/8$, and $0 \le \delta \le 5/6$. Then the following estimates hold, with arbitrarily small fixed losses. Set $d_{\min} = 1/100$.*

(1) *For elements with $q_u \ll Z^{d_{\min}}$, including unit elements, the normalized exponent relative to $C(\beta_*)$ is at most $13h/75 - l_y/2 + 2d_{\min}$.*

(2) *For* $d_{\min} \le d \le 1/2$ *and* $\delta \ge 1/50$, *use* $R = 76/75 - 2\delta/3$ *in* (18.4). *Its slope in* d *is* $101/150 - \delta/6 > 0$; *it is enough to bound* $E_B(1/2)$.
(3) *For* $1/2 \le d \le h$ *use* R_*. *The slope* $R_* + \delta/2 - z_0$ *is at least* $4/25$ *and at most* 2; *it is enough to bound* $E_B(h)$.
(4) *The lowest zero class* $a = a_0$, *without a detector, has* $R = 1$ *and* $\bar{g} \le 1/100$. *Its endpoint exponent is at most* $-1/4 + b/6 + 5\ell/4 + (1/2 + 3\ell/2)/50$.
(5) *Extending the third range to* $h + \zeta$ *costs at most* 2ζ, *provided* $\ell/(h + \zeta) > 7/37$. *Beyond* $h + \zeta$, *an absolute contour with sufficiently large fixed* $\operatorname{Re} z = z_\infty$ *gives any prescribed power saving.*

Proof. For the small-norm range retain the global reciprocal line $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$ and put $\operatorname{Re} w = 1/2$, $\operatorname{Re} z = z_0$. Use the first domain with margin $B - 1/2 > 0$. Formula (17.5) and the local table imply $G_p = O(1)$ off u, $G_p = O(q_p^{1/2})$ on u. Consequently the full correction is $\ll U^{\varepsilon} Z^{\ell z_0}$: primes not dividing u contribute $P_i^{z_0}$, and primes dividing u contribute at most $U^{\varepsilon} P_i^{z_0 - 1/2}$. The nonprincipal numerator costs $U^{3/5+\varepsilon}$ on this line, by the growth estimate used in [Ope26a, Lemma 10.6]. Counting $O(U)$ original elements and the factor U^{-z_0} give $U^{63/50+\varepsilon}$. The exterior factor relative to $C(\beta_*)$ is $Z^{13h/75 - l_y/2 + O(\epsilon_{\mathrm{b}})}$. Since $63/50 < 2$, the first claim follows.

For the intermediate norm range use the count without selected prime factors in Proposition 18.1, allowing an extra $1/75$. For the moderate norm range use its count with selected prime factors. Differentiating (18.4) gives the slopes asserted; for the latter use $R_* \ge 1 - \delta$ and $\delta \le 5/6$. The lowest zero class is bounded trivially by $O(U)$ and $\bar{g} \le \delta/2$. The extension beyond h follows from the slope upper bound. Finally, on $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (2, 2, z_\infty)$, absolute convergence gives

$$\mathcal{R}(U; Z) \ll Z^{l_x/2 + 1 + l_y + h z_\infty + \varepsilon} U^{1 + \varepsilon - z_\infty}.$$

Summing $U > Z^{h+\zeta}$ yields an exponent containing $-\zeta z_\infty$. Choose z_∞ after ζ is fixed. All statements concern the full correction, including its local zeros. □

For the rational geometry the saving in the small-norm range is at least $78699/1000000$, and the saving for the lowest zero class is at least $1093/192000$. For the intermediate norm range, direct substitution gives

$$\max_{1/50 \le \delta \le 5/6} E_{B_{\mathrm{r}}}(1/2) \le -\frac{49}{14400} + \frac{177}{2000000} = -\frac{59657}{18000000}.$$

Together with Lemma 19.1, these are strict, parameter-independent margins. Choose

$$0 < \zeta < \min\{5\ell - h,\ (19/320000)/64\}.$$

This also ensures $\ell/(h + \zeta) > 1/5 > 7/37$.

20. Combining the estimates and the continuation contradiction

We assemble the direct, principal and nonprincipal estimates of Sections 16–19 to prove the rational extension by Proposition 3.1. We formulate the comparison with its uniformity explicit so that Parts III and IV can apply it with their respective parameters and counts.

Proposition 20.1 (Comparison from strict exponent margins)**.** *Assume the balanced hypotheses* (15.3). *Suppose also that:*

(1) *the direct bound* (16.2) *with* $l_x/2 + b/12 = C_b(B)$;
(2) *the principal estimates of Proposition 17.2;*

(3) *the exponents for the small and intermediate norm ranges and the lowest zero class in Lemma 19.2 are negative, and* $E_B(h) \le 0$ *for the count with selected prime factors.*

The chosen count has the form $\#\mathcal{B} \ll U^{R+\varepsilon}(1+T_1)^A$ *with the common-character, coefficient, mask, support and height hypotheses of Section 9. Assume its losses can be made arbitrarily small by fixed length decrements and a target-independent prime mesh, and that* $0 \le \partial_d E_B \le 2$ *in the moderate norm range. If* $\Delta = \beta_* - B > 0$*, then there is a number* $\sigma > 0$*, chosen before the target* η*, such that*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_b(\beta_*)-\sigma}. \tag{20.1}$$

The principal factor is holomorphic on $\operatorname{Re} s > B$ *and satisfies (17.2).*

Proof. Replacing B by β_* in the central exponent subtracts the same Δ in every class. Choose a positive margin smaller than $\Delta/4$, the strict savings for the small and intermediate norm ranges and the lowest zero class, and $l_y/80, h/2400$. Fix ζ smaller still so that 2ζ consumes less than a quarter of that margin and all prime-length inequalities remain strict. Choose the requested moment loss, strict prime-length reserves and maximum individual prime lengths inside that margin. The chosen count and its underlying moments supply the strict reserves and the upper bound on each prime length. Fix an even K and the actual ℓ_i before the target. Now choose $0 < \kappa_P < B \min_i \ell_i$, decrease the common margin if necessary, and finally choose ϵ_{b} and the remaining smooth-weight and rounding losses so that their finite sum is smaller than a quarter of this margin. All these real choices precede η.

The class count is finite apart from logarithmically many dyadic intervals and admissible polynomial configurations, whose losses are absorbed in the chosen powers. Lemmas 19.2 and 17.1, the chosen count, and Propositions 18.2 and 17.2 therefore give a common $\mu > 0$ with

$$|J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}. \tag{20.2}$$

The internal smoothness orders, A_η and B_η are fixed before N. They may depend on the target; μ does not. If $\tau_{\mathrm{det}} > 0$ is the permissible detector height exponent, choose

$$0 < \tau_\eta \le \min\{d_{\min}/100, \tau_{\mathrm{det}}, \mu/[4(A_\eta+1)]\}, \qquad T_1 = Z^{\tau_\eta}.$$

Then choose N so large that $B_\eta - N\tau_\eta < C_b(\beta_*) - \mu/2$. Equation (20.2) proves (20.1) with $\sigma = \mu/2$. The dependence of the analysis height on η is harmless: J_η, f_η were fixed in (17.6). □

Proof of Theorem 15.1. Suppose $\Delta_{\mathrm{r}} = \beta_* - B_{\mathrm{r}} > 0$. Part I already gives $\beta_* \le 7/8$, so (18.1) holds. The geometric inequalities (15.4), the direct bound, Lemma 19.1 and the remaining-range estimates verify Proposition 20.1. For any $0 < \omega_0 < \Delta_{\mathrm{r}}$, the normalized direct bound is

$$|J_\eta(Z)| \ll Z^{C_b(B_{\mathrm{r}})+\omega_0}.$$

Apply Proposition 3.1 with $\sigma_0 = B_{\mathrm{r}}$ and $c = -11/16$. Its principal factor is holomorphic and within $1/2$ of one by (17.2); its comparison hypothesis is (20.1). The criterion implies $\beta_* \le B_{\mathrm{r}}$, contradicting $\Delta_{\mathrm{r}} > 0$. The transfer proof of Corollary 14.1 applies with this new positive boundary and gives the Dirichlet assertion. □

Part III. An algebraic extension of the bound

21. Algebraic parameters and the scope of the result

We state the stronger bound obtained by varying b and ℓ together and specify the algebraic parameters used in its proof. The following sections verify these choices using

the analytic construction of Part II and the character and amplitude notation of Part I. Whereas Part II improved 7/8 by 1/40000 with $b = b_{\mathrm{r}} = 1/8$ fixed, the joint variation gives $7/8 - B_{\mathrm{new}} = 4.293020083\ldots \times 10^{-5}$.

Theorem 21.1 (Algebraic extension)**.** *Every finite-order Hecke L-function over F, and every Dirichlet L-function, is zero-free in*

$$\operatorname{Re} s > B_{\mathrm{new}} := \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\ldots. \tag{21.1}$$

Principal poles are allowed and the boundary is excluded.

Let $r_{921} = \sqrt{921}$, and set

(21.2)

$$\delta_{\mathrm{c}} = \frac{49 - r_{921}}{48}, \qquad \ell_{\mathrm{new}} = \frac{33 + 8r_{921}}{1653}, \qquad b_{\mathrm{new}} = -\frac{4}{29} + \frac{230r_{921}}{26709},$$
$$l_{x,\mathrm{new}} = \frac{1 - b_{\mathrm{new}} - \ell_{\mathrm{new}}}{2}, \qquad l_{y,\mathrm{new}} = \frac{1 + b_{\mathrm{new}} - \ell_{\mathrm{new}}}{2}, \qquad h_{\mathrm{new}} = \frac{1 + b_{\mathrm{new}} + 3\ell_{\mathrm{new}}}{2},$$
$$B_{\mathrm{new}} = \frac{11}{12} - \frac{\ell_{\mathrm{new}}}{4}, \qquad C_{\mathrm{new}}(s) = s - \frac{4 + b_{\mathrm{new}}}{6}.$$

The subscript new denotes the algebraic extension and its associated lengths and comparison exponent. In particular, b_{new} is a real length difference, whereas b_* in Part I is an arithmetic element. Below we write $b = b_{\mathrm{new}}$, $\ell = \ell_{\mathrm{new}}$, $l_x = l_{x,\mathrm{new}}$, $l_y = l_{y,\mathrm{new}}$ and $h = h_{\mathrm{new}}$ when this geometry is understood. The approximate values, used only for orientation, are

ℓ_{new}	$0.166838387470\ldots$	b_{new}	$0.123405436989\ldots$
$l_{x,\mathrm{new}}$	$0.354878087771\ldots$	$l_{y,\mathrm{new}}$	$0.478283524759\ldots$
h_{new}	$0.811960299699\ldots$	δ_{c}	$0.388583712271\ldots$

Every sign and equality below uses the exact values in (21.2).

Theorem 21.1 is an analytic deduction from the full source inputs. Proposition 25.1 separately proves an algebraic lower bound under three explicit constraints. Part IV proves necessity of these constraints for the explicitly defined uniform exponent comparison system, also allowing the moment parameter to vary.

22. The direct estimate with free total length

We extend the direct estimate of Section 16 by allowing $M = l_x + l_y$ to vary independently of ℓ. Its individual exponent constraints will be used in Section 25 and in the full optimality proof of Part IV.

Assume

$$l_x > \ell, \quad l_y > \ell, \quad b = l_y - l_x > 0, \quad h = 1 - l_x + \ell > 0.$$

All lengths lie in fixed bounded ranges. The signal exponent is

$$C(s) = s - 5/6 + l_x/3 + \ell/6. \tag{22.1}$$

In particular the additive constant changes when b changes; the Part I expression $s - 11/16$ cannot be retained at the algebraic scales.

Lemma 22.1 (General reflected estimate)**.** *Under the common-weight and coefficient hypotheses of* [Ope26a, Lemma 14.3]*, let $B^{\mathcal{I}}_{a,\sigma}$ be the completed coefficients defined in Section 16, now with outer element-range exponent m and remaining prime-length exponent l. Thus $X'Y' \asymp Z^m$ and $Q = q_{b_*}X'Y'$; the remaining arithmetic, coefficient and*

smooth-weight data are fixed as in that definition. Their squared norm satisfies

$$\sum_{q_a \ll Q} |B^{\mathcal{I}}_{a,\sigma}|^2 \ll Z^{m+k(m,l)+\varepsilon},$$

$$k(m,l) = \max\left\{0, \frac{1+3l-2m}{4}, m+l-1, \frac{3l-1}{2}\right\}. \tag{22.2}$$

The estimate includes both signs of $m+l-1$.

Proof. Use the local lengths defined in the proof of Lemma 16.1, now with $H \le m - O$, $z_a \le l$. Suppress arbitrarily small annular and logarithmic offsets. Then

$$O, S_0, B_0, v, l_b, e_\lambda \ge 0, \quad 2A_0 \le O, \quad 0 \le N_0 \le A_0,$$

$$T_d = 2H + 2A_0 + 2z_a - 1 - l - N_0 - 3B_0, \qquad v + 3l_b + e_\lambda \le T_d,$$

$$\begin{aligned} E_{\mathrm{ref}} &= O/2 + \max(H, v+l_b) - S_0 - B_0 + z_a - u_{\mathrm{hyb}} - l_b - 2e_\lambda/3 \\ &\quad - (T_d - v - 3l_b - e_\lambda)_+/2, \qquad u_{\mathrm{hyb}} = \min\{v, z_a, (v+z_a)/3\}. \end{aligned}$$

There are four exhaustive cases. If the maximum is H and $u_{\mathrm{hyb}} = z_a$, then $E_{\mathrm{ref}} \le O/2 + H \le m$. If the maximum is H and $u_{\mathrm{hyb}} \ne z_a$, then $u_{\mathrm{hyb}} \ge v/2$ and the dual terms save at least $T_d/4$. Substitution gives

$$\begin{aligned} E_{\mathrm{ref}} &\le O/2 + H/2 - A_0/2 + z_a/2 + (1 + l + N_0 + 3B_0)/4 - S_0 - B_0 \\ &\le m/2 + (1+3l)/4. \end{aligned}$$

If the maximum is $v + l_b$ and $u_{\mathrm{hyb}} = z_a$, then

$$E_{\mathrm{ref}} \le O/2 + v \le O/2 + T_d \le 2m + l - 1.$$

If the maximum is $v + l_b$ and $u_{\mathrm{hyb}} \ne z_a$, then

$$E_{\mathrm{ref}} \le O/2 + v/2 + z_a \le O/2 + T_d/2 + z_a \le m + (3l-1)/2.$$

Their maximum is $m + k(m,l)$.

Restore all signed $O(1/\log Z)$ offsets, annular widths and losses from restricting the summation ranges. If each is bounded by ε_0, a common bound for losses $20\varepsilon_0$ bounds their effect in these four linear inequalities. The homogeneous joint-weight separation precedes squaring; its Gaussian translates have summable seminorms and its discarded whole annuli have arbitrary power saving. The remaining dyadic intervals are logarithmically many. Choose ε_0 inside the requested ε. This proves the bound with the source's coefficient classes and normalization of the outer element sum. □

Proposition 22.2 (Full direct exponent)**.** *The modified average satisfies*

$$|I_{\eta,\mathrm{mod}}(Z)| \ll Z^{L_{\mathrm{low}}+\varepsilon}, \qquad L_{\mathrm{low}} = \frac{l_x}{2} + \max\{0, b/12, b - l_y/2\} + \frac{1}{2}k(M,\ell). \tag{22.3}$$

Consequently its sufficient comparison boundary is

$$B_{\mathrm{low}} = L_{\mathrm{low}} + 5/6 - l_x/3 - \ell/6. \tag{22.4}$$

On balanced scales this simplifies to

$$B_{\mathrm{low}} = \frac{11}{12} - \frac{\ell}{4} + \frac{(8b+3\ell-3)_+}{12} + \frac{(5\ell-1)_+}{8}. \tag{22.5}$$

Proof. Keep all three terms in (7.5); their Cauchy contribution is $\max\{0, b/12, b - l_y/2\}$. For a rescaled subset of length $d_{\mathcal{I}}$, Lemma 22.1 applies with $m = M - 2d_{\mathcal{I}}$ and $l = \ell - d_{\mathcal{I}}$. After tuple counting and rescaling as in Proposition 16.2, its total exponent is

$$F(d_{\mathcal{I}}) = l_x/2 - d_{\mathcal{I}} + \max\{0, b/12, b - l_y/2 + d_{\mathcal{I}}/2\} + \tfrac{1}{2}k_{\mathcal{I}},$$

$$k_{\mathcal{I}} = \max\left\{0, \frac{1+3\ell-2M+d_{\mathcal{I}}}{4}, M+\ell-1-3d_{\mathcal{I}}, \frac{3\ell-1-3d_{\mathcal{I}}}{2}\right\}.$$

For $d_2 > d_1$, the additive maximum increases by at most $(d_2-d_1)/2$, and $k_{\mathcal{I}}$ by at most $(d_2-d_1)/4$. Thus $F(d_2)-F(d_1) \le -3(d_2-d_1)/8$, so the empty subset controls the direct exponent. Subtract (22.1) to obtain (22.4). With $M = 1-\ell$, the extra reflected maximum is $(5\ell-1)_+/4$, and the extra Gram maximum is $(8b+3\ell-3)_+/12$, proving (22.5). □

23. Comparison of the character-count estimates

We compare the character-count estimates available at the algebraic parameters, allowing for the supply of selected prime factors. This identifies the count used in Section 24 and records the alternative choices examined in the optimality proof of Part IV.

Let $z_{\sup} = \ell/d$ be the available prime length at the element norm scale $U = Z^d$ and let $c_\kappa = 1/(6\kappa)$, $3/4 \le \kappa \le 1$. For $1 \le t \le 3/2$, $t-1/2 \le r \le 1$ and $m = t-r$, the inverse and unweighted savings are

$$\begin{aligned} F_I(r) &= r + 2x\min\{(1-r)/2, z_{\sup}\},\\ F_S(r) &= 2m + 2x\min\{c_\kappa(1-2m), z_{\sup}\}. \end{aligned} \tag{23.1}$$

The count exponent for short detecting polynomials is $1-\delta\min_r\max\{F_I(r), F_S(r)\}$. The two savings are increasing and decreasing respectively, and their crossing is one of

$$\frac{2t}{3}, \qquad \frac{(2-4c_\kappa x)t + 2c_\kappa x - 2xz_{\sup}}{3-4c_\kappa x},$$
$$\frac{2t+2xz_{\sup}-x}{3-x}, \qquad \frac{(2-4c_\kappa x)t + (2c_\kappa-1)x}{3-x-4c_\kappa x}. \tag{23.2}$$

Each expression is used only when its corresponding capped or uncapped branches are valid. If a crossing lies outside the interval, take its appropriate endpoint. The count bound for long detecting polynomials remains $R_{\mathrm{long}}(t)$ from Proposition 18.1.

Removing an upper bound on available prime length can only increase a saving, and increasing κ can only decrease the prime-length bound for the unweighted moment. Thus neither operation improves upon the exponent R_* obtained with no additional prime-length restriction and $\kappa = 3/4$, after optimizing t. With no selected primes, balancing the short- and long-polynomial bounds without selected prime factors instead gives

$$t_N = \frac{\alpha}{\alpha-\delta/3}, \qquad R_N = 1 - \frac{2\alpha\delta}{3\alpha-\delta} \ge R_*.$$

This includes the no-prime alternative in the comparison. Strict moment reserves, coefficient restrictions and height factors are handled exactly as in Proposition 18.1. These count exponents describe the sufficient bounds furnished by the moment estimates; they are not lower bounds on the numbers of elements.

24. An exact polynomial proof of the endpoint

We derive the algebraic parameters from the critical class of the count in Section 23, then prove the endpoint inequality on the full zero-amplitude rectangle. The exact quadratic identity supplies the central nonprincipal estimate for the final comparison in Section 26.

At $x = 1/2$, formula (18.2) reduces to

$$R_*(\delta, 1/2) = \frac{185-288\delta+96\delta^2}{185-138\delta}. \tag{24.1}$$

The equation $R_* = 2/3$ has smaller root δ_c, since

$$288\delta^2 - 588\delta + 185 = 0.$$

At that root the coefficient of b in (18.5) vanishes, and $E_B(h) = 0$ forces

$$\ell = \frac{5 - 6\delta_c}{9 + 18\delta_c} = \ell_{\text{new}}.$$

The stationary condition in δ gives b_{new}. These equations locate the candidate; the following identity proves its validity over the full rectangle.

Lemma 24.1 (Algebraic endpoint inequality)**.** *At (21.2), for $(\delta, x) \in [0, 5/6] \times [0, 1/2]$, $\bar{g} = x\delta$ and $R = R_*$,*

$$E_{B_{\text{new}}}(h_{\text{new}}) \le 0. \tag{24.2}$$

Equality occurs only at $(\delta, x) = (\delta_c, 1/2)$.

Proof. Put $y = 1/2 - x \ge 0$ and

$$p(y) = 7 + 18y + 8y^2, \quad j(y, \delta) = 185 + 170y + (-138 + 12y + 96y^2)\delta,$$

so $P_x = p(y)/9$, $\mathcal{J} = j/108$. Write $C_0 = -1/4 + b_{\text{new}}/6 + 5\ell_{\text{new}}/4$. Substitution in (18.5) gives

$$\mathscr{Q} := 216\mathcal{J}(-E_{B_{\text{new}}}(h_{\text{new}})) = -2jC_0 + j(b_{\text{new}} + 2\ell_{\text{new}}y)\delta - 12h_{\text{new}}(5/6 - \delta)\delta p(y). \tag{24.3}$$

Write $\mathscr{Q} = \mathscr{A}(y)\delta^2 + \mathscr{D}(y)\delta + \mathscr{G}(y)$; these three coefficient polynomials are defined by this identity. For $\mathscr{A}(y) = \sum_{i=0}^{3} A_i y^i$, direct expansion yields

$$\begin{aligned} A_0 &= 31824/551 - 36688r_{921}/169157, \\ A_1 &= 50916/551 + 214096r_{921}/169157, \\ A_2 &= 17352/551 + 347296r_{921}/169157, \\ A_3 &= 2112/551 + 512r_{921}/551. \end{aligned}$$

Every A_i is positive. The exact discriminant identity is

$$4\mathscr{A}(y)\mathscr{G}(y) - \mathscr{D}(y)^2 = y(d_1 + d_2y + d_3y^2 + d_4y^3), \tag{24.4}$$

where

$$\begin{aligned} d_1 &= (3861265032800 - 122715443200r_{921})/279616521, \\ d_2 &= 2386327618300/279616521 - 19093999200r_{921}/93205507, \\ d_3 &= -3389811480400/279616521 + 46823488000r_{921}/93205507, \\ d_4 &= -40221718800/4905553 + 4408611200r_{921}/14716659. \end{aligned}$$

Expansion with $r_{921}^2 = 921$ verifies the identity coefficient by coefficient; the accompanying exact arithmetic checker verifies this identity and the associated parameter inequalities. A rational isolating interval, obtained by squaring its endpoints, is

$$30.347981810 \ < \ r_{921} \ < \ 30.347981811. \tag{24.5}$$

Substitution into these linear expressions gives

$$490 < d_1 < 491, \quad 2317 < d_2 < 2318, \quad 3122 < d_3 < 3123, \quad 892 < d_4 < 893.$$

The same bracket proves $A_0 > 0$; positivity of the other A_i is immediate. All decimal endpoints here are terminating rational numbers. Complete the square:

$$4\mathscr{A}(y)\mathscr{Q} = (2\mathscr{A}(y)\delta + \mathscr{D}(y))^2 + y(d_1 + d_2y + d_3y^2 + d_4y^3) \ge 0.$$

Since $\mathcal{J} \ge 35/54$ on the required rectangle, (24.3) proves the asserted sign. For $y > 0$ it is strict. At $y = 0$, the exact polynomial is $A_0(\delta - \delta_c)^2$, proving the equality assertion. □

25. An algebraic lower bound for the exponent constraints

We show that three affine comparison constraints force $B \geq B_{\mathrm{new}}$, even when the total averaging length is free. This is the algebraic part of the optimality argument; Section 34 will justify the constraints within the full specified comparison system and allow κ to vary.

Put $M = l_x + l_y$. The two terms 0 and $M + \ell - 1$ in the reflected majorant give the direct comparison constraints

$$B \geq L_0 := 5/6 + M/12 - \ell/6, \qquad B \geq L_K := 1/3 + 7M/12 + \ell/3. \tag{25.1}$$

At $(\delta, x) = (\delta_{\mathrm{c}}, 1/2)$, a count exponent $R \geq 2/3$ in (18.4), with $h > 0$, also requires

$$B \geq H_{\mathrm{c}} := 1 + \delta_{\mathrm{c}} - (1 + \delta_{\mathrm{c}})M/2 + \delta_{\mathrm{c}}\ell. \tag{25.2}$$

The following proposition is a statement about these three affine constraints. It assumes the constraints explicitly.

Proposition 25.1 (Algebraic lower bound)**.** *Let $M, \ell, B \in \mathbb{R}$ and $\delta_{\mathrm{c}} = (49 - \sqrt{921})/48$. If*

$$B \geq L_0, \qquad B \geq L_K, \qquad B \geq H_{\mathrm{c}}, \tag{25.3}$$

with L_0, L_K, H_{c} defined above, then $B \geq B_{\mathrm{new}}$. Equality is attained by $M = 1 - \ell_{\mathrm{new}}$ and $\ell = \ell_{\mathrm{new}}$.

Proof. The positive weights

$$w_0 = \frac{4 + 18\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}}, \quad w_K = \frac{2}{9 + 18\delta_{\mathrm{c}}}, \quad w_H = \frac{3}{9 + 18\delta_{\mathrm{c}}}$$

sum to one and satisfy

$$w_0/12 + 7w_K/12 - (1 + \delta_{\mathrm{c}})w_H/2 = 0, \qquad -w_0/6 + w_K/3 + \delta_{\mathrm{c}}w_H = 0.$$

Their weighted sum cancels M and ℓ, giving

$$B \geq \max\{L_0, L_K, H_{\mathrm{c}}\} \geq w_0L_0 + w_KL_K + w_HH_{\mathrm{c}} = \frac{7 + 18\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}} = \frac{1507 - 2r_{921}}{1653} = B_{\mathrm{new}}.$$

The cancellation proves the lower bound for every real M, ℓ; no numerical enumeration of parameter choices is involved. At $M = 1 - \ell_{\mathrm{new}}$ and $\ell = \ell_{\mathrm{new}}$, direct substitution gives $L_0 = L_K = H_{\mathrm{c}} = B_{\mathrm{new}}$. □

Scope of optimality. Part IV proves necessity of the three constraints in the explicitly defined uniform exponent comparison system. Restricting its operative moment parameter to $\kappa \geq 3/4$, monotonicity of $F(\kappa)$ in Proposition 34.1 gives $B \geq F(3/4) = B_{\mathrm{new}}$. The present geometry attains that bound. This proves optimality for that comparison problem in the restricted moment range, including free total length and the available unmarked choices; it does not exclude sharper analytic inputs.

26. The auxiliary estimates and the final contradiction

We combine the endpoint inequality of Section 24 with the direct estimate and principal-term construction from Part II. Checking the remaining strict margins permits another application of Proposition 3.1 and proves Theorem 21.1.

At (21.2), exact substitution gives

$$\begin{aligned} &b_{\mathrm{new}} > 0, \quad l_{x,\mathrm{new}} - \ell_{\mathrm{new}} > 0, \quad l_{y,\mathrm{new}} - \ell_{\mathrm{new}} > 0, \quad 1 - 3\ell_{\mathrm{new}} > 0, \quad \ell_{\mathrm{new}} < 1/5, \\ &l_{y,\mathrm{new}} - \ell_{\mathrm{new}} - 11b_{\mathrm{new}}/6 > 0, \qquad \ell_{\mathrm{new}}/h_{\mathrm{new}} > 1/5 > 7/37, \qquad 5\ell_{\mathrm{new}} - h_{\mathrm{new}} > 0. \end{aligned} \tag{26.1}$$

For example these inequalities follow by replacing r_{921} by the rational endpoints in (24.5) after reducing the expressions to $a+cr_{921}$; the denominators are positive. There is therefore no extra direct penalty in (22.5), and

$$|J_\eta(Z)| \ll Z^{C_{\mathrm{new}}(B_{\mathrm{new}})+\varepsilon}, \qquad C_{\mathrm{new}}(B_{\mathrm{new}}) = l_{x,\mathrm{new}}/2 + b_{\mathrm{new}}/12. \tag{26.2}$$

Equation (26.1) verifies the general hypotheses (15.3); the ranges in Lemma 19.2 therefore apply at these lengths. For the lowest zero class $\delta_0 = 1/50$ the saving is

$$m_F = -C_0 - (1/2 + 3\ell_{\mathrm{new}}/2)\delta_0 > 0.$$

For the intermediate norm range use $R = 76/75 - 2\delta/3$; the upper endpoint exponent is bounded by

$$E_{B_{\mathrm{new}}}(1/2) \le \frac{(225\ell_{\mathrm{new}} + 50 - 75b_{\mathrm{new}})\delta + 78\ell_{\mathrm{new}} - 49b_{\mathrm{new}} - 73}{300} < 0 \quad (1/50 \le \delta \le 5/6).$$

It is affine in δ, and both endpoint values are negative by (24.5). For the small-norm range the saving is

$$m_S = l_{y,\mathrm{new}}/2 - 13h_{\mathrm{new}}/75 - 2/100 > 0.$$

For the moderate norm range, $E_{B_{\mathrm{new}}}(d) \le E_{B_{\mathrm{new}}}(h_{\mathrm{new}}) \le 0$ by Lemma 24.1; for $h_{\mathrm{new}} < d \le h_{\mathrm{new}} + \zeta$ the cost is at most 2ζ. The large-norm range is handled by the same absolute contour as in Lemma 19.2. The only non-strict central inequality is the isolated critical class $(\delta_{\mathrm{c}}, 1/2)$; it is harmless under a positive contradiction gap.

The principal domain also needs no new argument:

$$437/500 < B_{\mathrm{new}} < B_{\mathrm{r}} < 7/8.$$

Lemma 17.1 and Proposition 17.2 therefore apply with $B = B_{\mathrm{new}}$ and the new lengths. In particular, the principal factor has no zeros on $\operatorname{Re} s > B_{\mathrm{new}}$ and is the same factor appearing in the new signal. The first-domain margin in the small-norm range is $B_{\mathrm{new}} - 1/2 > 0$.

The non-strict endpoint is covered by the same comparison proposition. Indeed, under the contradiction hypothesis $\Delta_{\mathrm{new}} = \beta_* - B_{\mathrm{new}} > 0$,

$$C_{\mathrm{new}}(B_{\mathrm{new}}) + E_{B_{\mathrm{new}}}(h_{\mathrm{new}}) - C_{\mathrm{new}}(\beta_*) \le -\Delta_{\mathrm{new}}.$$

The finitely many contour, moment, subdivision into exponent intervals and normalization losses have fixed finite coefficients. Proposition 20.1 chooses them inside this gap before fixing the target.

Proof of Theorem 21.1. If $\Delta_{\mathrm{new}} > 0$, the direct bound (26.2), the extended Euler domain, the strict auxiliary estimates and the endpoint identity verify all hypotheses of Proposition 20.1. Thus for some $\sigma > 0$ independent of the target,

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_{\mathrm{new}}(\beta_*)-\sigma}.$$

The internal height degrees are allowed to depend on η; choose $T_1 = Z^{\tau_\eta}$ and then the tail order as in (20.2). The modified average and signal are unchanged by this analysis choice.

Apply Proposition 3.1 with $\sigma_0 = B_{\mathrm{new}}$, $c = -(4+b_{\mathrm{new}})/6$ and, for example, $\omega_0 = \Delta_{\mathrm{new}}/2$. Its hypotheses now all hold: the low bound is at $C_{\mathrm{new}}(B_{\mathrm{new}}) + \omega_0$, the comparison saves σ, and the holomorphic principal factor is within $1/2$ of one. The criterion implies $\beta_* \le B_{\mathrm{new}}$, contradicting $\Delta_{\mathrm{new}} > 0$. Finally apply the norm-pullback argument of Corollary 14.1 with $B = B_{\mathrm{new}} > 0$. This gives all Dirichlet L-functions, including ζ, and completes the argument of Part III. □

Part IV. A moment-parameter extension and the optimal comparison bound

27. Statement of the limiting zero-free bound

Part III gives a bound strong enough to permit $\kappa < 3/4$, once the unweighted moment has been extended to that range. We state the resulting limiting bound here; Sections 28–33 prove it, and Section 34 proves its optimality within the specified comparison system.

Theorem 21.1 gives

$$\beta_* \le B_{\mathrm{new}} < 7/8, \qquad 2B_{\mathrm{new}} - 1 < 3/4.$$

We extend [Ope26a, Lemma 18.1] to $13/18 \le \kappa \le 1$ and iterate the resulting length optimization.

Theorem 27.1 (Limiting zero-free bound)**.** *Let*

$$\xi_\infty = \frac{20}{33} + \frac{\sqrt{1391}}{33}\cos\left(\frac{1}{3}\arccos\frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right), \tag{27.1}$$

where arccos $\in [0,\pi]$. *Equivalently, ξ_∞ is the unique root in $(0.38858, 0.38859)$ of $1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0$. Then*

$$\beta_* \le B_\infty := \frac{18\xi_\infty + 7}{18\xi_\infty + 9} = 0.874957019420098946\ldots.$$

Every finite-order Hecke L-function over F, and every Dirichlet L-function, including ζ, is zero-free for $\operatorname{Re} s > B_\infty$*; principal poles are allowed.*

The separate scalar optimality statement is Theorem 34.4, following the definition of its certificate class $\mathscr{A}$ in Section 34.2.

We retain the original modified average, two unweighted factors and sixth-power inverse amplification, with $\alpha = 5/6$. The notation is

$$\delta = 2a - 1, \quad x = \bar{g}/\delta, \quad b = l_y - l_x, \quad h = 1 - l_x + \ell;$$

t denotes the detector cutoff and ξ parametrizes the geometry. The exact scalar verifier and the consolidated Lean formalization are available in the accompanying repository [Liu26].

28. Extending the unweighted moment below three quarters

We extend the proof of [Ope26a, Lemma 18.1] from $[3/4, 1]$ to $[13/18, 1]$; its printed statement alone does not give this range. This is the analytic input needed for the variable-parameter count in Section 29, with its zero-free premise supplied at each step by the preceding bound.

Proposition 28.1 (Extended unweighted product moment)**.** *Retain the notation, row families, fixed twists, common coprimality conditions and bounded length ranges of (9.4). Distinct prime slots have disjoint underlying supports before masks and zero extensions. Then (9.4) holds uniformly for*

$$\frac{13}{18} \le \kappa \le 1, \tag{28.1}$$

in the cases

(28.2)

	length condition	*excluded inducing characters*
$z_{\mathrm{p}} = 0$	n_1, n_2 *arbitrary bounded*	*principal*
$z_{\mathrm{p}} > 0$	$n_1 + n_2 + 6\kappa z_{\mathrm{p}} \le M$	Θ.

If $z_{\mathrm{p}} > 0$ and $\kappa < 1$, assume $\beta_* \le (1+\kappa)/2$. The mesh ℓ_{mesh} depends only on the bounded real ranges and ε, not on the slot count. Constants and finite smooth-seminorm

and polynomial-height orders may depend on the fixed slot count and arithmetic data; they are uniform over moving rows, moduli, admissible common masks and frozen outer labels in the prescribed ranges.

Proof. Put $z = z_{\mathrm{p}}$ and $A = n_1 + n_2 + z$. We check the κ-dependent steps of the increasing-width induction in [Ope26a, Section 18]. Natural characters retain their conductor, redundant and fixed $\mathcal{S}$-zeros; cancellation on units does not remove a displayed prime from q_0. The zero-slot induction is unchanged.

Prime bound and masks. For $\kappa < 1$ put $s_\kappa = (1+\kappa)/2$. By [Ope26a, Lemma 4.9], whose real boundary ranges over $[1/2, 1]$,

$$s_\kappa \ge \beta_* \quad \Longrightarrow \quad \frac{L'}{L}(s_\kappa + e + it, \psi) \ll_e \log\{2Q_\psi(3+|t|)^2\}.$$

For each slot coefficient character $\vartheta \in \Theta$,

$$\psi \notin \Theta, \quad \psi\vartheta \text{ principal} \quad \Longrightarrow \quad \psi = \vartheta^{-1} \in \Theta,$$

a contradiction. Mellin displacement of the smooth von Mangoldt annulus to $s_\kappa + e$, followed by division of its weight by $\log(Py)$, gives the prime bound. Prime powers and bounded scales are admissible since

$$s_\kappa \ge 31/36, \qquad P^{1/2+\varepsilon_1} \le P^{s_\kappa+\varepsilon_1} \quad (P \ge 1).$$

Central normalization gives

$$|Q_{\boldsymbol{\omega}}|^2 \le C Z^{\kappa z+\varepsilon_1} p_j(\boldsymbol{W})^c (1+|\boldsymbol{\omega}|)^h. \tag{28.3}$$

Here p_j is a finite smooth seminorm, $\boldsymbol{\omega}$ comprises normalized norm-twist heights, and j, c, h are finite orders. The contour loss is $2ez$ with bounded total z; derivatives of profiles or normalized twists affect only the finite orders. For $\kappa = 1$ use absolute prime counting.

Exact mask deletion and reflection are independent of κ [Ope26a, Section 18, fixed masks and reflection]. The two plain factors have divisor coefficients $q_d^{-1/2}$; a masked prime slot is its natural slot minus the primes dividing the mask. Their inverse-root Euler masses are $O(Z^{\varepsilon_1})$ with arbitrary fixed loss. The retained lengths satisfy

$$A' \le A, \quad z' \le z, \qquad A' + (6\kappa-1)z' \le A + (6\kappa-1)z \le M.$$

If $z' = 0$, use the unrestricted zero-slot assertion. Reflection retains the conductor and redundant-radical union $\ll Z^M$, uniformly in the moving labels. It remains to prove the natural assertion.

Floor and comparison. Let $\rho > 0$ be the width floor and $\epsilon_{\mathrm{fr}} > 0$ the support-padding allowance, independent of the zero-class δ. At $M \le \rho$, absolute counting costs at most $\rho + \epsilon_{\mathrm{fr}}$ for the padded zero-slot core. For $z > 0$,

$$A \ge z, \qquad 6\kappa z \le M, \qquad z \le \frac{3M}{13}, \qquad \kappa z \le \frac{M}{6}. \tag{28.4}$$

Thus the additional terminal cost is $\le \rho/6$; the source's bound $z \le 2M/9$ is not needed.

For $M > \rho$, first prove $A \le 5M/6$. If $A > 5M/6$, set $L = M/4$. Then

$$6\kappa - 1 \ge 10/3, \qquad z < \frac{M}{6(6\kappa-1)} \le \frac{M}{20}, \qquad A - z > \frac{47M}{60} > 2L.$$

If an original plain length is below L, reflect the other factor. Otherwise subtract the comparison of lengths $L, A - z - L$ and reflect its longer factor only to bound that comparison separately. Centered reflection, including clipping and paired box multipliers,

with aggregate error ξ_{ref} gives

$$A_{\mathrm{comp}} \le \frac{3}{2}M - A + 2z + \xi_{\mathrm{ref}}. \tag{28.5}$$

Hence

$$\begin{aligned} A_{\mathrm{comp}} &\le \frac{23}{30}M + \xi_{\mathrm{ref}}, \\ A_{\mathrm{comp}} + (6\kappa - 1)z &\le \frac{5}{2}M - 2A + 2z + \xi_{\mathrm{ref}} \le \frac{14}{15}M + \xi_{\mathrm{ref}}. \end{aligned} \tag{28.6}$$

Both reserves are $\ge M/15 - \xi_{\mathrm{ref}} > 0$ if $\xi_{\mathrm{ref}} \le \rho/30$. For $z = 0$ the reserve is $M/6 - \xi_{\mathrm{ref}}$. The comparison therefore calls the earlier uncentered stage at the same width.

Transforms and children. The first Poisson transform, complete support extraction, sixth-power Gauss-row amplification, second complete correlation and full Möbius allocation of [Ope26a, Section 18] contain no κ. Retain the signed rectangle (9.5) until a positive squared norm is formed. The common character, mask and norm power on each separated side are preserved by [Ope26a, Lemma 18.2]; all displayed moving primes remain counted, and frozen masks remain common to both plain variables and both rectangles. The width ledger is

$$M' = M + J - g - g_2 + t_2 \le M - \sigma, \qquad 0 \le M'_{\mathrm{act}} - M' \le \delta_{\mathrm{fr},2}.$$

Here J is the first-transform length defect, g, g_2 the two complete-support savings, t_2 the second common-support length, and $\sigma > 0$ the width step, as in [Ope26a, Section 18].

After frequency enclosure, allocation, clipping and mask deletion,

$$\begin{aligned} F_{\mathrm{act}} &= (A_{\mathrm{act}} - M'_{\mathrm{act}} + (6\kappa - 1)z_{\mathrm{act}})_+ \\ &\le 6\Delta_{\mathrm{child}} + \delta_{\mathrm{fr},1} + 2\theta_N, \qquad z_{\mathrm{act}} \le z. \end{aligned}$$

The support loss $\Delta_{\mathrm{child}} \ge 0$ and aggregate error θ_N are those of the source's ledger; its inequality uses only the parent condition and $6\kappa - 1 \ge 0$. Remove whole slots until repair or exhaustion. Removing length d lowers the affine expression by $6\kappa d$ and costs κd. With $\eta = \ell_{\mathrm{mesh}}$, the first prefix reaching $F_{\mathrm{act}}/(6\kappa)$ has one possible overshoot, so

$$d_z \le \min\left\{ z_{\mathrm{act}}, \frac{F_{\mathrm{act}}}{6\kappa} + \eta \right\}, \qquad \kappa d_z \le \frac{F_{\mathrm{act}}}{6} + \eta. \tag{28.7}$$

An exhausted slot list uses the completed smaller-width zero-slot assertion; otherwise use the repaired positive-slot assertion there. Surviving slots retain their coefficients and disjoint supports. Both frequency errors are included before this repair, and Cauchy–Schwarz averages the two child errors.

For inducing characters in Θ, use the unchanged exceptional count and [Ope26a, Lemma 18.3]: the uncentered parent has $A \le M$, and the centered main terms cancel by equality of the scale products. The threshold $5M/6$ is independent of κ.

Quantifiers and finite induction. From the bounded ranges and ε, choose $\rho, \sigma, \epsilon_{\mathrm{fr}} > 0$ with $\epsilon_{\mathrm{fr}} \ll \min(\rho, \sigma)$ and

$$T_{\mathrm{term}} := \rho + \epsilon_{\mathrm{fr}} + \rho/6 + 3\sigma + 5\sigma/3 < \varepsilon/4, \qquad D = 2 + \lceil 2M_{\max}/\sigma \rceil.$$

Since $0 < 6\kappa - 1 \le 5$,

$$|\Delta(A - M + (6\kappa - 1)z)| \le |\Delta A| + |\Delta M| + 5|\Delta z|.$$

A numerical stage constant C_* covers these fixed aggregate operations. Before specifying the slot count, choose

$$\begin{aligned} \xi_{\mathrm{ref}} &< \min\{\epsilon_{\mathrm{fr}}/2, \rho/30, \sigma/(4C_*), \varepsilon/(16C_*D)\}, \\ \eta &< \min\{\sigma/6, \varepsilon/(16C_*D)\}, \qquad \varepsilon_0 < \varepsilon/(16C_*D). \end{aligned}$$

Now fix the slot system and its aggregate support bound H_N; impose $\log Z \gg H_N/\xi_{\mathrm{ref}}$ and the reflection and localization thresholds. Local divisor and prime shares sum to ε_0 per stage. Shares and thresholds may depend on N, while η does not. For the simultaneous induction over surviving subsets, take

$$\mathcal{E}_d = T_{\mathrm{term}} + (d+1)C_*(\eta + \xi_{\mathrm{ref}} + \varepsilon_0), \quad 0 \le d \le D-2, \qquad \mathcal{E}_{D-2} < \varepsilon.$$

Strict children drop the actual width by $\ge \sigma/2$; comparisons use only the preceding uncentered stage in their band, and all zero-slot bands precede positive-slot bands. Hence there is no same-band cycle.

Choose internal kernel, reflection, prime and terminal seminorm and height orders backwards through depth D, then external Fourier-tail orders. Full weighted Fourier integrals retain the height factors of (28.3): larger orders raise seminorms and constants, not powers $Z^{J\xi_{\mathrm{ref}}}$. Masks remain of polynomial norm through this finite depth. This proves the natural assertion; initial exact mask deletion gives the proposition. □

Zero-free premise. The required bound $\beta_* \le (1+\kappa)/2$ must precede a smaller fixed κ. It may instead hold by equality when $\kappa = 2\beta_* - 1$ in a contradiction argument. No assertion is made below $\kappa = 13/18$.

29. The same-witness count with a variable moment parameter

We combine the extended moment of Section 28 with the unchanged inverse moment and simultaneous zero detector. Keeping κ variable yields the count exponent from which Section 30 derives the improved length parameters.

Put $x = \bar{g}/\delta \in [0, 1/2]$ and

$$c = \frac{2}{3\kappa}, \qquad D_{\kappa,x} = 3 - (1+c)x, \qquad P_{\kappa,x} = (2-cx)(1-x). \tag{29.1}$$

At $\kappa = 3/4$, $(D_{\kappa,x}, P_{\kappa,x}) = (D_x, P_x)$. The detecting cutoff is t; d remains the element-norm exponent.

Proposition 29.1 (Variable-parameter count)**.** *Retain the detector, natural-row, coefficient, smooth-weight and height hypotheses of Proposition 18.1, with the common witness pair (5.5). Assume*

$$13/18 \le \kappa \le 1, \qquad \beta_* \le (1+\kappa)/2, \qquad \ell/d \ge 1/5 + \rho_{\mathrm{sup}}, \quad \rho_{\mathrm{sup}} > 0.$$

For $0 < \delta \le \alpha = 5/6$ and $0 \le x \le 1/2$, put

$$\mathcal{J}_\kappa = (\alpha-\delta)D_{\kappa,x} + \delta P_{\kappa,x}, \quad t_\kappa = 1 + \frac{\delta P_{\kappa,x}}{2\mathcal{J}_\kappa}, \quad R_\kappa = 1 - \delta + \frac{(\alpha-\delta)\delta P_{\kappa,x}}{2\mathcal{J}_\kappa}. \tag{29.2}$$

For each $\varepsilon > 0$, sufficiently fine fixed prime lengths give

$$\#\mathcal{B} \ll U^{R_\kappa+\varepsilon}(1+T_1)^A.$$

Proof. Apply Proposition 28.1 and the inverse moment to the same character and witness. For $m = t - r$, the capacities are

$$z_M = (1-r)/2, \qquad z_S = (1-2m)/(6\kappa).$$

The savings divided by δ are

$$F_I(r) = x + (1-x)r, \qquad F_S(r) = 2(t-r) + \frac{c}{2}x\{1 - 2(t-r)\}.$$

Their crossing is

$$r_\kappa(t,x) = \frac{(2-cx)t + (c/2-1)x}{3-(1+c)x}. \tag{29.3}$$

For $1 \le t \le 3/2$ and $0 \le x \le 1/2$,

$$\partial_t r_\kappa = \frac{2 - cx}{D_{\kappa,x}} > 0, \qquad \partial_x r_\kappa = \frac{(2-c)(2t-3)}{2D_{\kappa,x}^2} \le 0,$$

$$\partial_t (t - r_\kappa) = \frac{1-x}{D_{\kappa,x}} > 0, \qquad \partial_x (t - r_\kappa) = -\partial_x r_\kappa \ge 0,$$

$$r_\kappa(3/2, x) = 1, \qquad \frac{6-c}{10-2c} \le r_\kappa \le 1, \qquad \frac{1}{3} \le t - r_\kappa \le \frac{1}{2}.$$

Use the inverse moment for $r \ge r_\kappa$ and the plain moment for $r \le r_\kappa$. Their capacities satisfy

$$z_M \le \frac{4-c}{20-4c} < \frac{1}{5}, \qquad z_S \le \frac{c}{12} < \frac{1}{5}.$$

Decrease each selected length by a fixed $\nu > 0$ and choose the whole-slot mesh so that its rounding loss fits ε. Then

$$1 - r - 2z \ge 2\nu,$$

$$3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) \ge 8\nu + \frac{1}{5-c} > 0.$$

Nonpositive remaining capacities use the unmarked moments.

Substitution of (29.3) gives the short count

$$R_{\kappa,\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_{\kappa,x}}{D_{\kappa,x}}(3/2 - t).$$

The long count and the balancing identities are

$$R_{\mathrm{long}}(t) = 1 - \alpha + (\alpha - \delta)t,$$

$$R_{\kappa,\mathrm{short}}(t_\kappa) = R_{\mathrm{long}}(t_\kappa) = R_\kappa,$$

$$D_{\kappa,x} - P_{\kappa,x} = 1 + x - cx^2 > 0, \qquad 1 \le t_\kappa \le 3/2.$$

At $\delta = \alpha$, $t_\kappa = 3/2$ and $R_\kappa = 1 - \alpha$; use the unmarked amplified inverse estimate, as in Proposition 18.1. All coefficient classes, presentations and smooth separations are unchanged. □

30. A family of balanced geometries

We choose the averaging lengths by balancing the direct estimate with the critical class of the variable-parameter count. The resulting one-parameter family contains the Part III geometry; Section 31 verifies its endpoint inequality for all zero and amplitude classes.

For the geometry parameter ξ in

$$I = [38858/100000, 38859/100000] \tag{30.1}$$

define

$$\kappa(\xi) = \frac{18\xi^2 - 57\xi + 20}{6(36\xi^2 - 78\xi + 25)}, \qquad \ell(\xi) = \frac{5 - 6\xi}{9 + 18\xi}, \qquad B(\xi) = \frac{18\xi + 7}{18\xi + 9},$$

$$b(\xi) = \frac{8(3\xi - 1)(12\xi^2 - 40\xi + 15)}{9(2\xi + 1)(24\xi^2 - 20\xi + 5)},$$

$$l_x(\xi) = \frac{1 - b(\xi) - \ell(\xi)}{2}, \qquad l_y(\xi) = \frac{1 + b(\xi) - \ell(\xi)}{2}, \qquad h(\xi) = \frac{1 + b(\xi) + 3\ell(\xi)}{2}, \tag{30.2}$$

$$C_\xi(s) = s - \frac{4 + b(\xi)}{6}, \qquad B(\xi) = \frac{11}{12} - \frac{\ell(\xi)}{4}.$$

Suppress the argument at fixed ξ. For $\xi_0 = (49 - \sqrt{921})/48$, (30.2) recovers (21.2) and $\kappa(\xi_0) = 3/4$.

Lemma 30.1 (Critical class and balanced parameters)**.** *For the geometry* (30.2)*, at* $(\delta, x) = (\xi, 1/2)$*,*

$$R_\kappa = 2/3, \qquad E_{B(\xi)}(h) = 0, \qquad \partial_\delta E_{B(\xi)}(h) = 0.$$

The equation $R_\kappa(\delta, 1/2) = 2/3$ *is equivalent to*

$$(216\kappa - 18)\delta^2 - (468\kappa - 57)\delta + 150\kappa - 20 = 0. \tag{30.3}$$

The solution $\delta = \xi \in I$ *is its smaller root.*

Proof. At $x = 1/2$, $D = (5 - c)/2$, $P = 1 - c/4$, and

$$R_\kappa - \frac{2}{3} = \frac{(216\kappa - 18)\delta^2 - (468\kappa - 57)\delta + 150\kappa - 20}{216\kappa \mathcal{J}_\kappa}.$$

Thus (30.3) gives $\kappa(\xi)$. The sum of its roots exceeds two; since $\xi < 39/100$, the other root exceeds one. Substitution in (18.5) gives

$$R_\kappa(\xi, 1/2) = 2/3 \quad \Longrightarrow \quad \partial_b E_B(h) = R_\kappa/2 - 1/3 = 0,$$
$$E_B(h) = 0 \quad \Longleftrightarrow \quad \ell = \frac{5 - 6\xi}{9 + 18\xi},$$
$$\partial_\delta E_B(h) = 0 \quad \Longleftrightarrow \quad b = \frac{8(3\xi - 1)(12\xi^2 - 40\xi + 15)}{9(2\xi + 1)(24\xi^2 - 20\xi + 5)}.$$

The uniform sign is proved below. □

31. An exact uniform endpoint certificate

We prove the endpoint inequality uniformly along the family in Section 30, rather than only at its critical class. The polynomial identity and the auxiliary reserve bounds below supply the scalar estimates needed for the analytic comparison in Section 32.

Lemma 31.1 (Variable-parameter endpoint)**.** *For every* $\xi \in I$*, use* (30.2) *and* $R = R_{\kappa(\xi)}$*. Then*

$$E_{B(\xi)}(h) \le 0 \qquad (0 \le \delta \le 5/6, \quad 0 \le x \le 1/2). \tag{31.1}$$

Equality holds only at $(\delta, x) = (\xi, 1/2)$*.*

Proof. Put $y = 1/2 - x$, $c = 2/(3\kappa(\xi))$, and

$$D = (5 - c)/2 + (1 + c)y, \qquad P = 1 - c/4 + 2y + cy^2, \qquad \mathcal{J} = (\alpha - \delta)D + \delta P.$$

By (18.5),

$$E = C_0 + (1/2 + \ell)\delta + \ell x\delta - h(1 - R_\kappa), \qquad C_0 = -1/4 + b/6 + 5\ell/4.$$

Hence

$$Q = -2\mathcal{J}E = -2C_0\mathcal{J} + (b + 2\ell y)\delta\mathcal{J} - h(\alpha - \delta)\delta P. \tag{31.2}$$

Write $Q = A(y)\delta^2 + B_q(y)\delta + C_q(y)$ and $\mathscr{D}(y) = 4A(y)C_q(y) - B_q(y)^2$. Set

$$D_0 = (5 - c)/2, \quad D_1 = 1 + c, \quad P_0 = 1 - c/4, \quad T_0 = -3/2 + c/4, \quad T_1 = 1 - c, \quad T_2 = c.$$

The coefficients are

$$A_0 = bT_0 + hP_0, \qquad A_1 = bT_1 + 2\ell T_0 + 2h,$$
$$A_2 = bc + 2\ell T_1 + hc, \qquad A_3 = 2\ell c,$$
$$B_0 = -2C_0T_0 + \alpha D_0 b - \alpha h P_0,$$
$$B_1 = -2C_0T_1 + \alpha D_1 b + 2\alpha\ell D_0 - 2\alpha h,$$
$$B_2 = -2C_0 c + 2\alpha\ell D_1 - \alpha hc,$$
$$C_{q,0} = -2C_0\alpha D_0, \qquad C_{q,1} = -2C_0\alpha D_1.$$

The coefficients d_j of $\mathscr{D}$ are

$$d_0 = 4A_0C_{q,0} - B_0^2 = 0,$$
$$d_1 = 4(A_0C_{q,1} + A_1C_{q,0}) - 2B_0B_1,$$
$$d_2 = 4(A_1C_{q,1} + A_2C_{q,0}) - B_1^2 - 2B_0B_2,$$
$$d_3 = 4(A_2C_{q,1} + A_3C_{q,0}) - 2B_1B_2,$$
$$d_4 = 4A_3C_{q,1} - B_2^2.$$

The identity $d_0 = 0$ and the following strict bounds hold on I:

quantity	lower bound	quantity	lower bound
A_0	47/100	d_1	41/1000
A_1	121/100	d_2	198/1000
A_2	86/100	d_3	267/1000
A_3	29/100	d_4	76/1000

For $F = N/D$, cancel common factors and normalize the common sign. For a polynomial of degree n, write

$$N(L + (U-L)z) = \sum_{j=0}^{n} a_j z^j, \qquad \mathfrak{b}_i(N) = \sum_{j=0}^{i} a_j \frac{\binom{i}{j}}{\binom{n}{j}},$$
$$L = 38858/100000, \qquad U = 38859/100000.$$

The Bernstein identity (19.3) gives $N(L + (U-L)z) = \sum_i \mathfrak{b}_i(N)\mathsf{B}_{i,n}(z)$. For every table entry, $\mathfrak{b}_i(N), \mathfrak{b}_i(D) > 0$, so

$$F(\xi) \geq \frac{\min_i \mathfrak{b}_i(N)}{\max_i \mathfrak{b}_i(D)}.$$

The exact quotients exceed the table bounds; their coefficients and (31.2) are verified by `combine_v2-2_supplement/check_kappa_endpoint.py`. Therefore

$$A(y) > 0 \quad (y \geq 0), \qquad \mathscr{D}(y) > 0 \quad (y > 0),$$
$$4A(y)Q = (2A(y)\delta + B_q(y))^2 + \mathscr{D}(y) \geq 0, \qquad \mathcal{J} > 0 \implies E \leq 0,$$
$$Q(\delta, 0) = A_0(\delta - \xi)^2.$$

Thus equality holds precisely at $(\delta, x) = (\xi, 1/2)$. □

Lemma 31.2 (Uniform strict side reserves)**.** *All geometries* (30.2) *satisfy*

$$b > 0, \quad 0 < \ell < 1/5, \quad l_x - b - \ell > 0, \quad l_y - \ell > 11b/6,$$
$$h < 1, \qquad 5\ell > h, \qquad 437/500 < B < 7/8.$$

The reserve $l_x - b - \ell > 0$ is deliberately stronger than $l_x - \ell > 0$, since $b > 0$. The floor, intermediate and small-norm estimates have uniform strict reserves; the detecting and rounded count formulas have uniformly positive denominators.

Proof. The rational Bernstein certificate gives lower bound $1/25000$ on I for each expression in

$$\begin{gathered}
36\xi^2-78\xi+25,\quad \kappa',\quad B',\\
\kappa'-1000B',\quad \kappa-13/18,\quad 4/5-\kappa,\quad -p',\\
b,\quad \ell,\quad 1/5-\ell,\quad l_x-b-\ell,\quad l_y-\ell,\\
l_y-\ell-11b/6,\quad 1-h,\quad 5\ell-h,\\
B-43/50,\quad B-437/500,\quad 7/8-B,\\
-C_0-(1/2+3\ell/2)/50,\\
l_y/2-13h/75-1/50-(7/8-B),\\
-\{(225\ell+50-75b)\delta+78\ell-49b-73\}/300\quad(\delta=1/50,19/25),\\
c,\quad 1-c,\quad 1/5-(4-c)/(20-4c),\quad 1/5-c/12,\\
(\alpha-19/25)(5-c)/2+(19/25)(1-c/4)-3/5,\\
3-(1+c)51/100-2,\qquad (2-c51/100)(49/100)-3/4.
\end{gathered}\tag{31.3}$$

Here $p(\xi)=1188\xi^3-2160\xi^2+171\xi+190$. The 29 side bounds and eight endpoint bounds are recorded in the exact checker. Put

$$\begin{aligned}
m_F&=-C_0-(1/2+3\ell/2)/50,\\
m_S&=l_y/2-13h/75-1/50,\\
m_I(\delta)&=-\{(225\ell+50-75b)\delta+78\ell-49b-73\}/300.
\end{aligned}$$

Then

$$\begin{gathered}
m_F>1/25000,\qquad m_S-(7/8-B)>1/25000,\\
m_I(1/50),m_I(19/25)>1/25000\quad\Longrightarrow\quad m_I(\delta)>1/25000\quad(1/50\le\delta\le19/25),
\end{gathered}$$

since m_I is affine. These are the floor, small-norm and intermediate reserves of Lemma 19.2. The initial bound and small fixed class widths give

$$\delta\le2\beta_*-1+O(\epsilon_{\mathrm{b}})\le3/4+O(\epsilon_{\mathrm{b}})<19/25.$$

On the enlarged rounding box $0\le x\le51/100$, (31.3) gives $D>2$, $P>3/4$, and $D-P=1+x-cx^2>0$. Hence

$$\begin{aligned}
\mathcal{J}&=(\alpha-\delta)D+\delta P\ge2(\alpha-\delta)+(3/4)\delta\ge43/60>3/5,\\
3/2-t_\kappa&=\frac{(\alpha-\delta)D}{2\mathcal{J}}\ge\frac{\alpha-19/25}{2\alpha}=11/250>0.
\end{aligned}$$

The rational expressions have uniformly bounded derivatives on this compact box, so fixed rounding and class errors are absorbed in the prescribed losses. The detecting cutoff retains its original range. □

32. The one-step zero-free bound via the continuation criterion

We turn the variable-parameter count and uniform endpoint certificate into one application of Proposition 3.1. The conclusion is a new bound $\beta_*\le B(\xi)$ from a previously established moment premise; Section 33 repeatedly applies this implication.

Fix $\xi\in I$, $\kappa(\xi)\le3/4$, and use (30.2). The moment antecedent is

$$\beta_*\le\frac{1+\kappa(\xi)}{2}.\tag{32.1}$$

Theorem 13.1 also gives $\beta_*\le7/8$.

Proposition 32.1 (One bootstrap step)**.** *For $\xi \in I$ and $\kappa(\xi) \le 3/4$,*

$$\beta_* \le \frac{1+\kappa(\xi)}{2} \quad \Longrightarrow \quad \beta_* \le B(\xi).$$

Proof. Assume (32.1) and $\Delta := \beta_* - B > 0$. Retain the physical modified average, coefficients, zero extensions, disjoint supports and finite character expansions of Part I.

The direct estimate. For $d_{\mathcal{I}} = \sum_{i\in\mathcal{I}} \ell_i \le \ell$, Lemma 31.2 gives

$$l_x - d_{\mathcal{I}} > 0, \qquad l_y - d_{\mathcal{I}} - 11b/6 > 0,$$
$$M' = 1 - \ell - 2d_{\mathcal{I}} \ge 1 - 3\ell > 0, \qquad P_a = q_{b_*}^{-1} Z^b, \qquad P_a^2/Y' \ll P_a^{1/6}.$$

Thus $Q, Y', P_a \ge 1$ for large Z, and the reflected and Gram estimates apply to every rescaled subset. Its net penalty is

$$-d_{\mathcal{I}} + (d_{\mathcal{I}} + 5\ell - 1)_+ /8 \le -7d_{\mathcal{I}}/8 \le 0.$$

Proposition 16.2, with the positive even-slot normalizer, yields

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{C_b(B)+\varepsilon}, \qquad C_b(B) = l_x/2 + b/12, \qquad C_b(s) = s - (4+b)/6. \tag{32.2}$$

Small norms. Take $(s_r, w_r, z_r) = (\beta_* + \epsilon_{\mathrm{b}}, 1/2, 17/50)$, with first-domain margin less than $B - 1/2$. For $p \nmid u$, the error exponents in (17.5) are

$$-s_r, \quad -51/25, \quad 49/25 - 5s_r, \quad -77/50 < 0.$$

For $p \mid u$, the boundary exponents are

$$-1/2, \quad 3/2 - 2s_r, \quad (3/2 - 2s_r, 2 - 3s_r), \quad 2 - 3s_r, \quad 3 - 5s_r < 0.$$

Since $s_r \ge B > 437/500$, the quotient-free formula gives

$$G_p \ll \begin{cases} 1, & p \nmid u, \\ q_p^{1/2}, & p \mid u, \end{cases} \qquad \text{full tuple sum} \ll U^{\varepsilon} Z^{17\ell/50}.$$

The numerator growth, row count and Mellin factor contribute $U^{3/5+\varepsilon} U U^{-17/50} = U^{63/50+\varepsilon}$. Hence, for $d \le d_{\min} := 1/100$, the relative exponent is at most

$$\frac{13h}{75} - \frac{l_y}{2} + 2d_{\min} + O(\epsilon_{\mathrm{b}} + \varepsilon) = -m_S + O(\epsilon_{\mathrm{b}} + \varepsilon), \qquad m_S := \frac{l_y}{2} - \frac{13h}{75} - \frac{1}{50} > 0.$$

This includes unit rows and principal denominators; the numerator is nonprincipal. Lemma 31.2 gives $m_S > 7/8 - B$.

Floor and intermediate norms. The floor class has

$$a_0 = 51/100, \quad \delta_0 = 1/50, \quad R = 1, \quad \bar{g} \le \delta_0/2, \quad E_B(h) \le -m_F < 0,$$

where $m_F = -C_0 - (1/2 + 3\ell/2)/50$ and $C_0 = -1/4 + b/6 + 5\ell/4$. Its positive d-slope controls $d \le h$. For $d_{\min} \le d \le 1/2$, the unchanged unselected count gives

$$R = 76/75 - 2\delta/3, \qquad \partial_d E_B = 101/150 - \delta/6 > 0,$$

$$E_B(d) \le E_B(1/2) \le \frac{(225\ell + 50 - 75b)\delta + 78\ell - 49b - 73}{300} < 0.$$

The last affine expression is negative at $\delta = 1/50, 19/25$ by Lemma 31.2; all actual classes satisfy

$$\delta \le 2\beta_* - 1 + O(\epsilon_{\mathrm{b}}) \le 3/4 + O(\epsilon_{\mathrm{b}}) < 19/25 < \alpha.$$

Moderate and large norms. For $1/2 \le d \le h$, Proposition 29.1 uses the same character and witness in both moments. The length selected on the appropriate side of the crossing satisfies

$$z_{\mathrm{sel}} < 1/5 < \ell/h.$$

A fixed decrement, followed by a sufficiently fine whole-slot mesh, preserves both inverse-width reserves and the original coefficients and supports. Lemma 31.1 gives

$$E_B(h) \le 0, \qquad 0 < 33/50 - \delta/2 \le \partial_d E_B = R + \delta/2 - 17/50 < 2.$$

Thus $E_B(d) \le 0$ for $d \le h$. Choose $\zeta > 0$ with $\ell/(h+\zeta) > 1/5$; extending to $h+\zeta$ costs at most 2ζ. On $(s_r, w_r, z_r) = (2, 2, z_\infty)$,

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{l_x/2+1+l_y+(h+\zeta)(1+\varepsilon)-\zeta z_\infty+\varepsilon}.$$

Choose $z_\infty > 2$ after ζ to obtain any prescribed saving. In the central bins, normalization at β_* subtracts Δ; the uniform denominator bound absorbs bin overflow and rounding within this gap.

The principal term. Since $437/500 < B < 7/8$, Lemma 17.1 gives

$$H_\eta(s) = 1 + O(P_0^{-4/5}), \qquad |H_\eta(s) - 1| \le 1/2 \quad (\mathrm{Re}\, s \ge B).$$

Choose P_0 before the target; adjoining its conductor primes preserves the bound. The only principal frequency is $u = 1$, with residues at $w = 1, z = 1/6$ and normalizer c_S, including the factor $1/6$. Proposition 17.2 supplies the signal (17.6) and reserves

$$l_y/20, \qquad h/600, \qquad 0 < \kappa_P < B \min_i \ell_i.$$

Keep the approximation error on its global line and move only the main Gaussian integral to the right.

Application of Proposition 3.1. Let $m_I > 0$ be the intermediate reserve and choose

$$0 < \mu_0 < \min\{\Delta/4, m_F/4, m_S/4, m_I, l_y/80, h/2400\}, \qquad 2\zeta < \mu_0.$$

The order of the remaining fixed choices is

$$\text{moment loss} \longrightarrow \text{slot decrements} \longrightarrow \max_i \ell_i \longrightarrow \text{even slot system}$$
$$\longrightarrow \kappa_P \longrightarrow \text{class, amplitude and weight losses} \longrightarrow \eta.$$

Decrease the common margin as necessary. Logarithmically many configurations are absorbed in these losses. The proof of Proposition 20.1 uses the chosen count only through its exponent R, the displayed slope bounds and the uniform loss and mesh hypotheses, all verified above for R_κ. Proposition 20.1, with the count of Proposition 29.1, now gives

$$|J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu}(1+T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}, \qquad \mu > 0,$$

where μ is target-independent. Fix the arithmetic data and internal Fourier orders before A_η, B_η, and then choose

$$T_1 = Z^{\tau_\eta}, \qquad 0 < \tau_\eta \le \min\{d_{\min}/100, \tau_{\det}, \mu/[4(A_\eta+1)]\},$$
$$B_\eta - N\tau_\eta < C_b(\beta_*) - \mu/2 \quad \Longrightarrow \quad |J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu/2}.$$

The external order N changes seminorm constants, not the fixed internal degrees; J_η, f_η are independent of this height choice. Apply Proposition 3.1 with

$$(\sigma_0, c, \omega_0, \sigma) = \big(B, -(4+b)/6, \Delta/2, \mu/2\big).$$

The direct bound (32.2) and principal contraction imply continuation across a zero sufficiently near β_*. This contradiction proves $\beta_* \le B$. □

33. Proof of the new zero-free bound by iteration

Proposition 32.1 applies the continuation criterion of Proposition 3.1 at each finite step. The limit below proves Theorem 27.1; Section 34 proves its optimality in the stated comparison system.

Put

$$p(\xi) = 1188\xi^3 - 2160\xi^2 + 171\xi + 190. \tag{33.1}$$

The endpoint signs and $p' < 0$ on I give a unique zero $\xi_\infty \in I$.

Lemma 33.1 (Explicit root)**.** *The root ξ_∞ is given by (27.1).*

Proof. With $u = 66\xi - 40$,

$$242p\left(\frac{u+40}{66}\right) = u^3 - 4173u - 56940.$$

Set

$$\theta = \frac{1}{3}\arccos\frac{28470}{1391^{3/2}} \in (0, \pi/6).$$

For $u = 2\sqrt{1391}\cos(\theta - 2\pi/3)$,

$$u^3 - 4173u = 2 \cdot 1391^{3/2}\cos(3\theta) = 56940.$$

Moreover,

$$-\sqrt{1391} < u < 0 \quad\Longrightarrow\quad 0 < \frac{40 - \sqrt{1391}}{66} < \xi < \frac{20}{33} < 1.$$

Since $p(0) > 0 > p(1)$ and p has a root in each of $(-\infty, 0)$, $(0, 1)$ and $(1, \infty)$, this is its unique root in $(0, 1)$, hence the isolated root in I. □

Define

$$\kappa_\infty = \kappa(\xi_\infty), \quad B_\infty = B(\xi_\infty), \quad b_\infty = b(\xi_\infty), \quad \ell_\infty = \ell(\xi_\infty). \tag{33.2}$$

The identity

$$\kappa(\xi) - (2B(\xi) - 1) = -\frac{p(\xi)}{18(2\xi+1)(36\xi^2 - 78\xi + 25)} \tag{33.3}$$

gives $\kappa_\infty = 2B_\infty - 1$.

Proof of Theorem 27.1. On I,

$$\kappa'(\xi) = \frac{9(24\xi^2 - 20\xi + 5)}{2(36\xi^2 - 78\xi + 25)^2} > 0, \qquad B'(\xi) > 0, \qquad \kappa' - 1000B' > 0.$$

Let $\xi(\kappa)$ be the inverse map and set $g(\kappa) = 2B(\xi(\kappa)) - 1$. Then

$$0 < g'(\kappa) = \frac{2B'(\xi(\kappa))}{\kappa'(\xi(\kappa))} < 1/500,$$

$$\kappa_\infty < \kappa \le 3/4 \quad\Longrightarrow\quad \kappa_\infty < g(\kappa) < \kappa,$$

by (33.3) and $p' < 0$. Starting with the known 7/8 bound, define

$$\kappa_0 = 3/4, \qquad \xi_n = \xi(\kappa_n), \qquad B_n = B(\xi_n), \qquad \kappa_{n+1} = g(\kappa_n) = 2B_n - 1.$$

Proposition 32.1 gives

$$\beta_* \le (1 + \kappa_n)/2 \quad\Longrightarrow\quad \beta_* \le B_n = (1 + \kappa_{n+1})/2.$$

Thus $B_0 = B_{\text{new}}$ and

$$\kappa_n \downarrow \kappa_\infty, \qquad B_n \downarrow B_\infty, \qquad \beta_* \le \inf_{n\ge 0} B_n = B_\infty.$$

Only these real bounds are passed to the limit; the analytic constants and prime systems may depend on n. Corollary 14.1 supplies the Dirichlet transfer. □

The limiting parameters are

B_∞	$0.874957019420098946\ldots$	κ_∞	$0.749914038840197892\ldots$
b_∞	$0.123405685419541002\ldots$	ℓ_∞	$0.166838588986270882\ldots.$

Substituting $\xi = (9B-7)/(18(1-B))$ into (33.1) gives

$$7884B_\infty^3 - 18819B_\infty^2 + 14643B_\infty - 3686 = 0. \tag{33.4}$$

The branch is specified by (27.1).

Corollary 33.2 (Rational checkpoint)**.** *Every L-function in Theorem 27.1 is zero-free for*

$$\operatorname{Re} s > B_{\mathrm{cp}} := \frac{34998281}{40000000} = 0.874957025.$$

Proof. For $\xi_1 = (9B_{\mathrm{cp}} - 7)/(18(1 - B_{\mathrm{cp}})) \in I$,

$$p(\xi_1) < 0 \quad\Longrightarrow\quad \xi_\infty < \xi_1 \quad\Longrightarrow\quad B_\infty < B_{\mathrm{cp}}.$$

□

The two improvements are

$$B_{\mathrm{new}} - B_{\mathrm{cp}} = 4.4799168740\ldots \times 10^{-8},$$
$$B_{\mathrm{new}} - B_\infty = 5.03790697938897\ldots \times 10^{-8}.$$

Both require changing the lengths: fixing $\ell = \ell_{\mathrm{new}}$ leaves $11/12 - \ell/4 = B_{\mathrm{new}}$.

34. Proof of optimality in the full comparison system

The preceding section proves that B_∞ is a valid bound; we now prove that no smaller boundary is admitted by the specified exponent comparison system. We first obtain an affine lower bound, then show that the permitted moment, length, detector and contour choices cannot evade its constraints, leading to Theorem 34.4.

For $13/18 \le \kappa \le 1$, let $\xi(\kappa)$ be the smaller root of (30.3), and set

$$I_0 = [19/50, 39/100], \qquad F(\kappa) = \frac{18\xi(\kappa) + 7}{18\xi(\kappa) + 9}.$$

Writing the critical quadratic as $q_\kappa(\xi)$, one has

	$\kappa = 13/18$	$\kappa = 1$
$\xi = 19/50$	$2776/1875$	$1507/625$
$\xi = 39/100$	$-4003/15000$	$-871/5000.$

Since q_κ is affine in κ, its leading coefficient is positive and $q_\kappa(1) = 19 - 102\kappa < 0$,

$$19/50 < \xi(\kappa) < 39/100 < 1 < \text{the second root}.$$

On I_0, put $D_0(\xi) = 36\xi^2 - 78\xi + 25$. Then

$$D_0' < 0, \qquad D_0 \ge D_0(39/100) = 139/2500 > 0,$$
$$\kappa'(\xi) = \frac{9(24\xi^2 - 20\xi + 5)}{2D_0(\xi)^2} > 0, \qquad B'(\xi) = \frac{36}{(18\xi + 9)^2} > 0,$$
$$p''(\xi) < 0, \qquad p'(\xi) \le p'(19/50) = -597474/625 < 0.$$

Thus $\xi(\kappa)$ inverts the rational function (30.2) on this larger algebraic interval. The analytic endpoint certificate remains on I.

For arbitrary $M, \ell \in \mathbb{R}$, define

$$L_0 = 5/6 + M/12 - \ell/6, \qquad L_K = 1/3 + 7M/12 + \ell/3, \tag{34.1}$$
$$H_\xi = 1 + \xi - (1+\xi)M/2 + \xi\ell.$$

Proposition 34.1 (Variable-moment algebraic lower bound)**.** *Let* $13/18 \le \kappa \le 1$ *and* $M, \ell, B \in \mathbb{R}$*. If*

$$B \ge L_0, \qquad B \ge L_K, \qquad B \ge H_{\xi(\kappa)}, \tag{34.2}$$

then $B \ge F(\kappa)$*, with equality at*

$$\ell = \frac{5 - 6\xi(\kappa)}{9 + 18\xi(\kappa)}, \qquad M = 1 - \ell.$$

Under the additional endpoint compatibility condition

$$B \le (1+\kappa)/2, \tag{34.3}$$

one has $\kappa \ge \kappa_\infty$ *and* $B \ge B_\infty$*. Equality is attained by* (33.2)*.*

Proof. For $\xi = \xi(\kappa)$, the weights

$$(w_0, w_K, w_H) = \frac{(4 + 18\xi, 2, 3)}{9 + 18\xi}$$

are positive and satisfy

$$w_0 + w_K + w_H = 1,$$
$$w_0/12 + 7w_K/12 - (1+\xi)w_H/2 = 0,$$
$$-w_0/6 + w_K/3 + \xi w_H = 0.$$

Consequently

$$B \ge \max\{L_0, L_K, H_\xi\} \ge w_0 L_0 + w_K L_K + w_H H_\xi = \frac{18\xi + 7}{18\xi + 9} = F(\kappa). \tag{34.4}$$

At the displayed M, ℓ, $L_0 = L_K = H_\xi = F(\kappa)$. Moreover (33.3) gives

$$F(\kappa) \le B \le (1+\kappa)/2 \Longrightarrow g(\kappa) = 2F(\kappa) - 1 \le \kappa$$
$$\Longleftrightarrow p(\xi) \le 0 \Longleftrightarrow \xi \ge \xi_\infty \Longleftrightarrow \kappa \ge \kappa_\infty.$$

Monotonicity yields $B \ge F(\kappa) \ge F(\kappa_\infty) = B_\infty$. The exact dual identities are checked in the supplement. □

Endpoint compatibility is a scalar condition; the moment requires an actual preceding family bound. For a known $\beta_* \le \widehat{B}$, the relations are

$$\kappa \ge 2\widehat{B} - 1, \qquad B \ge F(\kappa) \ge F(2\widehat{B} - 1).$$

If $2\widehat{B} - 1 \in \kappa(I)$ and is at most $3/4$, Proposition 32.1 realizes the last boundary. In particular $F(2B_\infty - 1) = B_\infty$.

34.1. **The sharp count obstruction in the stated moment system.** We optimize the supplied marginal exponents with zero losses and unlimited prime supply. The only witness constraints are

$$t - 1/2 \le r \le t, \qquad 0 \le m \le 1/2, \qquad r + m \ge t, \qquad 1 \le t \le 3/2. \tag{34.5}$$

Here κ is the smallest authorized operative parameter; $\kappa \mapsto (1 - 2m)/(6\kappa)$ is decreasing. Fix

$$\delta = \xi(\kappa) \in I_0, \qquad x = 1/2, \qquad c = 2/(3\kappa) \in [2/3, 12/13], \qquad \alpha = 5/6.$$

The optimized width-one exponents are

$$\begin{aligned} e_I(r) &= 1-\delta(1+r)/2 && (r\le 1),\\ e_S(m) &= 1-\delta\{2m+(c/4)(1-2m)\} && (0\le m\le 1/2), \end{aligned} \tag{34.6}$$

and the amplified long exponent is

$$e_L(r) = 1-\alpha+(\alpha-\delta)r \qquad (r\ge 1). \tag{34.7}$$

The unselected alternatives satisfy

$$1-\delta r \ge e_I(r) \quad (r\le 1), \qquad \{z>0 : r+2z\le 1\} = \varnothing \quad (r>1).$$

For the witness domain $\mathcal{W}_t$ in (34.5), put

$$e_{\mathrm{inv}}(r) = \begin{cases} e_I(r), & r\le 1,\\ e_L(r), & r\ge 1, \end{cases} \qquad R_{\mathrm{cert}}(t) = \sup_{(r,m)\in\mathcal{W}_t} \min\{e_{\mathrm{inv}}(r), e_S(m)\}.$$

These are supplied exponents, not lower bounds for actual counts.

Lemma 34.2 (Critical minimax count)**.** *For the moment choices and witness domain above, the smallest uniform certified count exponent is*

$$\inf_{1\le t\le 3/2} R_{\mathrm{cert}}(t) = R_\kappa(\delta, 1/2) = 2/3.$$

Prime caps, unselected choices and marginal interpolation cannot lower it.

Proof. On $m = t-r$, the crossing in (29.3) satisfies

$$t-1/2 \le r_\kappa(t,1/2) \le 1, \qquad 1/3 \le t - r_\kappa(t,1/2) \le 1/2,$$
$$e_I' = -\delta/2 < 0, \qquad \frac{d}{dr} e_S(t-r) = \delta(2-c/2) > 0.$$

Hence this admissible split imposes

$$\begin{aligned} R_{\mathrm{cert}}(t) &\ge \min\{e_I(r_\kappa), e_S(t-r_\kappa)\}\\ &= R_{\kappa,\mathrm{short}}(t) = 1-\delta+\frac{\delta P_{\kappa,1/2}}{D_{\kappa,1/2}}(3/2-t). \end{aligned}$$

At $t=3/2$, $(r,m)=(1,1/2)$; the unmarked inverse and zero-slot plain estimates both give $1-\delta$. No vanishing marked reserve is used.

The second admissible split $(r,m)=(t,0)$ imposes $R_{\mathrm{cert}}(t)\ge R_{\mathrm{long}}(t) = e_L(t)$, because

$$e_S(0) - e_L(t) \ge -5/12 + \delta(3/2-c/4) \ge -5/12 + (19/50)(5/4) = 7/120 > 0.$$

If a discrete presentation omits $m=0$, take $(r,m)=(t-\epsilon,\epsilon)$ and pass to the scalar limit. The short/long derivation in Proposition 29.1 gives the reverse uniform bound. Thus

$$\begin{aligned} R_{\mathrm{cert}}(t) &= \max\{R_{\kappa,\mathrm{short}}(t), R_{\mathrm{long}}(t)\},\\ R_{\kappa,\mathrm{short}}' < 0, &\qquad R_{\mathrm{long}}' = \alpha-\delta > 0,\\ R_{\kappa,\mathrm{short}}(t_\kappa) &= R_{\mathrm{long}}(t_\kappa) = R_\kappa(\delta,1/2) = 2/3. \end{aligned}$$

Restoring prime caps weakens the marginal savings. At each split, for $0\le\theta\le 1$,

$$\theta e_I + (1-\theta)e_S \ge \min(e_I, e_S).$$

The same minimum includes unmarked alternatives; hence neither choice nor Hölder interpolation lowers the minimax. □

Enlarging the ordinary moment width also cannot help. For $M' \ge 1$, write

$$I' = M' - \delta r - \delta z, \qquad S' = M' - 2\delta m - \delta z.$$

Their affine width conditions give

$$\begin{aligned} &r \le 1,\ z > z_I = (1-r)/2: && I' - e_I(r) \ge (2-\delta)(z - z_I) > 0,\\ &r > 1: && I' \ge r(1-\delta) \ge e_L(r),\\ &z > z_S = (1-2m)/(6\kappa): && S' - e_S(m) \ge (6\kappa - \delta)(z - z_S) > 0. \end{aligned}$$

For selected lengths below these capacities, $M' \ge 1$ and smaller amplitude savings likewise give no improvement. Other displayed unselected exponents satisfy

$$R_{\mathrm{I}}(\delta) \ge 1 - \delta/2 \ge 161/200 > 2/3,$$
$$R_N = 1 - \frac{2\alpha\delta}{3\alpha - \delta} \ge 1 - \frac{2\alpha(39/100)}{3\alpha - 39/100} > 2/3.$$

At $x = 1/2$, each slot has maximal gain $\delta/2$ per unit length. Redistribution, finite unions and separate detector choices introduce no stronger exponent: each detector permits the two adverse splits above. A relation between different witnesses, or a new joint moment, would be an additional input.

Lemma 34.3 (Necessary constraints for a putative improvement)**.** *Suppose a construction in the stated comparison system has $B < B_\infty$, with $l_x, l_y > \ell > 0$, $b = l_y - l_x > 0$, $M = l_x + l_y$ and $h = 1 - l_x + \ell > 0$. Use the full direct majorant (22.4) and uniform nonprincipal comparisons for every permitted class. If $13/18 \le \kappa \le 1$ and the supplied critical count exponent is at least $2/3$, then necessarily*

$$B \ge L_0, \qquad B \ge L_K, \qquad B \ge H_{\xi(\kappa)}.$$

For an endpoint-compatible certificate, these inequalities contradict Proposition 34.1.

Proof. For $G = \max\{0, b/12, b - l_y/2\} \ge b/12$,

$$\begin{aligned} B \ge B_{\mathrm{low}} &= 5/6 + l_x/6 + G - \ell/6 + k(M,\ell)/2\\ &\ge L_0 + k(M,\ell)/2 \ge \max\{L_0, L_K\}, \end{aligned}$$

using the actual branches $k \ge 0$ and $k \ge M + \ell - 1$. The scale inequalities consequently imply

$$M - 2\ell > b > 0, \qquad B \ge L_0 > 5/6,$$
$$M - 2\ell \le 12B - 10, \qquad h \ge 6 - 6B + b/2 > 3/4, \qquad h < 1,$$
$$M < 13/14, \qquad l_y < M, \qquad (1+\xi)/2 < 139/200 < 5/6 < B.$$

Thus $d = h$ is a moderate norm and the critical zero class is below any upper cutoff supplied by $\beta_* > B$.

At $(d, \delta, x) = (h, \xi, 1/2)$, direct substitution gives

$$E_B(h) = H_\xi - B + h(R - 2/3). \tag{34.8}$$

All z_0 terms cancel, and the branch H_ξ is independent of b. Since $h > 0$ and $R \ge 2/3$, $E_B(h) \le 0$ requires $B \ge H_\xi$. Raising the floor to $\delta_f \ge \xi$ instead gives

$$E_{B,\mathrm{floor}}(h) = H_{\delta_f} - B + h/3,$$
$$H_{\delta_f} - H_\xi = (\delta_f - \xi)(1 - M/2 + \ell) \ge 0, \qquad 1 - M/2 + \ell > 3/4.$$

Its trivial-count comparison therefore requires the stronger $B \ge H_\xi + h/3$. A floor below ξ leaves the critical class.

Global reciprocal contours. For these comparisons assume $B < \beta_* \le 7/8$. Put $s_r = \beta_* + \gamma$, $\gamma > 0$, $w_r = \omega$ and $z_r = \zeta$. For $\omega > 0$, $\zeta > 1/6$ and $s_r + \omega > 1$, the off-row defects

$$-s_r - \omega,\ -s_r - 6\zeta,\ -\omega - 6\zeta,\ 4 - 6s_r - 6\zeta,\ 1 - s_r - \omega - 6\zeta$$

are < -1 and the on-row defects are negative. The local table and quotient-free selected formula give

$$G_p = O(1)\ (p \nmid u), \qquad G_p \ll q_p^{\max(1-\omega,0)}\ (p \mid u),$$
$$\text{full positive tuple sum} \ll U^{\varepsilon} Z^{\ell\zeta}, \qquad P_i^{\zeta - \min(\omega,1)} \le P_i^{\zeta}.$$

For $0 < \omega \le 11/10$, the strip bound [Ope26a, Lemma 4.8] contributes $U^{3/5+\varepsilon}$, giving the relative dyad exponent

$$\gamma + h(\zeta - 1/6) + l_y(\omega - 1) + d(8/5 - \zeta) + O(\varepsilon).$$

At $d = h$, it is bounded below by

$$\begin{aligned}\gamma + 43h/30 + l_y(\omega - 1) &\ge 43h/30 - \beta_* l_y + \gamma(1 - l_y)\\ &> 43/40 - (7/8)(13/14) = 21/80 > 0.\end{aligned}$$

For $\omega > 11/10$, Euler convergence gives instead $\gamma + 5h/6 + l_y(\omega - 1) > 0$.

For $-1/100 \le \omega \le 0$, admissibility requires $s_r + \omega > 1$ and $\zeta \ge 17/50$. Even after suppressing puncture and deletion costs, the endpoint is $> 21/80$; the selected-divisor factor adds $-\omega\ell \ge 0$ and deleted numerator factors add only nonnegative conductor powers.

Absolute and principal contours. For any eligible fixed $s_r, w_r > 1$ and sufficiently large z_r, Euler convergence and the same local table give an endpoint exponent relative to $C(B)$ of

$$(s_r - B) + l_y(w_r - 1) + 5h/6 > 1 - B + 5h/6 > 3/4.$$

It remains positive relative to $C(\beta_*)$, since $\beta_* - B < 1/24$. At $d = h$ all z_r terms cancel. Principal contour changes affect only error reserves: the residues $w = 1$, $z = 1/6$ and the signal exponent are fixed. Thus the displayed central, unselected, global and absolute comparisons cannot evade $B \ge H_\xi$. Only the stated local, strip-growth and Euler-convergence estimates are included; new estimates or residue extractions change the system. □

34.2. **Conclusion of the optimality proof.** Let $\mathscr{A}$ be the set of self-compatible endpoint certificates for the loss-free comparison with

$$l_x, l_y > \ell > 0, \quad b = l_y - l_x > 0, \quad M = l_x + l_y, \quad h = 1 - l_x + \ell > 0,$$
$$13/18 \le \kappa \le 1, \qquad B \le (1 + \kappa)/2,$$
$$B \ge B_{\mathrm{low}}, \qquad E_B(d) \le 0 \quad \text{for every permitted zero and amplitude class.}$$

The supplied count choices are those of Section 34.1; the contour, floor and norm choices are those covered above. Prime supply is unlimited and all positive losses are suppressed. Analytic applications retain the source's coefficient, mask, support and domain conditions. Uniform scalar comparisons include classes whose nonemptiness is unknown; a fixed positive gap $\beta_* - B$ does not replace the requirement $E_B(d) \le 0$.

Theorem 34.4 (Optimality of the displayed exponent comparison)**.** *For the self-compatible endpoint certificates $\mathfrak{C} \in \mathscr{A}$,*

$$\inf_{\mathfrak{C} \in \mathscr{A}} B(\mathfrak{C}) = B_\infty.$$

Every certificate has $B(\mathfrak{C}) \ge B_\infty$; equality is attained in the loss-free system by (33.2). This includes free M, all detecting cutoffs in $[1, 3/2]$, the specified marked and unmarked choices, and the auxiliary choices above.

Proof. If $B < B_\infty$, Lemmas 34.2 and 34.3 give $B \geq \max(L_0, L_K, H_\xi)$. Proposition 34.1 and endpoint compatibility then give $B \geq B_\infty$, a contradiction. Conversely, at (33.2),

$$B_{\mathrm{low}} = B_\infty = L_0 = L_K = H_{\xi_\infty}, \qquad B_\infty = (1+\kappa_\infty)/2.$$

Lemma 31.1 gives every central comparison; Lemma 31.2 supplies the remaining strict reserves. Hence this geometry belongs to the loss-free system. Its analytic conclusion is the limit of the authorized finite steps in Theorem 27.1, not a moment application with an unproved endpoint premise. □

Corollary 34.5 (Arbitrary successive admissible constructions)**.** *Allow an arbitrary different geometry, cutoff, operative parameter and eligible auxiliary choices at every improving step of this system. If each moment premise comes from the preceding proved boundary, every output boundary and every real limit is at least B_∞. The sequence in Section 33 attains this infimum.*

Proof. For an authorized improving step,

$$\beta_* \leq \widehat{B}, \qquad \widehat{B} \leq (1+\kappa)/2, \qquad B \leq \widehat{B} \Longrightarrow B \leq (1+\kappa)/2.$$

The theorem applies independently to each step; its closed lower bound persists under real limits. □

This is optimality of supplied uniform upper-bound exponents. Information excluding a critical class, a relation between detector witnesses, or a stronger analytic estimate changes the system.

34.3. **Rational lengths and additional parameters.**

Proposition 34.6 (Rational one-step certificate)**.** *Put*

$$(\kappa_{\mathrm{cp}}, b_{\mathrm{cp}}, \ell_{\mathrm{cp}}) = \left(\frac{74991414}{10^8}, \frac{617}{5000}, \frac{5005157}{30000000}\right), \qquad 11/12 - \ell_{\mathrm{cp}}/4 = B_{\mathrm{cp}}.$$

With the coefficient, mask, support and smooth-weight hypotheses of the original modified average, these parameters give

$$\beta_* \leq B_{\mathrm{new}} \quad \Longrightarrow \quad \beta_* \leq B_{\mathrm{cp}}.$$

Thus the rational checkpoint also follows in one step from Part III.

Proof. Suppress the subscript cp and set $l_x = (1-b-\ell)/2$, $l_y = l_x + b$, $h = 1 - l_x + \ell$ and $c = 2/(3\kappa)$. The moment premise follows by positive squaring:

$$q := \frac{1507 - 1653(1+\kappa)/2}{2} = \frac{6069596329}{200000000} > 0,$$

$$921 - q^2 = \frac{402989723759}{40000000000000000} > 0 \quad \Longrightarrow \quad B_{\mathrm{new}} < (1+\kappa)/2, \qquad 13/18 < \kappa < 3/4.$$

Use (31.2) at this rational triple, without imposing the critical-class equations, and write

$$Q = A(y)\delta^2 + B_q(y)\delta + C_q(y), \quad A(y) = \sum_{j=0}^{3} A_j y^j, \quad 4A(y)C_q(y) - B_q(y)^2 = \sum_{j=0}^{4} d_j y^j.$$

Let $\mathcal{S}_{\mathrm{cp}}$ consist of the 24 expressions in (31.3) after omitting $36\xi^2 - 78\xi + 25$, κ', B', $\kappa' - 1000B'$ and $-p'$, and substituting the rational parameters above into the remaining expressions. Exact substitution gives the *rational checkpoint certificate* $\mathcal{C}_{\mathrm{cp}}$:

$$A_j > \frac{1}{4}\ (0 \leq j \leq 3), \qquad d_j > \frac{1}{10^7}\ (0 \leq j \leq 4), \qquad s > \frac{1}{25000}\ (s \in \mathcal{S}_{\mathrm{cp}}). \tag{34.9}$$

The accompanying exact-arithmetic certificate file [Liu26] records these lower bounds and all 33 exact values in its `rational_checkpoint` entry. Consequently, for $y = 1/2 - x \geq 0$,

$$4A(y)Q = (2A(y)\delta + B_q(y))^2 + \sum_{j=0}^{4} d_j y^j > 0, \qquad E_B(h) = -Q/(2\mathcal{J}_\kappa) < 0$$

on $[0, 5/6] \times [0, 1/2]$.

The side certificate gives the balanced direct hypotheses, $\ell/h > 1/5$, the principal domain and all floor, intermediate, small-norm, capacity and rounding reserves used in Section 32. Proposition 29.1 therefore applies with the preceding moment premise, and its exponent satisfies $0 < \partial_d E_B < 2$ on the actual class range. If $\Delta = \beta_* - B > 0$, the same small-, intermediate- and large-norm estimates, with $\mathcal{C}_{\mathrm{cp}}$ in place of the two ξ-family certificates, verify Proposition 20.1. Together with the direct and principal estimates, this gives

$$|J_\eta(Z)| \ll_\eta Z^{C_b(B)+\Delta/2}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_b(\beta_*)-\sigma}, \qquad \sigma > 0,$$

with σ independent of η and $|H_\eta - 1| \leq 1/2$ on $\operatorname{Re} s > B$. Proposition 3.1 is the desired contradiction; Corollary 14.1 transfers the bound to Dirichlet L-functions. □

At the limit,

$$\ell_\infty = 11/3 - 4B_\infty, \qquad 53835237b_\infty^3 + 381756141b_\infty^2 + 94706196b_\infty - 17602208 = 0.$$

The B-cubic (33.4) and this b-cubic reduce modulo 7 to

$$2X^3 - 3X^2 - X + 3, \qquad X^3 - 3X^2 - 3X - 1,$$

with value lists

$$(3, 1, 5, 6, 2, 5, 6), \qquad (6, 1, 3, 4, 3, 6, 5).$$

Their leading coefficients are nonzero and neither has a root; Gauss's lemma proves irreducibility. Thus

$$[\mathbb{Q}(B_\infty) : \mathbb{Q}] = [\mathbb{Q}(\ell_\infty) : \mathbb{Q}] = [\mathbb{Q}(b_\infty) : \mathbb{Q}] = 3.$$

The supplement also verifies the eliminations.

Rational parameters have the same infimum. Given $\epsilon > 0$, choose a finite bootstrap output $B_n < B_\infty + \epsilon/2$ and a slightly larger rational target $B_n < \widetilde{B} < \min\{B_\infty + \epsilon, (1 + \kappa_n)/2\}$. This makes the direct and central inequalities strict. On the compact class rectangle their exponents are continuous; a rational $\widetilde{\kappa} > \kappa_n$ sufficiently close to κ_n preserves the preceding premise, and sufficiently close rational lengths preserve all strict comparisons and side reserves. Hence

$$\inf_{\mathscr{A} \text{ with rational parameters}} B = \inf_{\mathscr{A}} B = B_\infty.$$

The first infimum is not attained: positivity of the dual weights forces $L_0 = L_K = H_\xi = B_\infty$ at equality, so $\ell = \ell_\infty \notin \mathbb{Q}$.

The total length M is already free, balance occurs at equality, and t is already minimized. The variables δ, x, d are class coordinates requiring uniform control. Prime caps, meshes and smoothing losses can only weaken the relaxed comparison; the central and remote contour coordinates cancel at $d = h$. A stronger analytic input can change the obstruction. For example, with the present D, P, formally vary α and put

$$R = 1 - \delta + \frac{(\alpha - \delta)\delta P}{2\mathcal{J}}, \qquad \mathcal{J} = (\alpha - \delta)D + \delta P.$$

Then

$$\partial_\alpha R = \frac{\delta^2 P^2}{2\mathcal{J}^2} > 0, \qquad \partial_\alpha e_L(r) = r - 1 > 0 \quad (r > 1).$$

A smaller *proved* amplification exponent than 5/6 improves these envelopes. Stronger marked capacities, a joint witness moment, better reflected or Gram estimates, or restrictions on actual critical classes could likewise alter the comparison. Such changes require new results with the original coefficient, mask and height hypotheses; they are not additional free parameters in the theorem.

Appendix A. Theta coefficient conventions

We match the theta coefficients used in Section 4 with the cited cusp expansions, recording the conjugations and fixed normalizing factors explicitly. This identifies the same arithmetic normalization throughout Parts I–IV.

The coefficient formulas, support and normalization are those of [DR24, equations (5.7)–(5.16) and Appendix A], with the conjugation convention in [Ope26a, Proposition 5.1, proof, cusp expansion and infinity coefficient]. Precisely, if d_i^{DR} denotes the coefficient in [DR24, equation (5.12)], then our three functions are

$$(d_0, d_1, d_2)(\mu) = \big(\overline{d_1^{\mathrm{DR}}(-\mu)}, \overline{d_{19}^{\mathrm{DR}}(-\mu)}, \overline{d_{10}^{\mathrm{DR}}(-\mu)}\big);$$

the corresponding coefficient formulas are rows $1, 19, 10$ of [DR24, Appendix A], reproducing [Pat77, Table III, p. 151]. The matrices H_j are the representatives $\gamma_1, \gamma_{19}, \gamma_{10}$ specified in [DR24, Section 5.1, before equation (5.9)]. These locators refer to the version specified in the bibliography.

For clarity, write τ^{DR} for the infinity coefficient in [DR24, equation (5.7)], and $g^{\mathrm{DR}}(1,c) = \sum_{x \bmod c}(x/c)_3 \check{e}(x/c)$ for its unnormalized cubic Gauss sum. The fourth case of that formula, with its integer exponent equal to zero, gives

$$\tau^{\mathrm{DR}}(\pm\lambda^{-3}cn^3) = 3^{5/2}\frac{|n|}{|c|}\overline{g^{\mathrm{DR}}(1,c)}.$$

[DR24, equation (5.16), p. 19] uses the conjugated coefficient at $-\mu$, while [DR24, equation (5.7), p. 17] conjugates the Gauss sum in the infinity coefficient; hence, since $|c| = \sqrt{q_c}$,

$$d_0(\lambda^{-3}cn^3) = \overline{\tau^{\mathrm{DR}}(-\lambda^{-3}cn^3)} = 3^{5/2}|n|\widetilde{g}_3(c).$$

Finally, [DR24, equations (5.6), (5.15), (5.16)] gives the constant term $3^{5/2}v^{2/3}/2$ for the representative γ_1, and zero for γ_{19}, γ_{10}, exactly as in Section 4.1.

Appendix B. Index of cited results

The first table indexes the inputs and deductions in Part I. The second records how the same inputs are used at the new parameters in Parts II and III. The third records the additional inputs and deductions in Part IV. Each citation identifies the proved result used in that step.

Reference	Function in the argument
[Ope26a, Proposition 2.1]	Holomorphic continuation with a uniform exponent.
[Ope26a, Lemmas 4.1–4.4]	Ray characters, Gauss identities and reciprocity phases.
[Ope26a, Lemmas 4.5–4.10]	Smooth separation, transform decay and Hecke bounds.
[Ope26a, Proposition 5.1; Lemmas 5.2–5.8]	Completed theta functional equation and unweighted mean square.
[Ope26a, Lemmas 6.1–6.2; Proposition 6.3]	Balanced additive mean square and $Z^{1/4+\varepsilon}$ estimate.
[Ope26a, Lemma 7.1]	Exact scalar Euler quotient and holomorphic correction.

Reference	Function in the argument
[Ope26a, Lemmas 8.1–8.2; Proposition 8.3]	Zero-free rectangles and simultaneous large character polynomials.
[Ope26a, Lemma 9.1; Proposition 9.2]	Sextic sieve and preliminary character count.
[Ope26a, Lemmas 10.2–10.6]	Contour tails, exponent accounting and principal residues.
[Ope26a, Lemma 11.1; Proposition 11.2; Theorem 3.1]	Height choice, first comparison and 11/12 conclusion.
[Ope26a, Lemmas 13.1–13.4]	Fixed-ray prime asymptotics and complete correlations.
[Ope26a, Corollary 14.1; Lemmas 14.2–14.3]	Divisibility-restricted theta reflection and mean square.
[Ope26a, Lemma 15.1; Propositions 15.2–15.3]	Modified completed and additive norms; $Z^{3/16+\varepsilon}$.
[Ope26a, Proposition 16.1; Lemma 16.2]	Joint local errors and absolute tuple bounds.
[Ope26a, Lemmas 17.1–17.3, 17.5–17.6; Corollary 17.4]	Möbius-weighted moment, finite induction and sixth-power amplification.
[Ope26a, Lemmas 18.1–18.3]	Unweighted fourth moment and cancellation when the products of scales are equal.
[Ope26a, Lemma 19.1; Proposition 19.2]	Prime sum bounds and refined character counts.
[Ope26a, Lemmas 20.1–20.2; Proposition 20.3]	Final exponent inequality and uniform parameter choices.
[Ope26a, Proposition 11.3]	Transfer to Dirichlet functions.

Input	Use in Parts II and III
[Ope26a, Lemma 14.3]	General reflected exponents: Lemmas 16.1 and 22.1.
[Ope26a, Proposition 15.2]	Direct bounds with the full additive majorant: Propositions 16.2 and 22.2.
[Ope26a, Lemma 7.1]	Extended principal Euler region: Lemma 17.1.
[Ope26a, Lemma 13.1]; [Ope26a, Lemmas 10.2–10.6]	Normalization, changed-scale residues and tails: Proposition 17.2 and Lemma 19.2.
[Ope26a, Proposition 8.3]; [Ope26a, Lemmas 17.1, 17.6, 18.1, 19.1]	Fixed-κ count with strict reserves: Proposition 18.1.
[Ope26a, Proposition 16.1]; [Ope26a, Lemma 16.2]	Joint conductor deficit and changed-scale high exponent: Proposition 18.2.
[Ope26a, Lemmas 4.5–4.10, 10.2–10.4, 11.1]	Smooth separation and combination of height estimates: Proposition 20.1.
Proposition 3.1 and Corollary 14.1	Continuation contradiction and Dirichlet transfer at both new boundaries.

Input	Use in Part IV
[Ope26a, Lemma 4.9]; [Ope26a, Section 18, proof of Lemma 18.1]	Prime displacement, width induction, and the extended moment: Proposition 28.1. The source proof is extended, rather than its printed parameter range invoked unchanged.
[Ope26a, Proposition 8.3]; [Ope26a, Lemmas 17.1, 17.6, 19.1]	The original common detector and inverse amplification, combined with the extended plain moment in Proposition 29.1.
(18.5) and the exact Bernstein calculation of Section 31	Critical geometry, the universal endpoint inequality, and strict side reserves: Lemmas 30.1, 31.1, and 31.2.
Propositions 16.2, 17.2, 18.2; Lemma 17.1	Changed-scale direct and high representations, principal residues, and every norm range in Proposition 32.1.
Proposition 3.1; Corollary 14.1	One-step continuation contradiction, real-bound iteration, and Dirichlet transfer in Theorem 27.1.

References

[Ber12] S. Bernstein, Démonstration du théorème de Weierstrass fondée sur le calcul des probabilités, *Communications de la Société mathématique de Kharkow*, 2nd series, **13** (1912), no. 1, 1–2. Original paper and bibliographic record.

[BGL14] V. Blomer, L. Goldmakher, and B. Louvel, *L-functions with n-th-order twists*, Int. Math. Res. Not. IMRN **2014**, no. 7, 1925–1955. doi:10.1093/imrn/rns257.

[DR24] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, Ann. of Math. (2) **200** (2024), no. 3, 967–1057. doi:10.4007/annals.2024.200.3.3. Equation and appendix locators in this paper refer to arXiv:2109.07463v3, 14 May 2024.

[Far12] R. T. Farouki, The Bernstein polynomial basis: A centennial retrospective, *Computer Aided Geometric Design* **29** (2012), no. 6, 379–419. doi:10.1016/j.cagd.2012.03.001.

[GZ20] P. Gao and L. Zhao, *Moments of quadratic Hecke L-functions of imaginary quadratic number fields*, J. Number Theory **209** (2020), 359–377. doi:10.1016/j.jnt.2019.09.002. The equation locator refers to arXiv:1707.00091v3.

[GZ24] P. Gao and L. Zhao, *Moments and one level density of sextic Hecke L-functions*, Funct. Approx. Comment. Math. **70** (2024), no. 1, 7–28. doi:10.7169/facm/2057. The lemma locator refers to arXiv:2201.01885v2.

[GL13] L. Goldmakher and B. Louvel, *A quadratic large sieve inequality over number fields*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 2, 193–212. doi:10.1017/S0305004112000370.

[GM26] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) **203** (2026), no. 2, 623–675. doi:10.4007/annals.2026.203.2.6. The theorem locator refers to arXiv:2405.20552v2, 7 April 2026.

[HB95] D. R. Heath-Brown, *A mean value estimate for real character sums*, Acta Arith. **72** (1995), no. 3, 235–275. doi:10.4064/aa-72-3-235-275.

[HB00] D. R. Heath-Brown, *Kummer's conjecture for cubic Gauss sums*, Israel J. Math. **120**, part A (2000), 97–124. doi:10.1007/s11856-000-1273-y.

[Hux68] M. N. Huxley, *The large sieve inequality for algebraic number fields*, Mathematika **15** (1968), no. 2, 178–187. doi:10.1112/S0025579300002540.

[IK04] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004. doi:10.1090/coll/053.

[Kub69] T. Kubota, *On automorphic functions and the reciprocity law in a number field*, Lectures in Mathematics, Department of Mathematics, Kyoto University, no. 2, Kinokuniya Book-Store, Tokyo, 1969. Kyoto University repository.

[Liu26] B. Liu, *Lean formalizations of the rational and algebraic zero-free extensions*, accompanying source-code repository, 2026, Lean 4.34.1. GitHub repository for the paper and Lean formalizations. Paths below are relative to the repository root. The consolidated entry for Parts I–III is `combine_v2-2_lean/PartsI_III.lean`, with public namespaces `CombinedPaper`, `RHZeroFreeExtension` and `RHV2`; the Part IV entry is

`combine_v2-2_lean/PartIV.lean`, with final conclusions in namespace `CombineV22`. The latter namespace exports `beta_star_le`, `hecke_nonzero`, `dirichlet_nonzero` and `zeta_nonzero`. The verifier `combine_v2-2_lean/check.py` replays the eight consolidated OAI foundation files and both manuscript files with `-trust=0`, then checks the explicit final signatures. The audited transitive axioms are `propext`, `Classical.choice` and `Quot.sound`.

[MP24] J. Maynard and K. Pratt, *Half-isolated zeros and zero-density estimates*, Int. Math. Res. Not. IMRN **2024**, no. 19, 12978–13014. doi:10.1093/imrn/rnae191. The appendix locator refers to the arXiv manuscript.

[M+17] A. Meurer et al., *SymPy: symbolic computing in Python*, PeerJ Comput. Sci. **3** (2017), e103. doi:10.7717/peerj-cs.103.

[Mil20] J. S. Milne, *Class Field Theory*, course notes, version 4.03, 6 August 2020. Author's notes.

[Ope26a] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\operatorname{Re} s > 7/8$, 30 September 2026. Preprint, 30 September version. Public manuscript. All result and equation locators for this source refer to this dated version.

[Ope26b] OpenAI, *The Quasi-Riemann Hypothesis (alternate* 11/12 *proof)*, preprint, 5 October 2026. Public manuscript.

[Ope26c] OpenAI, *The quasi-Riemann hypothesis: scope of the Lean formalization*, repository documentation, accessed 7 October 2026. Scope and comparator links.

[Pat77] S. J. Patterson, *A cubic analogue of the theta series*, J. Reine Angew. Math. **296** (1977), 125–161. doi:10.1515/crll.1977.296.125.

[Rie16] M. Riesz, *Sur l'hypothèse de Riemann*, Acta Math. **40** (1916), 185–190. Original paper and bibliographic record.

[TZ19] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, Algebra Number Theory **13** (2019), no. 5, 1039–1068. doi:10.2140/ant.2019.13.1039.

Department of Mathematics, Purdue University, West Lafayette, IN, 47907, USA
Email address: `liu2053@purdue.edu`